The LisbOn KInetics Global Model
Reference manual

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The following is the reference manual of the LisbOn KInetics Global Model (LoKI-GM) simulation tool [1,2], intended to model non-equilibrium low-temperature plasmas (LTPs), produced from different gases or gas mixtures under a wide range of working conditions.

LoKI-GM comprises two modules, that can run self-consistently coupled or as standalone tools:

- a Boltzmann solver (LoKI-B) for the electron Boltzmann equation [3,4];
- a Chemical solver (LoKI-C) for the global kinetic model(s) of pure gases or gas mixtures [1].

LoKI-GM leverages on the scientific heritage in the field of non-equilibrium plasma kinetics of the Portuguese group N-Plasmas Reactive: Modeling and Engineering, N-PRiME [5] of *Instituto de Plasmas e Fusão Nuclear* with *Instituto Superior Técnico*, and it was developed resorting to well-grounded scientific foundations established years ago [6,7]. Several numerical improvements and updates have recently been implemented in LoKI-GM, and further developments are ongoing, as part of a fruitful collaboration with the *Departamento de Física* of *Facultad de Ciencias* with *Universidad de Córdoba* (regarding LoKI-C) and the group Elementary Processes in Gas Discharges, EPG [8] of the Eindhoven University of Technology (regarding LoKI-B).

The LisbOn KInetics Global Model is an open-source tool, distributed under the *GNU general public license*, and developed with an ontology that prioritizes the clear separation between tool and data. Its implementation follows a flexible and extensible object-oriented design, using the following programming languages:

- MATLAB® [9] for LoKI-B and LoKI-C, leveraging its matrix-based architecture;
- C++ [10] for LoKI-B++, taking advantage of the higher performance of a compiled language, and linked with Eigen [11], a C++ template library for linear algebra.

LoKI-B++ is a C++ counterpart of MATLAB's LoKI-B, replicating essentially all of its original features. Unless otherwise specified, the acronym LoKI-B in this manual refers to either version of the code. For simplicity, and unless otherwise specified, the acronym LoKI in this manual refers to the code LoKI-GM.

LoKI-B, LoKI-B++ and LoKI-C are part of the LoKI tool suite and are freely available at `https://github.com/LoKI-suite`, for users to perform LTPs simulations, and for modelers / developers who are invited to continue testing the tool and/or to contribute for its development and improvement.

Contents

1 Introduction **3**

1.1 General features . . . 3

1.2 This reference manual . . . 3

1.3 How to contact us . . . 4

1.4 How to reference the code . . . 4

1.5 Acknowledgments . . . 4

2 The LisbOn KInetics Global Model **5**

2.1 General description . . . 5

2.2 The electron Boltzmann equation . . . 8

2.3 The time-dependent electron Boltzmann equation . . . 11

2.4 Prescribed EEDFs . . . 12

2.5 The plasma chemistry global model . . . 13

2.5.1 Rate coefficients for volume and surface reactions . . . 14

2.5.2 Rate coefficients for the transport of species . . . 15

2.6 Thermal model for the gas/plasma . . . 19

3 Simulation input/output **22**

3.1 Output of LoKI-B . . . 22

3.1.1 Swarm parameters . . . 22

3.1.2 Power-transfer terms . . . 25

3.2 Output of LoKI-C . . . 27

4 Workflow of LoKI-GM **29**

4.1 Steady-state simulations . . . 29

4.2 Quasi-stationary simulations . . . 31

4.3 Post-discharge simulations . . . 32

4.4 Closure conditions . . . 32

5 Numerical solution **33**

5.1 Numerical solution of LoKI-B . . . 33

5.1.1 Discretisation of the EBE . . . 33

5.1.2 The particle balance equation . . . 38

5.1.3 The power balance equation . . . 42

5.1.4 Conservation conditions for the electron-electron collision operator . . . 48

5.1.5 Solution of LoKI-B . . . 50

5.2 Numerical solution of LoKI-C . . . 52

6 Validity limits and control (for LoKI-B) **53**

References **55**

1 Introduction

Overall, LoKI-GM provides the combined chemical and transport description of plasma charged/neutral species, both in volume and surface phases, for user-defined working conditions: mixture compositions, pressure, dimensions and excitation features (glow (DC/HF), post-discharge, pulsed) [1].

1.1 General features

LoKI-GM comprises two modules, that can run self-consistently coupled or as standalone tools:

- a Boltzmann solver (LoKI-B) for the electron Boltzmann equation [3, 4];
- a Chemical solver (LoKI-C) for the global kinetic model(s) of pure gases or gas mixtures [1].

LoKI-B solves the space independent form of the two-term electron Boltzmann equation (EBE) to calculate the isotropic and the anisotropic parts of the electron distribution function (the former usually termed the *electron energy distribution function*, EEDF), and the associated electron macroscopic parameters. LoKI-B applies to non-magnetised non-equilibrium LTPs, excited by DC/HF electric fields or time-dependent (non-oscillatory) electric fields (a new feature introduced in LoKI-B_v2.0.0, when LoKI-B runs independently) [4] from different gases or gas mixtures. The tool uses a stationary description for DC fields, a Fourier time-expansion description for HF fields, and a time-dependent description for time-varying fields.

LoKI-B was developed as a response to the need of having an electron Boltzmann solver easily addressing the simulation of the electron kinetics in any complex gas mixture (of atomic / molecular species), describing first and second-kind electron collisions with any target state (electronic, vibrational and rotational), characterized by any user-prescribed population. LoKI-B includes electron-electron collisions, it handles rotational collisions adopting either a discrete formulation or a more convenient continuous approximation, and it accounts for variations in the electron population caused by non-conservative events (ionisation and attachment) by adopting either a space-homogeneous exponential temporal growth or a time-constant exponential spatial growth of the electron density.

LoKI-C solves the system of zero-dimensional (volume average) rate balance equations for the most relevant charged and neutral species in the plasma. LoKI-C receives as input data the kinetic schemes for the gas/plasma/surface system under study, via an intuitive csv-like input file, and gives as output the particle densities of the different gas/plasma/surface species, the corresponding creation/destruction reaction rates, and the reduced electric field (and any related quantity, such as the discharge current or the discharge power-density). The tool uses several modules to describe the mechanisms (collisional, radiative and transport) controlling the creation/destruction of species, namely various transport models for the charged particles (ambipolar diffusion, effective diffusion, and Liberman and Lee's low-pressure model) and for the neutral particles (multi-component diffusion of species following Wilke's approach, and considering wall recombination). LoKI-C includes also a gas/plasma thermal model, for the self-consistent calculation of the gas temperature, and supports multi-component mean-field microkinetic mesoscopic models to handle surface kinetics in a fully coupled way with volume kinetics. The inclusion of several transport models and support for surface kinetics models (see sections 2.5.1 and 2.5.2) are distinguishing features of LoKI-C compared with other global chemistry models.

Essential information for quickly obtaining and running LoKI-GM, understanding the files included in its distribution and the structure of its output data, and learning how to prepare input files, is provided in the User Guide [12].

1.2 This reference manual

This reference manual begins by introducing the LisbOn KInetics Global Model (LoKI-GM) (see section 2). It then describes the general types of simulations supported and the ontology used to define the systems under study (see section 2.1). The formulations adopted to solve the EBE in LoKI-B are presented in Ssections 2.2 and 2.3, while the plasma chemistry global model implemented in LoKI-C is described in section 2.5. The thermal model used in LoKI-GM is presented in section 2.6.

The simulation input/output of LoKI-B and LoKI-C are detailed in sections 3.1 and 3.2, respectively. The overall workflow of LoKI-GM is described in section 4, distinguishing between *steady-state simulations*,

quasi-stationary simulations (based on either the *Local Field Approximation* or the *Local Mean Energy Approximation*), and *post-discharge simulations*, and highlighting the closure conditions implemented in the code.

Essential aspects of the numerical solution in LoKI-B and LoKI-C are provided in section 5. This includes the discretisation of the EBE (see section 5.1), the formulation and discretisation of the electron particle-balance and power-balance equations (see Sections 5.1.1 and 5.1.2), and the numerical methods used to solve the EBE in LoKI-B (see section 5.1.5) and the system of rate balance equations in LoKI-C (see section 5.2).

Finally, the validity limits and control of LoKI-B are discussed in section 6.

1.3 How to contact us

After downloading and installing LoKI-GM, and especially if you intend to interact with us, you are invited to send a short message
to loki@tecnico.ulisboa.pt
with **subject** LoKI-GM
just giving your **name** and **affiliation**.

Also, you are much welcome to report any problem or bug you may find using the code, or to send any feedback with suggestions for further developments.

1.4 How to reference the code

LoKI-GM is the result of the efforts of the Portuguese group N-Plasmas Reactive: Modeling and Engineering (N-PRiME) of *Instituto de Plasmas e Fusão Nuclear* with *Instituto Superior Técnico*, the *Departamento de Física* of *Facultad de Ciencias* with *Universidad de Córdoba* and the group Elementary Processes in Gas Discharges (EPG) of the Eindhoven University of Technology. These groups decided to share the outcome of its research on code development with the members of the Low-Temperature Plasmas community.

When using the code in your work, please give proper credits to the main developers, by adding the following citations.

For LoKI-B
- Tejero A *et al. The LisbOn KInetics Boltzmann solver* 2019 *Plasma Sources Sci. Technol.* **28** 043001
- Tejero A *et al. On the quasi-stationary approach to solve the electron Boltzmann equation in pulsed plasmas* 2021 *Plasma Sources Sci. Technol.* **30** 065008
- Tejero A *et al. The LisbOn Kinetics Boltzmann solver*, DOI: https://doi.org/10.57859/ulisboa-istipfn.000062, URL: https://github.com/LoKI-Suite/LoKI-GM

For LoKI-GM
- Alves L L *et al. LoKI-GM: a global model tool for plasma chemistry studies* 2026 arXiv preprint, arxiv:2607.27234. Manuscript submitted for publication to *Plasma Sources Sci. Technol.* (under review)
- Alves L L *et al. The LisbOn Kinetics Global Model*, DOI: https://doi.org/10.57859/ulisboa-istipfn.000062, URL: https://github.com/LoKI-Suite/LoKI-GM

1.5 Acknowledgments

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2 The LisbOn KInetics Global Model

The study of plasma chemistry is key for developing plasma-driven applications (*e.g.* material processing, plasma medicine, environmental control, energy storage, etc), since it can provide information on the plasma-enhanced production of reactive species, the analysis of chemical reaction pathways, and the transfer of energy between the different species. These aspects are extremely relevant for understanding the processes and suggesting optimisation routes. However, modelling the plasma chemistry can be challenging, because it addresses various type of particles (charged and neutral, in several excited states), intrinsically in non-equilibrium with each other, which undergo a large number of reactions (involving collisional, radiative and electrostatic interactions).

Global (zero-dimensional or spatially averaged) models are widely employed to study complex chemistries in LTPs. By adopting a spatially averaged description of the plasma, these models substantially reduce the computational cost while still enabling the identification of dominant reaction pathways and the creation and destruction mechanisms of the most relevant plasma species. In their most general formulation, global models involve the coupled solution of an electron Boltzmann solver, to describe the electron kinetics, and a Chemistry solver, to account for the kinetics of the heavy species, as implemented in the LoKI-GM simulation tool [1].

The scientific foundations for the numerical solution of the EBE (as adopted in LoKI-B) were established years ago [6, 7], and are based on the early works of Lorentz, Holstein, Allis and Delcroix [13–16]. These authors expand the electron distribution function in Legendre polynomials $P_l(\cos(\theta))$ around the angle θ, defining the spatial orientation of the velocity vector with respect to the polar direction of the total anisotropy (produced by electric fields and density gradients). The topical review [3] presents the basic features of LoKI-B (also included in this reference manual), along with examples of results obtained for different model and real gases, verifying the tool against analytical results, benchmarking it against numerical calculations, and validating the output by comparison with available measurements of swarm parameters.

In addition, the topical review [17], integrated in the collection *Foundations of low-temperature plasmas and their applications* published by *Plasma Sources Science and Technology*, gives a summary of the essentials concerning the two-term EBE and plasma global models, providing an extensive list of references for complementary in-depth reading. The users are advised to carefully read this literature, for obtaining details about these models and their numerical solution.

2.1 General description

LoKI-GM provides a self-consistent description of the chemical kinetics and transport properties of a plasma or the post-discharge of a plasma with electron density n_e, created from a general mixture of atomic/molecular gases at pressure p, temperature T_g and input flow Q_{sccm} (in sccm).

When describing *active plasmas*, the excitation assumes an electric field directed along the z-axis $\vec{E}(t) = -E(t)\vec{e}_z$, where $E(t)$ is defined as follows:

- for *steady-state simulations*, in which LoKI-C and LoKI-B are run in a coupled manner, the time-dependent electric field satisfies
$$E(t) = E_p \cos(\omega t) \quad , \tag{1}$$
where the oscillation frequency ω is either
  - $\omega = 0$, for a DC excitation;
  - $\omega \gg \nu_c$ (ν_c is the typical electron-neutral collision frequency), for a HF excitation.

  In this case, the electric-field amplitude (and any related quantity, such as the discharge current or the discharge power-density) is self-consistently calculated by imposing charge neutrality (see section 4);
- for *quasi-stationary simulations*, in which case LoKI-C runs as a standalone tool, the time-dependent electric field $E(t)$ (or any related quantity, such as the discharge current or the discharge power-density) is defined by the user. In this case, LoKI-GM uses precomputed lookup tables of electron macroscopic parameters, at certain electric-field values, which are then used in the chemistry solution. The lookup tables are used to obtain electron swarm parameters, rate coefficients, and power transfer as functions of the electric field (for the *Local Field Approximation*, LFA) or the electron mean energy (for the *Local Mean Energy Approximation*, LEA). This configuration imposes charge neutrality, and can be applied to pulsed discharges (see section 4);

The electric field (1) can also be applied to *post-discharge simulations*. These simulations typically begin with a coupled LoKI-C/LoKI-B steady-state calculation, followed by a standalone LoKI-C simulation performed at $E = 0$ while imposing charge neutrality, in order to describe the time-dependent post-discharge (see section 4). It is also possible to perform only the post-discharge simulation, starting from suitable initial conditions, by running LoKI-C in standalone mode.

LoKI-GM supports also electron kinetics simulations (including swarm analysis), in which case LoKI-B runs as a standalone tool for reduced electric fields specified as discrete value(s) (for *electron kinetics stationary simulations*, see section 2.2), or as a time-dependent (non-oscillatory, $\omega = 0$) function (for *electron kinetics time-dependent simulations* [4], see section 2.3).

The full time-dependent coupling between LoKI-B and LoKI-C is not yet supported [1].
Please refer to the User Guide [12] for details on how to configure the setup file for the different simulation types available in LoKI-GM.

LoKI handles simulations in any atomic / molecular gas mixture, considering collisions with target states featuring internal degrees of freedom (electronic, vibrational and rotational), organized as a system of "Russian puppets" [2]. The populations of these states are self-consistently calculated when using the LoKI simulation tool, and are prescribed by the user when running LoKI-B as standalone tool. Table 1 exemplifies the system of states for a N_2-N-He gas mixture. Here,

- **System:** is the ensemble of all the gases present in the simulation ($N_2 \cup N \cup He \cup \cdots$).
- **Gas:** refers to each type of atom or molecule present in the simulation (N_2, N, He, ...). Each gas k has mass M_k and individual density N_k, being composed by several electronic / vibrational / rotational levels.
- **State:** specific configurations of electronic (i), vibrational (v) and/or rotational (J) levels belonging to a given gas (e.g, N_2(X), N_2(X,v=0), N_2(X,v=0,J=2), N(^{4}S), He(2^3S), ...). Each gas k is composed by several electronic states k_i with absolute density N_{k_i}; each k_i-state can include several vibrational states k_{i_v} with absolute density $N_{k_{i_v}}$, each one composed by several rotational states $k_{i_{v_J}}$ with absolute density $N_{k_{i_{v_J}}}$.
- **Parent (state):** state of a certain gas comprising a set of child states. Parent states can be electronic or vibrational, but not rotational (*e.g.*, N_2(X) is the parent of N_2(X,v=0,1,2,...)). Each parent level comprises all of its children.
- **Child (states):** ensemble of states that share the same parent. Child states can be vibrational or rotational, but not electronic (*e.g.*, N_2(X,v=0,J=0,1,2,...) are the children of N_2(X,v=0)).
- **Sibling (states):** relation between the states that share the same parent (*e.g.*, N_2(X,v=0,1,2,...) are siblings).
- **Childless (states):** ensemble of the innermost states (without children) (*e.g.*, N_2(X,v=0,J=0) is childless).

The previous terminology, allows introducing several definitions and relations for the particle densities with the different species (gases / states) in the system.

- **Total gas density:** total particle density of the heavy species in the system, evaluated as:

$$N = \frac{p}{k_B T_g} \quad , \tag{2}$$

  where p is the *total* pressure, k_B is the Boltzmann constant, and where the gas temperature T_g is assumed equal for all k-gases in the system.
- **Individual gas density:** total particle density of a gas in the system. For gas k this is given by N_k, with $\sum_k N_k = N$.

[1] LoKI-GM contains already a tentative implementation (mostly intended for post-discharge scenarios) of a solution for the system of equations (24) and (57) (LoKI-C), coupled to equations (19a)-(19b) (LoKI-B). By default, this feature is disabled in the current version of LoKI. User caution is advised if/when the feature is activated, since preliminary numerical tests revealed convergence difficulties.

[2] LoKI handles also surface states, when the plasma kinetics is extended to include plasma-surface interactions.

Table 1: Example of a gas mixture (childless states are displayed in red).

	Gases	States		
System	**N_2**	**N_2(X)** Parent of → ← Childs of	**N_2(X,v=0)** Parent of → ← Childs of	**N_2(X,v=0,J=0)**
				N_2(X,v=0,J=1)
				N_2(X,v=0,J=2)
				⋮
			N_2(X,v=1)	
			N_2(X,v=2)	
			⋮	
		N_2(A)		
		N_2(B)		
		⋮		
	N	**N(^{4}S)**		
		N(^{2}D)		
		N(^{2}P)		
		⋮		
	He	**He(1^1S)**		
		He(2^3S)		
		He(2^1S)		
		⋮		
	⋮			

- **Gas fraction:** fraction of a gas in the system. For gas k this is given by $\chi_k \equiv N_k/N$.
- **Absolute density:** particle (absolute) density of a state. For a generic state i of gas k this is represented by N_{k_i}.
- **Population:** density of a state relative to its siblings. For a generic state i of gas k this is represented by $\xi_{k_i} \equiv N_{k_i}/N_{\text{parent of } k_i} = N_{k_i}/(\sum_{i\in\text{siblings}} N_{k_i})$.
- **Density:** particle density of a state relative to the total gas density. For a generic state i of gas k this is represented by $\delta_{k_i} \equiv N_{k_i}/N$.

From the previous definitions, it is straightforward to obtain the following normalisation conditions

$$\sum_{k\in\text{system}} \chi_k = 1 \tag{3a}$$

$$\sum_{i\in\text{siblings}} \xi_i = 1 \tag{3b}$$

$$\sum_{i\in\text{childless}} \delta_i = 1 \ , \tag{3c}$$

concluding also that the following relationships hold

$$\delta_i = \xi_i \delta_{i\text{'s parent}} \tag{4a}$$

$$\delta_{k_i} = \xi_{k_i} \chi_k \quad . \tag{4b}$$

LoKI uses SI units for all physical quantities, except the energies that are expressed in eV (electron-volt), the reduced fields that are expressed in Td (Townsend; 1 Td = 10^{-21} V m^2), and the gas flows that are expressed in sccm (standard cubic centimeter per minute).

2.2 The electron Boltzmann equation

In the classical two-term approximation, the electron distribution function can be written assuming a separation between the Legendre expansion of the energy distribution and the space-time profile of the electron density $n_e(z,t)$ as follows

$$\begin{aligned} F(z,u,t) &\simeq \left\{ f^0(u) + \left[\vec{f}^1(u) \cdot \frac{\vec{v}}{v} \right] \cos(\omega t) \right\} n_e(z,t) \\ &= \left[f^0(u) + f^1(u) \cos(\theta) \cos(\omega t) \right] n_e(z,t) \quad , \end{aligned} \tag{5}$$

where $u = m_e v^2/(2e)$ is the electron kinetic energy in eV (with m_e and e the electron mass and charge, respectively); $f^0(u) \equiv f(u)$ is the isotropic part of the electron distribution function, identified with the EEDF in the framework of the two-term approximation, and satisfying the normalisation condition $\int_0^\infty f(u)\sqrt{u}du = 1$; and $f^1(u)$ is the (real part of the) first-anisotropy of the electron distribution function.

Under the previous conditions, the EBE including a non-conservative collisional operator can be written as (with $\gamma \equiv \sqrt{2e/m_e}$) [3]

$$\frac{1}{\gamma\sqrt{u}} \frac{\langle \nu_{\text{eff}} \rangle}{N} u f(u) + \frac{1}{N} \frac{1}{\gamma} \frac{dG(u)}{du} = S(u) \tag{7a}$$

$$f^1(u) = -\frac{\zeta(E/N)}{\Omega_{\text{PT}}(u)} \frac{df(u)}{du} \quad , \tag{7b}$$

or

$$\frac{1}{3} \frac{\alpha_{\text{eff}}}{N} u f^1(u) + \frac{1}{N} \frac{1}{\gamma} \frac{dG(u)}{du} = S(u) \tag{8a}$$

$$f^1(u) = -\frac{(\alpha_{\text{eff}}/N)}{\Omega_{\text{SST}}(u)} f(u) - \frac{(E/N)}{\Omega_{\text{SST}}(u)} \frac{df(u)}{du} \quad , \tag{8b}$$

where (7a) and (8a) or (7b) and (8b) correspond to the isotropic and anisotropic components of the EBE, respectively, in the following approximations:

- **temporal growth model:** the space-homogeneous equations (7a)-(7b), which can be used in either the DC or the HF cases, assume an exponential temporal growth of the electron density with a growth constant $\langle \nu_{\text{eff}} \rangle \equiv \langle \nu_{\text{ion}} \rangle - \langle \nu_{\text{att}} \rangle \equiv \sum_{k,i} \delta_{k_i} \left[\langle \nu_{k_i,\text{ion}} \rangle - \langle \nu_{k_i,\text{att}} \rangle \right]$, corresponding to the electron net creation frequency for the system under study

$$\frac{\partial n_e(t)}{\partial t} = \langle \nu_{\text{eff}} \rangle n_e(t) \quad , \tag{9}$$

  with $\langle \nu_{\text{ion}} \rangle$ and $\langle \nu_{\text{att}} \rangle$ the corresponding electron mean frequencies for ionisation and attachment, respectively, and where the sums extend over all levels of all gases in the mixture (see section 3.1.1). The temporal growth model was used to simulate Pulsed Townsend (PT) discharges by Tagashira *et al* [18], using the formulation of Thomas [19] for the one dimensional continuity equation of electrons under the action of an uniform electric field;

[3] As mentioned, in a HF case the quantity $f^1(u)$ in (7b) represents the *real part of* the first-anisotropy of the electron distribution function. Indeed, in this case $f^1(u,t) = \tilde{f}^1(u) e^{j\omega t}$ with

$$\tilde{f}^1(u) = -\gamma\sqrt{u} \frac{E_p}{N\gamma\sqrt{u}\,\Omega_c(u) + j\omega} \frac{df(u)}{du} = -\frac{\zeta(E/N)}{\Omega_{\text{PT}}(u)} \frac{df(u)}{du} + j \frac{\zeta(E/N) \frac{\omega/N}{\Omega_c \gamma\sqrt{u}}}{\Omega_{\text{PT}}(u)} \frac{df(u)}{du} \quad . \tag{6}$$

- **spatial growth model:** the time-constant equations (8a)-(8b), which can only be used in the DC case, assume an exponential spatial growth of the electron density in the direction opposite to that of the applied electric field, with a growth constant $\alpha_{\text{eff}} \equiv \alpha - \eta \equiv \sum_{k,i} \delta_{k_i} \left(\alpha_{k_i} - \eta_{k_i}\right)$ corresponding to the effective electron Townsend coefficient

$$\frac{\partial n_e(z)}{\partial z} = \alpha_{\text{eff}}\, n_e(z) \quad , \tag{10}$$

with α and η the corresponding first and second Townsend coefficients for ionisation and attachment, respectively (see section 3.1.1). The spatial growth model is usually adopted when simulating Steady State Townsend (SST) discharges maintained with a DC electric field [18].

Note that the inclusion of growth models for the electron density is necessary if one intends a space and time independent description, yet including non-conservative collisional mechanisms such as electron ionisation and attachment. Note further that the current version of LoKI-B does not include electron recombination mechanisms.

In these equations:

- E/N is the *reduced electric field*, where $E \equiv E_p/\zeta$ (with $\zeta = 1\,\text{or}\,\sqrt{2}$ for the DC or the HF cases, respectively);
- the quantities Ω_{PT} and Ω_{SST} have the dimensions of a cross section, being defined as

$$\Omega_{\text{PT}}(u) \equiv \Omega_c(u) + \frac{\left[1/(\gamma^2 u)\right](\omega/N)^2}{\Omega_c(u)} \tag{11a}$$

$$\Omega_{\text{SST}}(u) \equiv \sigma_c(u) \quad , \tag{11b}$$

where $\Omega_c(u) \equiv \sigma_c(u) + [1/(\gamma\sqrt{u})]\,(\langle\nu_{\text{eff}}\rangle/N)$, with $\sigma_c(u) \equiv \sum_k \chi_k \sigma_{k,c}$ the electron-neutral total momentum-transfer cross section, for the system under study, and $\sigma_{k,c}$ the corresponding cross section for gas k;

- $G \equiv G_E + G_{\text{el}} + G_{\text{CAR}} + G_{ee}$ is the *upflux* function that contains power gain/loss contributions due to, in order, the applied electric-field, the elastic collisions, the rotational collisions (in the continuous approximation for rotations, CAR, with a Chapman-Cowling correction due to the gas temperature [20–23], written here for the case of homonuclear diatomic molecules), and the electron-electron collisions [22], respectively

$$\begin{aligned} G_E(u) &= N\gamma \frac{u}{3}\frac{1}{\zeta}\frac{E}{N} f^1(u) \\ &= -N\gamma \begin{cases} g_E^{\text{PT}}(u) \left(\dfrac{E}{N}\right)^2 \dfrac{df(u)}{du} & \text{, for a PT situation} \\ g_E^{\text{SST}}(u) \left[\dfrac{E}{N}\dfrac{\alpha_{\text{eff}}}{N} f(u) + \left(\dfrac{E}{N}\right)^2 \dfrac{df(u)}{du}\right] & \text{, for a SST situation} \end{cases} \end{aligned} \tag{12a}$$

$$\begin{aligned} G_{\text{el}}(u) &= -\sum_k 2\frac{m_e}{M_k}\nu_{k,c}^{\text{el}}(u) u^{3/2}\left[f(u) + \frac{k_B T_g}{e}\frac{df(u)}{du}\right] \\ &= -N\gamma\, g_{\text{el}}(u)\left[f(u) + \frac{k_B T_g}{e}\frac{df(u)}{du}\right] \end{aligned} \tag{12b}$$

$$\begin{aligned} G_{\text{CAR}}(u) &= -\sum_k 4 B_k \nu_{0,k} u^{1/2}\left[f(u) + \frac{k_B T_g}{e}\frac{df(u)}{du}\right] \\ &= -N\gamma\, g_{\text{CAR}}(u)\left[f(u) + \frac{k_B T_g}{e}\frac{df(u)}{du}\right] \end{aligned} \tag{12c}$$

$$\begin{aligned} G_{ee}(u) &= -2\nu_{ee}(u) u^{3/2}\left[I(u) f(u) + J(u)\frac{df(u)}{du}\right] \\ &= -N\gamma\, g_{ee}(u)\left[I(u) f(u) + J(u)\frac{df(u)}{du}\right] \quad , \end{aligned} \tag{12d}$$

with

$$g_E^{\mathrm{PT}}(u) = \frac{u}{3}\frac{1}{\Omega_{\mathrm{PT}}(u)} \tag{13a}$$

$$g_E^{\mathrm{SST}}(u) = \frac{u}{3}\frac{1}{\Omega_{\mathrm{SST}}(u)} \equiv D(u) \tag{13b}$$

$$g_{\mathrm{el}}(u) = 2u^2 \sum_k \frac{m_e}{M_k}\chi_k \sigma_{k,c}^{\mathrm{el}}(u) \tag{13c}$$

$$g_{\mathrm{CAR}}(u) = 4u \sum_k B_k \chi_k \sigma_{0,k}(u) \tag{13d}$$

$$g_{ee}(u) = \frac{2}{\gamma}\frac{\nu_{ee}(u)}{N} u^{3/2} \quad , \tag{13e}$$

and where $\nu_{k,c}^{\mathrm{el}} \equiv N_k \gamma \sqrt{u}\, \sigma_{k,c}^{\mathrm{el}}$ is the electron-neutral elastic momentum-transfer collision frequency for the gas k ($\sigma_{k,c}^{\mathrm{el}}$ being the corresponding cross section), $\nu_{0,k} \equiv N_k \gamma \sqrt{u}\, \sigma_{0,k}$ is the electron-neutral collision frequency for the rotational excitation of gas k ($\sigma_{0,k} \equiv 8\pi Q_k^2 a_0^2/15$ being the corresponding cross section, with a_0 the Bohr radius), B_k and Q_k are the rotational constant and the quadrupole-moment constant (in units of ea_0^2) of the gas k, respectively [4], $I(u)$ and $J(u)$ are the relevant Spitzer integrals [15, 24], and $\nu_{ee} \equiv 4\pi \left[e^2/\left(4\pi\varepsilon_0 m_e\right)\right]^2 \ln\Lambda_c / v^3 n_e$ is the electron-electron collision frequency (with ε_0 the vacuum permittivity, $\ln\Lambda_c$ the Coulomb logarithm ($\Lambda_c \equiv 12\pi n_e \lambda_D^3$) and λ_D the Debye length);

- $S \equiv \sum_{i,j>i} S_{i,j} + \sum_i S_{i,\mathrm{ion}} + \sum_i S_{i,\mathrm{att}}$ is the discrete collision operator that contains power loss/gain contributions due to inelastic/superelastic mechanisms, ionisation and attachment (as mentioned before, recombination is not yet considered), respectively

$$\begin{aligned} S_{i,j}(u) &= \delta_i \left[(u+V_{i,j})\sigma_{i,j}(u+V_{i,j})f(u+V_{i,j}) - u\sigma_{i,j}(u)f(u)\right] \\ &+ \delta_j \frac{g_i}{g_j}\left[u\sigma_{i,j}(u)f(u-V_{i,j}) - (u+V_{i,j})\sigma_{i,j}(u+V_{i,j})f(u)\right] \end{aligned} \tag{14a}$$

$$\begin{aligned} S_{i,\mathrm{ion}}(u) &= \delta_i \left\{ \int_{u+V_{i,\mathrm{ion}}}^{2u+V_{i,\mathrm{ion}}} u'\sigma_{i,\mathrm{ion}}^{\mathrm{sec}}(u', u'-V_{i,\mathrm{ion}}-u)f(u')du' \right. \\ &+ \left. \int_{2u+V_{i,\mathrm{ion}}}^{\infty} u'\sigma_{i,\mathrm{ion}}^{\mathrm{sec}}(u',u)f(u')du' - u\sigma_{i,\mathrm{ion}}(u)f(u) \right\} \end{aligned} \tag{14b}$$

$$S_{i,\mathrm{att}}(u) = -\delta_i u \sigma_{i,\mathrm{att}}(u) f(u) \quad . \tag{14c}$$

In (14a)-(14c), the subscripts i, j and the densities δ_i, δ_j now refer to any state of any gas in the system; $\sigma_{i,j}$ is the electron-collision cross section for the $i \to j$ excitation with energy threshold $V_{i,j}$ and statistical weights g_i and g_j, respectively (note that the third and fourth superelastic terms in (14a) were written resorting to the Klein-Rosseland relation [25]); $\sigma_{i,\mathrm{att}}(u)$ is the electron-collision cross section for the attachment of state i; $\sigma_{i,\mathrm{ion}}^{\mathrm{sec}}(u', u)$ is the single differential cross section (SDCS) for the ionisation of state i with energy threshold $V_{i,\mathrm{ion}}$; and

$$\sigma_{i,\mathrm{ion}}(u) = \int_0^{(u-V_{i,\mathrm{ion}})/2} \sigma_{i,\mathrm{ion}}^{\mathrm{sec}}(u,u')du' \tag{15}$$

is the integral cross section for the ionisation of state i. The SDCS takes into account the distribution of the energy available after an ionisation event, caused by a primary electron with energy u', yielding a secondary electron born in the event with energy u and a scattered electron with energy $u' - V_{i,\mathrm{ion}} - u$.

Note that the effect of rotational inelastic/superelastic mechanisms, in the electron kinetics of homonuclear diatomic molecules, can be considered by adopting either the CAR approach of (12c) or the discrete approach of the collision operator (14a), yielding similar results provided that the rotational constants Q_k and B_k are coherently matched to the rotational cross section set $\sigma_{i,j}$ [20].

LoKI and LoKI-B allow the user to choose between different descriptions of the ionisation mechanism, namely:

[4] In `LoKI-B_v1.0.0`, the rotational cross section $\sigma_{0,k}$ was wrongly calculated with a linear (not quadratic) dependence on the quadrupole-moment constant Q_k. The bug is now corrected. We acknowledge Jan van Dijk and Daan Boer (TU/e, Eindhoven, The Netherlands) for pointing this out.

- **conservative:** taking ionisation as conservative a mechanism, described by the inelastic part of the collisional operator (14a), in which case the first terms in the left-hand side of (7a) and (8a) should be set to zero;

- **usingSDCS:** taking ionisation as a non-conservative mechanism, described by the collisional operator (14b) with a SDCS, and using the EBE given by (7a)-(8b);

- **oneTakesAll:** taking ionisation as a non-conservative mechanism, and prescribing no-sharing of energy between the secondary electron, introduced at $u = 0$, and the scattered electron, introduced at energy $u' - V_{i,\mathrm{ion}}$. In this case the SDCS for the secondary electrons writes $\sigma^{\mathrm{sec}}_{i,\mathrm{ion}}(u',u) = \sigma_{i,\mathrm{ion}}(u')\delta(u)$ and the ionisation operator becomes

$$S_{i,\mathrm{ion}}(u) = \delta_i \left[(u+V_{i,\mathrm{ion}})\sigma_{i,\mathrm{ion}}(u+V_{i,\mathrm{ion}})f(u+V_{i,\mathrm{ion}}) - u\sigma_{i,\mathrm{ion}}(u)f(u) + \frac{1}{\gamma}\frac{\langle \nu_{i,\mathrm{ion}}\rangle}{N}\delta(u) \right] , \tag{16}$$

  corresponding to the inelastic part of (14a) plus an extra term that describes the introduction of new electrons at zero-energy;

- **equalSharing:** taking ionisation as a non-conservative mechanism, and prescribing equal-energy-sharing between the secondary and the scattered electrons, introduced at energy $(u' - V_{i,\mathrm{ion}})/2$. In this case the SDCS for the secondary electrons writes $\sigma^{\mathrm{sec}}_{i,\mathrm{ion}}(u',u) = \sigma_{i,\mathrm{ion}}(u')\delta[u - (u' - V_{i,\mathrm{ion}})/2]$ and the ionisation operator becomes

$$S_{i,\mathrm{ion}}(u) = \delta_i \left[4(2u+V_{i,\mathrm{ion}})\sigma_{i,\mathrm{ion}}(2u+V_{i,\mathrm{ion}})f(2u+V_{i,\mathrm{ion}}) - u\sigma_{i,\mathrm{ion}}(u)f(u) \right] , \tag{17}$$

  where the first term of $S_{i,\mathrm{ion}}(u)du$ describes the entrance of both the secondary and the scattered electrons at a total energy between $2u$ and $2u + 2du$, produced by a primary electron with energy $u' = 2u + V_{i,\mathrm{ion}}$, whereas the second term describes the exit of primary electrons with energy u after ionizing collisions.

Naturally, the attachment mechanism is taken **always** as non-conservative, meaning that, in the case of electronegative gases, the code adopts the electron-density growth-model (bf temporal or **spatial**) specified by the user.

Note that the current version of LoKI-B does not include any external sources of electrons, such as electron beam injection or photoionization. However, the model adopted in the code can be easily upgraded as to consider these effects: (i) on the right-hand side of the EBE, with the addition, to the source function S, of an energy-resolved gain-term; (ii) on the left-hand side of the EBE, with the addition of the corresponding energy-integral term, describing the growth of the electron density in a stationary situation. The inclusion of these external sources of electrons, as well as the handling of the non-conservative electron-collision mechanisms (particularly the attachment), must be carefully monitored in all cases, since they can be responsible for the development of significant anisotropies, in which case the two-term expansion of the EBE adopted here is no longer valid.

2.3 The time-dependent electron Boltzmann equation

The original capabilities of LoKI-B have been extended in order to obtain the solution of the EBE for plasma electrons under the action of a time-dependent (non-oscillatory, $\omega = 0$) electric-field along the z-axis [4]. In this case, assuming the classical two-term expansion and a space-independent exponential temporal growth of the electron density

$$\begin{aligned} F(u,t) &\simeq \left[f^0(u,t) + f^1(u,t)\cos(\theta) \right] n_e(t) \\ \frac{\partial n_e(t)}{\partial t} &= \langle \nu_{\mathrm{eff}} \rangle (t) n_e(t) \quad , \end{aligned} \tag{18}$$

the EBE is updated as follows (see equations (7a)-(7b)):

$$\frac{1}{N}\frac{1}{\gamma}\sqrt{u}\frac{\partial f(u,t)}{\partial t} + \frac{1}{\gamma\sqrt{u}}\frac{\langle \nu_{\mathrm{eff}} \rangle (t)}{N} u f(u,t) + \frac{1}{N}\frac{1}{\gamma}\frac{\partial G(u,t)}{\partial u} = S(u,t) \tag{19a}$$

$$f^1(u,t) = -\frac{(E(t)/N)}{\Omega_c(u,t)}\frac{\partial f(u,t)}{\partial u} \quad . \tag{19b}$$

In these equations, the first term on the left-hand side of (19a) accounts for the intrinsic time evolution of the isotropic component of the electron distribution function; the second term on the left-hand side of (19a) and the time-dependent collision frequency $\Omega_c(u,t)$ in (19b) account for the intrinsic time evolution of the electron density. In this case, the collision frequency is given by

$$\Omega_c(u,t) \equiv \sigma_c(u) + \frac{1}{\gamma\sqrt{u}}\frac{1}{N n_e(t)}\frac{dn_e(t)}{dt} = \sigma_c(u) + \frac{1}{\gamma\sqrt{u}}\frac{\langle\nu_{\text{eff}}\rangle(t)}{N} \quad . \tag{20}$$

It is assumed a steady-state form for equation (19b), describing the first anisotropy, because the characteristic evolution time of the EEDF τ is much larger than the characteristic evolution time of the anisotropic component, *i.e.*

$$\tau \simeq \frac{1}{\sum_k (m_e/M_k)\,\nu_{k,c}^{\text{el}}} \gg \frac{1}{\nu_{k,c}^{\text{el}}} \quad . \tag{21}$$

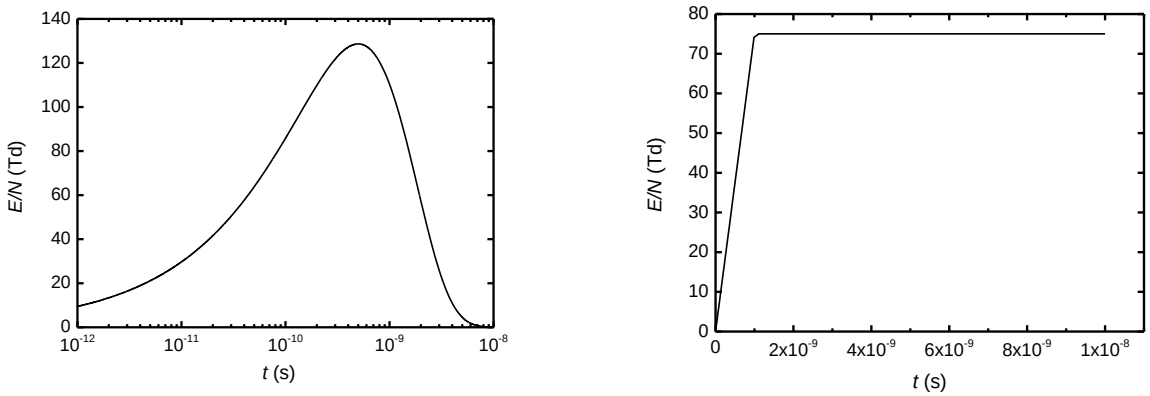


Figure 1: Example of time-dependent (non-oscillatory) reduced electric-fields that can be used in LoKI-B simulations: electric-field pulse (left) and electric-field step (right), calculated using (22a) and (22b), respectively.

When running LoKI-B independently, equations (19a)-(19b) are used to obtain the time-dependent solution of the EBE for user-prescribed electric-field forms, initial and final times, and sampling values of the field. As an example, figure 1 shows two forms of the reduced electric-field that can be used in the time-dependent solution of LoKI-B, corresponding to an electric field pulse given by

$$\frac{E(t)}{N}(\text{Td}) = \frac{E_0}{N}\sqrt{\frac{t(\text{s})}{\tau_{\text{rise}}}}\,\exp\left(-\frac{t(\text{s})}{\tau_{\text{rise}}}\right) \tag{22a}$$

with $E_0/N = 300$ Td and $\tau_{\text{rise}} = 10^{-9}$ s; and an electric-field step given by

$$\frac{E(t)}{N}\,(\text{Td}) = \begin{cases} \frac{E_{\max}}{N}\frac{t(\text{s})}{\tau_{\text{rise}}}\,, & 0 \le t(\text{s}) \le \tau_{\text{rise}} \\ \frac{E_{\max}}{N}\,, & t(\text{s}) > \tau_{\text{rise}} \end{cases} \quad , \tag{22b}$$

with $E_{\max}/N = 75$ Td and $\tau_{\text{rise}} = 10^{-9}$ s.

2.4 Prescribed EEDFs

LoKI calculations can also adopt a prescribed EEDF, corresponding to a generalized Maxwellian given by the expression [26–28]

$$f(u) = s\left(\frac{2}{3T_e}\right)^{3/2}\frac{\Gamma(\frac{5}{2s})^{3/2}}{\Gamma(\frac{3}{2s})^{5/2}}\,\exp\left(-\left[\frac{\Gamma(\frac{5}{2s})}{\Gamma(\frac{3}{2s})}\frac{2u}{3T_e}\right]^s\right) \quad , \tag{23}$$

where Γ represents the gamma-function, T_e is in eV and s is a positive real number that controls the "shape" of the EEDF. In particular, (23) yields a Maxwellian EEDF and a Druyvesteyn EEDF for $s = 1$ and $s = 2$, respectively.

This setting corresponds to a local thermodynamic equilibrium framework, where the electron energy (defined by the electron temperature, given as input parameter) is dissipated exclusively in electron-neutral *conservative* collisions. Thus, in this case LoKI-B considers neither an electron-density growth model nor time-dependent

calculations. Indeed, when adopting a prescribed EEDF, LoKI-B calculations are limited to evaluating (i) the EEDF for given T_e, according to (23), (ii) the electron rate coefficients for the set of cross sections defined as input data (see section 3.1.1), (iii) the most relevant electron transport parameters (see section 3.1.1), (iv) the power transferred in electron-neutral elastic/inelastic/superelastic collisions (see section 3.1.2), and (v) the *equivalent* reduced electric-field that would ensure a local equilibrium between the power gained from the field and lost in collisions (see section 3.1.2).

2.5 The plasma chemistry global model

LoKI-GM solves the spatially-average rate balance equations of the various gas/plasma/surface states, assuming that the creation/destruction of each state X_j is due to several r-reactions (as defined by the kinetic scheme), such as

$$\sum_l a_{l,r}^{(1)} \mathrm{X}_l \longrightarrow \sum_{l'} a_{l',r}^{(2)} \mathrm{X}_{l'} \quad ,$$

where $a_{l,r}^{(1)}$ and $a_{l',r}^{(2)}$ are the stoichiometric coefficients of states X_l and $\mathrm{X}_{l'}$, as they appear on the left- and right-hand sides of reaction r, respectively; and is also due to radial and/or axial transport within a cylinder of radius R and length L (see figure 2). Note that, when $L \to \infty$ the plasma container becomes an infinitely long

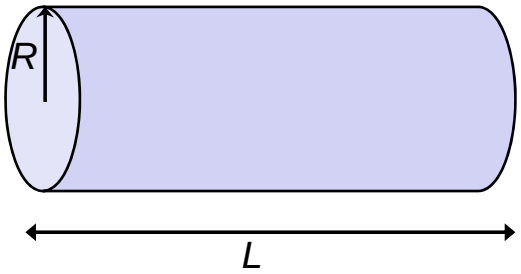


Figure 2: Schematic representation of a plasma container with radius R and length L.

cylinder with radius R, in which case the axial transport can be neglected, whereas when $R \to \infty$ it degenerates into a parallel-plate system with length L, equivalent to a slab geometry, in which case the radial transport can be neglected.

Under these conditions, the spatially-average rate balance equation of a generic j-state of a gas writes [17]:

$$\frac{\partial N_j}{\partial t} = S_j^{\mathrm{chem}} + S_j^{\mathrm{transp}} + S_j^{\mathrm{flow}} \quad , \tag{24}$$

where

$$S_j^{\mathrm{chem}} = \sum_{r=\mathrm{chem}} \left\{ \left[a_{jr}^{(2)} - a_{jr}^{(1)} \right] k_{r_j} \Pi_l N_l^{a_{lr}^{(1)}} \right\} \tag{25a}$$

$$S_j^{\mathrm{transp}} = \begin{cases} \displaystyle\sum_{r=\mathrm{transp}} a_{jr}^{(2)} \frac{n_l}{\tau_r} - \frac{N_j}{\tau_{\mathrm{transp}_j}} \quad , \text{ for neutral states} \\ \\ \displaystyle -\frac{N_j}{\tau_{\mathrm{transp}_j}} \quad , \text{ for charged species} \end{cases} \tag{25b}$$

$$S_j^{\mathrm{flow}} = -k_{\mathrm{outflow}} n_j \tag{25c}$$

represent the chemistry (volume and surface), the transport and the flow net-source terms, respectively.

In (25a), N_l is the particle density of state l and k_{r_j} is the rate coefficient of the r-reaction involving the j-state. The quantity N_l is written for **all states** as a *volume* density, and therefore it is corrected by the surface-to-volume factor $A/V = 2(R+L)/(RL)$ when applied to *surface* states: physisorbed/chemisorbed states X_f and X_s, respectively, and vacant physical/chemical sites F_v and S_v, respectively.

The characteristic transport times τ_{transp_j} in (25b) are presented and discussed in section 2.5.2.

The outflow rate coefficient k_{outflow} in (25c), assumed equal for all volume states, influences the calculation of the gas pressure given by the particle conservation equation

$$\sum_j n_j = N = \frac{p}{k_B T_g} \quad . \tag{26}$$

The expression of k_{outflow} depends on the flow model adopted, and leads to different numerical implementations of the chemistry solution (see section 4)

$$k_{\text{outflow}} \simeq \begin{cases} \dfrac{Q_{\text{sccm}}^{\text{outflow}}}{V} \dfrac{10^{-6}}{60} \dfrac{T_g(t)}{273.15} \dfrac{1.013 \times 10^5}{p(t)} \\ \quad \text{when specifying a numerical value} \\ \quad \text{for the outflow } Q_{\text{sccm}}^{\text{outflow}} \text{ (in sccm)} \\ \dfrac{1}{N(t)} \displaystyle\sum_j \left(S_j^{\text{chem}} + S_j^{\text{transp}} \right) + \dfrac{1}{T_g(t)} \dfrac{dT_g(t)}{dt} \\ \quad \text{when adopting an isobaric model that imposes} \\ \quad \text{the isobaric condition } dp/dt = 0 \text{ to (26) and (24)} \; . \end{cases} \tag{27}$$

In these equations, $N(t)$, $p(t)$ and $T_g(t)$ represent, respectively, the time-dependent total density, pressure and temperature of the gas.

2.5.1 Rate coefficients for volume and surface reactions

For volume reactions, k_{r_j} can be defined as [3, 29] (see User Guide [12] for details on the designation of the corresponding rate coefficient functions)

$$k_{r_j} = \begin{cases} \gamma \int_0^\infty u \sigma_{r_j}(u) f(u) du \;\; , \text{ for electron collisions (see section 3.1.1)} \\ \text{user-defined expressions} \quad , \quad \text{for heavy-species collisions} \\ A_{r_j} \beta_{r_j} \;\; , \text{ for radiative transitions} \quad , \end{cases} \tag{28}$$

where σ_{r_j} is the electron scattering cross section for the r-reaction with the j-target; A_{r_j} is the Einstein coefficient for radiative transition r from the j-state; and β_{r_j} is the corresponding escape factor describing the radiation imprisonment. Currently, LoKI-GM includes escape factor functions for steady-state simulations in cylindrical geometry.

The heavy-species collision rate coefficients can have any expression defined by the user. Typical examples are constant values, gas temperature power-laws or Arrhenius-type expressions (see User Guide [12])

$$p_{r_j} T^{q_{r_j}} \exp\left(\pm \frac{T_{\text{ref}}}{T} \right) \quad , \tag{29}$$

where p_{r_j} and q_{r_j} are rate coefficient parameters; T_{ref} is a reference temperature and T is a general temperature (*e.g.* the gas temperature T_g of the electron temperature T_e).

For two-body reactions of a microkinetic mesoscopic surface model, k_{r_j} can be defined as [29–31] (see User Guide [12] for details on the designation of the corresponding rate coefficient functions)

$$k_{r_j} = \begin{cases} \dfrac{k_{r_j}^0 \exp\left(-\frac{E_{r_j}}{k_B T} \right)}{[F] + [S]} \dfrac{1}{\tau_{r_j}} \;\; , \\ \quad \text{for adsorption-like and surface-diffusion-like reactions} \\ \exp\left(-\dfrac{E_{dr_j}}{k_B T_w} \right) \dfrac{1}{\tau_{r_j}} \;\; , \text{ for desorption reactions} \quad . \end{cases} \tag{30}$$

Examples of adsorption-like reactions are the physical adsorption, the chemical adsorption and the Eley-Rideal recombination, respectively

$$\begin{aligned} X_j + F_v &\longrightarrow X_f \\ X_j + S_v &\longrightarrow X_s \\ X_j + Y_s &\longrightarrow XY + S_v \quad ; \end{aligned}$$

examples of surface-diffusion-like reactions are surface transport and the Langmuir-Hinshelwood recombination, respectively

$$\begin{aligned} X_f + S_v &\longrightarrow X_s + F_v \\ X_f + Y_s &\longrightarrow XY + S_v + F_v \quad ; \end{aligned}$$

and desorption reactions correspond to

$$X_f \longrightarrow X_j + F_v \quad .$$

In (30), $[F]$ and $[S]$ are the *volume-equivalent* total densities of physical sites and chemical sites, respectively, defined from the corresponding surface densities (in m^{-2}) using $[F,S]_{\mathrm{surface}} = [F,S](V/A)$ (note that $[F]_{\mathrm{surface}} + [S]_{\mathrm{surface}}$ is defined in the *working conditions* of the code, via the variable `surfaceSiteDensity`; see User Guide [12]); $k^0_{r_j}$ is a dimensionless steric factor; E_{r_j} is the activation energy for adsorption and surface diffusion; E_{dr_j} is the activation energy for desorption; T_w is the wall temperature, adopted in the expression of some rate coefficients for wall reactions (see section 2.6 for more details about T_w); and τ_{r_j} are the characteristic times of the different surface mechanisms r involving the j-state, defined as

$$\tau_{r_j}^{-1} = \begin{cases} \tau_{\mathrm{transp}_j}^{-1} \quad , \text{ for adsorption-like reactions} \\ \\ \frac{3}{4}\nu_{D_j} \exp\left(-\frac{E_{D_j}}{k_B T_w}\right) \quad , \text{ for surface-diffusion-like reactions} \\ \\ \nu_{dr_j} \quad , \text{ for desorption reactions} \quad , \end{cases} \tag{31}$$

where $\tau_{\mathrm{transp}_j}^{-1}$ is the characteristic transport frequency of neutrals (see below); ν_{D_j} and ν_{dr_j} are the frequencies of surface diffusion and desorption, respectively; E_{D_j} is the energy barrier for surface diffusion; and the (optional) factor 3/4 is taken only for surface transport reactions as a correction to back diffusion [30], assuming that the remaining 1/4 of the species entering the collection zone can reach a chemisorption site.

2.5.2 Rate coefficients for the transport of species

LoKI-GM incorporates rate coefficients derived from multiple formulations to describe the transport of both neutral and charged species (see User Guide [12] for details on the implementation of transport models in LoKI and the designation of the corresponding rate coefficient functions).

The characteristic transport time of neutral state j, appearing in (25b) and (31), can be defined from the *heuristic Chantry model* as [32]

$$\tau_{\mathrm{transp}_j} \equiv \tau_{\mathrm{diff}_j} + \tau_{\mathrm{wall}_j} \tag{32a}$$

$$\tau_{\mathrm{diff}_j} \simeq \frac{\Lambda^2}{D_j} \tag{32b}$$

$$\tau_{\mathrm{wall}_j} \simeq \frac{1-\gamma_j/2}{\gamma_j \langle v_j \rangle} \frac{2RL}{L+R} \quad , \tag{32c}$$

by combining the limiting situations of *multi-component diffusion transport*, τ_{diff_j}, and *wall recombination/de-excitation*, τ_{wall_j}, assuming no deposition conditions. In these equations, D_j is the diffusion coefficient of the j-state (see more details below); Λ is a characteristic diffusion length, estimated assuming a separation of variables in the axial and radial directions, and further considering zero-density boundary conditions at the discharge walls

$$\Lambda^2 \equiv \left[\left(\frac{\pi}{L}\right)^2 + \left(\frac{2.405}{R}\right)^2 \right]^{-1} \quad ; \tag{33}$$

γ_j is the recombination probability of the j-state; and

$$\langle v_j \rangle = \sqrt{\frac{8 k_B T}{\pi M_j}} \tag{34}$$

is the thermal velocity of the j-state, with M_j the corresponding mass and T a gas-related temperature, to be defined within the transport/thermal model considered (see section 2.6 for details).

Note that equations (32c)-(33) allow retrieving the limiting cases of transport in pure slab and pure cylindrical geometries, by imposing $R \to \infty$ and $L \to \infty$, respectively. Note further that, in (31), the characteristic transport frequency of neutrals is usually defined in the limiting case of wall recombination as

$$\tau_{\text{transp}_j}^{-1} \simeq \tau_{\text{wall}_j}^{-1} \simeq \gamma_j \frac{\langle v_j \rangle}{4} \frac{A}{V} \quad , \tag{35}$$

which can be obtained from (32c), by neglecting $\gamma_j/2$.

A more evolved *multi-component transport* model for neutrals departs from the general flux boundary-condition [32]

$$\left[\frac{\nabla N_j}{N_j} \right]_{\text{wall}} = -\frac{\langle v_j \rangle}{4 D_j} \frac{\gamma_j}{1 - \gamma_j/2} \equiv -\lambda_j^{-1} \tag{36a}$$

applied to the density profile

$$N_j(r,z) = N_{j0}\, J_0\left(\frac{r}{\Lambda_{r,j}} \right) \cos\left(\left(\frac{z}{\Lambda_{z,j}} \right) \right) \quad , \tag{36b}$$

where J_0 is the 0th order Bessel function and $\Lambda_{r,j}$ and $\Lambda_{z,j}$ are diffusion lengths for the j-state along the radial and the axial directions, respectively. As a result we obtain (J_1 is the 1st order Bessel function)

$$z_j \,\text{tg}\,(z_j) = \frac{L}{2\lambda_j} \quad , \quad \text{with } 0 \le z_j = \tfrac{L}{2\Lambda_{z,j}} \le \tfrac{\pi}{2} \tag{37a}$$

$$x_j \frac{J_1(x_j)}{J_0(x_j)} = \frac{R}{\lambda_j} \quad , \quad \text{with } 0 \le x_j = \tfrac{R}{\Lambda_{r,j}} \le 2.405 \quad , \tag{37b}$$

which can be solved numerically, together with (36a), to update the characteristic transport time of neutrals (32a), according to

$$\tau_{\text{transp}_j} = \frac{\Lambda_j^2}{D_j} \tag{38a}$$

$$\frac{1}{\Lambda_j^2} \equiv \frac{1}{\Lambda_{r,j}^2} + \frac{1}{\Lambda_{z,j}^2} \quad . \tag{38b}$$

The multi-component diffusion coefficients D_j in (32a) and (38a), describing the collisional transport of the neutrals, are calculated adopting Wilke's model [33] as the limiting case where each neutral species j diffuses in a mixture of stagnant gases

$$D_j \simeq \frac{1 - \delta_j}{\sum_{i(i \neq j)} \frac{\delta_i}{D_{ji}}} \quad , \tag{39}$$

where D_{ji} is the binary diffusion coefficient of the j-state in the i-state, given by [34,35]

$$D_{ji} \equiv \frac{3}{16} \frac{\sqrt{4\pi k_B T_g/(2M_{ji})}}{N \pi \sigma_{h,ji}^2 \Omega_{ji}^{(1,1)}(T^\star)} \tag{40}$$

with

$$M_{ji} \equiv \frac{M_j M_i}{M_j + M_i} \tag{41a}$$

$$\sigma_{h,ji} \equiv \frac{1}{2}(\sigma_{h,j} + \sigma_{h,i}) \tag{41b}$$

$$T^\star \equiv \frac{k_B T_g}{\varepsilon_{ji}} \tag{41c}$$

$$\varepsilon_{ji} \equiv \sqrt{\varepsilon_j \varepsilon_i} \tag{41d}$$

$$\Omega_{ji}^{(1,1)}(T^\star) \simeq a_1 (T^\star)^{-a_2} + (T^\star + a_3)^{-a_4} \quad . \tag{41e}$$

In (41a)-(41d), $(\sigma_{h,j},\ \varepsilon_j)$ and $(\sigma_{h,i},\ \varepsilon_i)$ are the Lennard-Jones parameters for states j and i, respectively; and (41e) is a fitting formula for the collision integral of the same Lennard-Jones potential, where $a_1 = 1.0548$, $a_2 = 0.15504$, $a_3 = 0.55909$ and $a_4 = 2.1705$.

Simplified *binary diffusion* models can also be considered, neglecting the effects of the multi-component transport and replacing the corresponding diffusion coefficients by binary coefficients ($D_j \simeq D_{j,\text{parent gas }k}$), with the characteristic diffusion times given by

$$\tau_{\text{bin}-\text{diff}_j} \simeq \frac{\Lambda^2}{D_{j,\text{parent gas }k}} \quad . \tag{42}$$

These binary models can adopt several possible frameworks (see User Guide [12] for more details):

- `binaryDiff` : introducing a wall recombination/deactivation probability γ_j to define $1/\tau_{\text{transp}_j} \equiv \gamma_j/\tau_{\text{bin}-\text{diff}_j}$;
- `binaryDiffusion`: assuming a uniform distribution of wall recombination/deactivation probabilities across the m states considered, $\gamma_j \equiv 1/m$, to define $1/\tau_{\text{transp}_j} \equiv \gamma_j/\tau_{\text{bin}-\text{diff}_j}$;
- `binaryTransportChantry`: replacing τ_{diff_j} with $\tau_{\text{bin}-\text{diff}_j}$ in the heuristic model (32a)-(32c);
- `binaryTransport`: replacing D_j with the binary coefficient $D_{j,\text{parent gas }k}$ in the multi-component transport model (38a)-(38b).

Note that the previous transport models can be adapted for use in a surface chemistry model, by converting the corresponding rate coefficients into equivalents of surface binary-reactions (cf. in User Guide [12] the rate coefficient functions with designations terminated by `Surface`).

The characteristic transport time of charged species j, appearing in (25b), can be evaluated using two broad classes of models: ambipolar-based formulations and h-factor transport models. A comprehensive review of these approaches is given by Alves and Tejero-del-Caz [36], and only the essential aspects are summarized below.

In dense plasmas, characterized by $\lambda_D \ll \Lambda$ (with λ_D the Debye length), the transport of charged particles can be described adopting *ambipolar-based* models [37,38], which write

$$\tau^{\text{amb}}_{\text{transp}_j} = \frac{\Lambda^2}{D_{\text{amb}_j}} \quad , \tag{43}$$

where D_{amb_j} is the so-called ambipolar diffusion coefficient of charge species j. This general formulation can be realized in various models, which obtain different expressions for D_{amb_j} from the solution of the particle and momentum-transfer fluid equations, subject to suitable boundary conditions (usually zero-density at the walls), where the latter equation corresponds to an ambipolar diffusion flux proportional to the density gradient.

The ambipolar diffusion coefficient can be defined for high/low-pressure regimes, by expressing the results of the unified theory proposed by Self and Ewald [39] in the convenient form of an abacus for an effective (ambipolar) diffusion coefficient D_{eff_j} as a function of pressure, as proposed by Ferreira and Ricard [40] for a single type of positive ions, and later extended by Coche *et al.* [37] to plasmas with several positive ions and a single negative ion with low density N_n. The formulation for the *effective ambipolar diffusion model* can be summarized as follows:

- the abacus function proposed in [39,40] can be obtained using (the subscript SE refers to Self and Ewald results)

$$\left(\frac{D_{\text{eff}_e}}{D_{\text{amb}_e}}\right)_{\text{SE}} = \left[\text{tgh}\left(\frac{\Lambda}{\lambda_+}\right)^{0.36}_{\text{SE}}\right]^{2.9} \quad , \tag{44}$$

  where D_{eff_e} and D_{amb_e} are the effective and the classical ambipolar diffusion coefficients for electrons, respectively, and λ_+ is an effective positive-ion mean-free-path given by (assuming quasi-neutrality) [41]

$$D_{\text{amb}_e} \simeq \frac{D_e\mu_+ + D_+\mu_e}{\mu_+ + \mu_e} \tag{45}$$

$$\lambda_+ = \frac{M_+\mu_+}{e}\sqrt{\frac{3eT_e}{M_+}} \tag{46}$$

with μ_e and D_e the electron mobility and transverse free-diffusion coefficient (see section 3.1.1), respectively, and

$$M_+ \equiv \frac{N_+}{\sum_{j_p}(N_{j_p}/M_{j_p})} \tag{47a}$$

$$\mu_+ \equiv \frac{\sum_{j_p} N_{j_p}\mu_{j_p}}{N_+} \tag{47b}$$

$$D_+ \equiv \frac{\sum_{j_p} N_{j_p} D_{j_p}}{N_+} \tag{47c}$$

$$N_+ \equiv \sum_{j_p} N_{j_p} \quad , \tag{47d}$$

with μ_{j_p} and D_{j_p} the j_p-ion mobility and free-diffusion coefficient, respectively, and where the subscript j_p is for sums over positive ions only;

- the results of Self and Ewald (for a single positive-ion species and no negative ions) can be generalized using

$$\frac{D_{\mathrm{eff}_e}}{D_{\mathrm{amb}_e}} = \left(\frac{D_{\mathrm{eff}_e}}{D_{\mathrm{amb}_e}}\right)_{\mathrm{SE}} \frac{1+\alpha_n}{\varsigma} \tag{48a}$$

$$\frac{\Lambda}{\lambda_+} = \left(\frac{\Lambda}{\lambda_+}\right)_{\mathrm{SE}} \sqrt{\frac{1}{\varsigma}} \tag{48b}$$

with $\alpha_n \equiv N_n/n_e$ representing the plasma electronegativity and

$$\varsigma \equiv \frac{1+\alpha_n(T_e/T_g)}{1+\alpha_n} \quad ; \tag{49}$$

- the effective ambipolar diffusion coefficient for the positive ions $D_{\mathrm{eff}j_p}$ can be obtained using

$$\frac{D_{\mathrm{eff}j_p}}{D_{\mathrm{amb}j_p}} \simeq \frac{D_{\mathrm{eff}_e}}{D_{\mathrm{amb}_e}} \tag{50a}$$

$$D_{\mathrm{amb}j_p} \simeq D_{j_p} - \mu_{j_p}\frac{D_+ - D_e}{\mu_+ + \mu_e} \quad , \tag{50b}$$

where $D_{\mathrm{amb}j_p}$ represents the classical ambipolar diffusion coefficient for positive ion j_p.

In the case of intermediate to high pressures

$$\frac{D_{\mathrm{eff}j_p}}{D_{\mathrm{amb}j_p}} \simeq \frac{D_{\mathrm{eff}_e}}{D_{\mathrm{amb}_e}} \simeq 1 \tag{51}$$

and the ion transport can be described adopting the *classical ambipolar diffusion model*, also distributed with LoKI-GM (see User Guide [12]), taking $D_{\mathrm{eff}j_p} \simeq D_{\mathrm{amb}j_p}$.

The ambipolar diffusion theory was generalized by Rogoff [38], to account for the presence of negative ions, and the resulting framework was subsequently adopted by several authors, under different approximations. The Quantemol Global Model (QGM) [42] clusters the ion fluxes into two equations for single positive (+) and negative (−) components. The loss frequency of positive ions is formulated following GlobalKin [43], accounting for both ambipolar transport and thermal losses at the wall

$$\tau^{\mathrm{QGM}}_{\mathrm{transp}j_p} \equiv \tau_{\mathrm{amb}j_p} + \tau_{\mathrm{wall}j_p} \simeq \frac{V}{A}\frac{\Lambda}{D_{a_+}} + \frac{V}{A}\frac{4}{\gamma_{j_p}\langle v_{j_p}\rangle} \quad , \tag{52a}$$

where in general $\gamma_{j_p} = 1$ and

$$D_{a_+} \simeq D_+ \frac{1+T_r\,(1+2\alpha)}{1+\alpha T_r} \quad , \tag{52b}$$

with $T_r \equiv T_e/T_+$, $\alpha \equiv 1/\alpha_n$ and D_+ is defined as in (47c). In plasmas with moderate to low electronegativity, the ambipolar diffusion coefficients derived by Rogoff [38] may be used when solving the coupled particle and

momentum-transfer fluid equations. This formulation was initially proposed by Ferreira *et al.* [44], to describe the radial transport in a DC axially homogeneous positive column with three types of charged particles (+, − and electrons), and was later extended by Guerra and Loureiro [45] for application to an oxygen plasma with several positive ions. After a considerable number of approximations, one obtains

$$\tau^{\text{eigen}}_{\text{transp}j_p} \simeq \frac{\Lambda^2_{\text{eigen}}}{D_{\text{amb}j_p}} \quad , \tag{53}$$

where $D_{\text{amb}j_p} \simeq T_r D_{j_p}$ and the effective diffusion length Λ_{eigen} is calculated as an eigenvalue to the problem.

In low pressure discharges, the transport of charged-particles can be described using scaling laws for the density gradient, that rely on the flux of positive ions at the sheath edge, expressed for a single-ion j in an electropositive plasma as

$$\Gamma_j = N_{j_s} u_B \quad , \tag{54}$$

where N_{j_s} the positive-ion density at the sheath edge, and $u_B = \sqrt{k_B T_e / M_j}$ is the Bohm speed. This flux can be written as a function of the edge-to-center density ratio (designated *h-factor*), and can then be used in the volume-average ion-particle balance equation to express the corresponding effective loss frequency in terms of the h-factors [46–49]

$$\left(\tau^{\text{h-factor}}_{\text{transp}_j}\right)^{-1} = 2uB\left(\frac{h_{L_j}}{L} + \frac{h_{R_j}}{R}\right) \quad . \tag{55}$$

Here, the h-factors $h_{L_j} \equiv N_{j_s,\text{axial}}/N_{e_0}$ and $h_{R_j} \equiv N_{j_s,\text{radial}}/N_{e_0}$ are the edge-to-center positive-ion density ratios along the axial and the radial directions, respectively. The h-factors in (55) can be estimated by equating the flux of positive ions at the sheath edge (54) to its expression obtained from different formulations, for example: (i) drift-dominated models for electropositive plasmas, solved numerically at low-pressure [47, 48] or analytically for plane-parallel electropositive discharges [50]; (ii) drift-diffusion models, featuring the transition from electropositive to moderate [48, 49] or strong [51–54] electronegative plasmas.

The models expressed by (32a), (38a)-(38b), (42), for neutral particles, and by (43), (52a)-(52b), (53), (55), for charged particles, are distributed with LoKI-GM as alternative rate-coefficient functions for the transport of species.

2.6 Thermal model for the gas/plasma

The results of the plasma chemistry are significantly dependent on the gas temperature, because it affects several rate coefficients for heavy-species collisions causing changes also in the total gas density, at constant pressure. Consequently, global models are often complemented by the thermal balance equation for the gas/plasma, which self-consistently calculates the *average* temperature T_g of the heavy-species in the system, assumed thermalised.

The thermal model adopted in LoKI-GM solves the thermal balance equations in various regions of a cylindrical discharge, for a distributed source-term within the gas/plasma volume and a localized source-term at the discharge wall, considering two alternative boundary conditions [5]:

- *external*, in which case the thermal balance equations are solved in the gas/plasma system, in the inner region close to the lateral wall of the discharge cylindrical tube, and in the outer region of the cylindrical tube, to calculate the temperature T_g of the heavy-species in the gas/plasma system, the near-wall temperature T_{nw} and the wall temperature T_w, giving the external temperature T_{ext} as boundary condition;
- *wall*, in which case the thermal balance equations are solved in the gas/plasma system, and in the inner region close to the lateral wall of the discharge cylindrical tube, to calculate the temperature T_g of the heavy-species in the gas/plasma system and the near-wall temperature T_{nw}, giving the wall temperature T_w as boundary condition.

In the **gas/plasma system** the model considers a **radial conduction-flow** that sets a **one-dimensional** parabolic profile for the gas temperature (see figure 2), between the discharge centre ($r = 0$) and a radial position near the lateral wall of the discharge cylindrical tube ($r = R_{nw}$)

$$T_g(r) = T_0 - (T_0 - T_{nw})\frac{r^2}{R^2_{nw}} \quad \text{for } 0 \le r \le R_{nw}, \text{ with } T_0 \equiv T_g(0) \text{ and } T_{nw} \equiv T_g(R_{nw}) \quad , \tag{56}$$

[5] The present section and the corresponding implementation of the thermal module in LoKI-GM received a considerable contribution by T C Dias.

which is used to obtain the spatial-average gas/plasma thermal balance equation (in eV s^{-1} m^{-3}) [17, 55]

$$\sum_k N_k \frac{C_k}{N_A}\frac{\partial T_g}{\partial t} = -\frac{\lambda_{\rm mix}\left[T_g - T_{nw}\right]}{\Lambda_{\rm th}^2} + \left|\frac{\Theta_{\rm el}}{N}\right| N n_e + \sum_{r={\rm vol}} R_r \Delta H_r + f_w \sum_{r={\rm wall}} R_r \Delta H_r \quad . \tag{57}$$

Here, T_g is the spatial-average gas temperature, T_{nw} is the near-wall temperature, C_k is the molar specific-heat at constant pressure (or volume) of the gas k (in eV K^{-1} mol^{-1}); N_A is Avogadro's number; $\Lambda_{\rm th} = R_{nw}/\sqrt{8} \simeq R/\sqrt{8}$ is the thermal diffusion length; $\Theta_{\rm el}/N \equiv \Theta_{\rm el}^{\rm loss}/N + \Theta_{\rm el}^{\rm gain}/N$ is the net power density lost by the electrons in elastic collisions with the heavy-species (see section 3.1.2 for more details); R_r represents the rate of a **volume reaction** r = vol or a **wall reaction** r = wall, that involves heavy-species and yields an enthalpy exchange ΔH_r (in eV; $\Delta H_r \gtrless 0$ for exothermic and endothermic reactions, respectively). Note that the volume source-term can be written as $\sum_{r={\rm vol}} R_r \Delta H_r \equiv \sum_{r=e-h} R_r \Delta H_r + \sum_{r=h-h} R_r \Delta H_r$, by separating the contributions of electron-heavy-species collisions and heavy-species-heavy-species collisions, respectively; f_w is a general parameter representing the fraction of the enthalpy-exchange in wall-reactions that returns to the gas/plasma volume, such that

- $0 \leq f_w \leq 1$ (chosen by the user), for the *wall* boundary condition
- $f_w \equiv 0$ (set by LoKI-GM), for the *external* boundary condition;

and $\lambda_{\rm mix}$ is the thermal conductivity of the gas mixture (in eV s^{-1} K^{-1} m^{-1}) given by [34]

$$\lambda_{\rm mix} = \sum_k \lambda_k \left[1 + \sum_{k\neq l} G_{k,l}\frac{\xi_k}{\xi_l}\right]^{-1} \tag{58a}$$

$$G_{k,l} = \frac{1.065}{2\sqrt{2}}\left(1+\frac{M_k}{M_l}\right)^{-1/2}\left[1+\left(\frac{\lambda_k^0}{\lambda_l^0}\right)^{1/2}\left(\frac{M_k}{M_l}\right)^{1/4}\right]^2 \tag{58b}$$

$$\lambda_k^0 \equiv \frac{\lambda_k}{E_k} \quad , \tag{58c}$$

with λ_k the thermal conductivity of the gas k and E_k the corresponding Eucken factor. An improved value for the latter factor is given by [34]

$$E_k = 0.115 + 0.354\frac{C_{p,k}}{k_B N_A} \tag{58d}$$

with $C_{p,k}$ the molar specific-heat at constant pressure of the gas k (in eV K^{-1} mol^{-1}).

The temperature at the plasma centre can be expressed as a function of the average and the near-wall temperatures as follows:

$$T_g \equiv \frac{1}{R_{nw}^2/2}\int_0^{R_{nw}} T_g(r) r dr = \frac{T_0 + T_{nw}}{2} \Longrightarrow T_0 = 2T_g - T_{nw} \quad . \tag{59}$$

In equation (57), the term on the left-hand side represents the intrinsic time-variation of the gas thermal-power; whereas the terms on the right-hand side represent, in order, the power lost in conduction, the power gained in electron-neutral elastic collisions, and the net power gained due to the different volume/wall reactions involving heavy-species.

In the **near-wall region**, the thermal balance equation yields the equality between the power per unit area (in eV s^{-1} m^{-2}) flowing outward the gas/plasma system by **conduction**, $\Gamma_{\rm cond}(R_{nw})$, and flowing inward the region near the tube wall by **convection**, $\Gamma_{\rm conv,int}$

$$\begin{cases} \Gamma_{\rm cond}(R_{nw}) \simeq \frac{4\lambda_{\rm mix}}{R}(T_g - T_{nw}) \\ \\ \Gamma_{\rm conv,int} = h_{\rm int}(T_{nw} - T_w) \end{cases} , \quad \Gamma_{\rm cond}(R_{nw}) = \Gamma_{\rm conv,int} \Longrightarrow T_{nw} = \frac{\frac{4\lambda_{\rm mix}}{Rh_{\rm int}}T_g + T_w}{\frac{4\lambda_{\rm mix}}{Rh_{\rm int}} + 1} \quad , \tag{60}$$

where $h_{\rm int}$ is the convection coefficient (in eV s^{-1} K^{-1} m^{-2}) in the inner region near the tube wall.

In the **region outside the cylindrical tube** the model solves the thermal balance equation across the tube wall (neglecting the effects of conduction within the dielectric), with a localized source-term at the wall boundary due to the enthalpy-exchange in wall-reactions, and considering an **external convection flow** $\Gamma_{\text{conv,ext}}$

$$\begin{cases} \left(\Gamma_{\text{conv,ext}} - \Gamma_{\text{conv,int}}\right) A = \sum_{r=\text{wall}} R_r \Delta H_r V \\ \\ \Gamma_{\text{conv,ext}} = h_{\text{ext}} (T_w - T_{\text{ext}}) \end{cases}$$

$$\implies \quad T_w = \frac{h_{\text{ext}} T_{\text{ext}} + h_{\text{int}} T_{nw} + \sum_{r=\text{wall}} R_r \Delta H_r (R/2)}{h_{\text{ext}} + h_{\text{int}}} \tag{61a}$$

$$\implies \quad \begin{cases} T_w = \frac{\frac{4\lambda_{\text{mix}}}{Rh_{\text{ext}}} T_g + \left(\frac{4\lambda_{\text{mix}}}{Rh_{\text{int}}} + 1\right) T_{\text{ext}} + \left(\frac{4\lambda_{\text{mix}}}{Rh_{\text{int}} h_{\text{ext}}} + \frac{1}{h_{\text{ext}}}\right) \sum_{r=\text{wall}} R_r \Delta H_r (R/2)}{\frac{4\lambda_{\text{mix}}}{Rh_{\text{int}}} + \frac{4\lambda_{\text{mix}}}{Rh_{\text{ext}}} + 1} \\ \\ T_{nw} = \frac{\left(\frac{4\lambda_{\text{mix}}}{Rh_{\text{int}}} + \frac{4\lambda_{\text{mix}}}{Rh_{\text{ext}}}\right) T_g + T_{\text{ext}} + \frac{1}{h_{\text{ext}}} \sum_{r=\text{wall}} R_r \Delta H_r (R/2)}{\frac{4\lambda_{\text{mix}}}{Rh_{\text{int}}} + \frac{4\lambda_{\text{mix}}}{Rh_{\text{ext}}} + 1} \end{cases} , \tag{61b}$$

where equations (61b) were obtained combining (60)-(61a). Here, h_{ext} is the convection coefficient (in eV s^{-1} K^{-1} m^{-2}) in the region outside the cylindrical wall; $A = 2\pi RL$ and $V = \pi R^2 L$ are the area and the volume of the cylindrical tube, respectively. As expected, by setting $h_{\text{int}}, h_{\text{ext}} \gg 1$ in equations (60)-(61b), one obtains $T_{nw} \sim T_w \sim T_{\text{ext}}$.

In summary,

- when adopting *external*, the model solves (57) at $f_w \equiv 0$, using (61b) and the boundary condition T_{ext}, to obtain the time-dependent solutions $T_g(t)$, $T_{nw}(t)$ and $T_w(t)$ for source terms $\sum_{r=\text{vol}} R_r(t) \Delta H_r$ and $\sum_{r=\text{wall}} R_r(t) \Delta H_r$;
- when adopting *wall*, the model solves (57) with $0 \leq f_w \leq 1$, using (60) and the boundary condition T_w, to obtain the time-dependent solutions $T_g(t)$ and $T_{nw}(t)$ for source terms $\sum_{r=\text{vol}} R_r(t) \Delta H_r$ and $\sum_{r=\text{wall}} R_r(t) \Delta H_r$.

Note that the time-dependent solutions are obtained with the following approximations: (i) by articulating the time-dependent equation of conduction with steady-state equations for convection; (ii) by assuming a parabolic profile for the gas temperature, valid at all times. These approximations might hinder the convergence of the model depending on the temperature T adopted to calculate the thermal velocity $\langle v \rangle$ (see (34)).

When running the thermal model for the *wall* boundary condition, $T = T_w$ is a good choice to calculate $\langle v \rangle$, because T_w = constant in this case. However, when adopting the *external* boundary condition, $T = T_w(t)$ can lead to difficulties in model convergence due to a positive feedback loop in (61b) (when the source term at the wall depends on $\langle v \rangle$, and $\langle v \rangle$ depends on $T_w(t)$).

3 Simulation input/output

The general input to LoKI-GM specifies the working conditions, including the applied reduced electric field and frequency, gas pressure and temperature, electron density (or a related quantity such as discharge current or power density), and, in the case of gas mixtures, the fractions of the various components. For LoKI-B, the input data additionally comprise the population distributions of the electronic, vibrational, and rotational states of the atomic/molecular species considered (determined self-consistently when coupled with LoKI-C), as well as the relevant sets of electron-scattering cross sections obtained from the open-access database LXCat [56]. For LoKI-C, the required inputs further include the geometrical dimensions of the plasma reactor, gas inflow/outflow conditions, wall/external temperatures relevant to the gas/plasma thermal model, the surface site densities when a microkinetic mesoscopic surface model is employed, and the rate coefficients for all reactions considered in the kinetic scheme, including the electron rate coefficients evaluated in LoKI-B.

As output, LoKI-B provides the isotropic and anisotropic components of the electron distribution function, the corresponding electron macroscopic parameters (transport parameters and rate coefficients), and the electron power absorbed from the electric field and transferred to the various collisional channels. LoKI-C yields the densities of states and the reaction rates associated with the processes included in the kinetic scheme. Sections 3.1 and 3.2 provide details about these output quantities.

The supported input/output formats comprise both graphical user interfaces (GUI) and text-based files in ASCII, JSON [57] and HDF5 [58]. Currently, GUI-based input is available only for LoKI-B; however, future developments will extend this capability to LoKI-GM. In the HDF5 format, all output data are stored in a single, structured file. This file can be accessed and post-processed using Python scripts, through one of the available HDF5 interfaces [59], or by applying appropriate import filters in data analysis tools such as MATLAB or Origin [60].

3.1 Output of LoKI-B

On output, LoKI-B calculates the isotropic and the anisotropic parts of the electron distribution function, in addition to various electron macroscopic parameters obtained by integration over the EEDF (obtained from the solution to the EBE or adopting some other form prescribed by the user, *e.g.* a generalized Maxwellian EEDF), namely:

- swarm parameters (transport parameters and rate coefficients), which can be used to adjust complete sets of electron-scattering cross sections [61,62] or as input parameters in macroscopic (fluid / global) plasma models [17,63,64];
- power-transfer terms [17,24], providing the distribution of power among the different collisional channels and controlling the calculation errors (see section 5.1.3).

3.1.1 Swarm parameters

The electron mean energy (in eV) is defined as

$$\varepsilon \equiv \int_0^\infty u^{3/2} f(u) du \quad , \tag{62}$$

and is used to introduce the kinetic temperature (in eV)

$$T_e \equiv \frac{2}{3}\varepsilon \quad . \tag{63}$$

The DC reduced transverse free-diffusion coefficient (in $m^{-1}\ s^{-1}$) and the DC reduced mobility (in $V^{-1}\ m^{-1}\ s^{-1}$) are calculated from [7,17,24]

$$D_e N = \frac{\gamma}{3}\int_0^\infty \frac{u}{\Omega_x(u)} f(u) du \tag{64a}$$

$$\mu_e N = -\frac{\gamma}{3}\int_0^\infty \frac{u}{\Omega_x(u)} \frac{df(u)}{du} du \quad , \tag{64b}$$

with $x = c, \mathrm{SST}$ for temporal or spatial growth of the electron density, respectively (see (11a)-(11b)), and these quantities allow defining the electron characteristic energy (in eV)

$$u_k \equiv \frac{D_e N}{\mu_e N} \quad . \tag{65}$$

Similarly to (64a)-(64b), the DC reduced transverse diffusion coefficient and the DC reduced mobility for energy transport (in eV $\mathrm{m}^{-1}\ \mathrm{s}^{-1}$ and $\mathrm{m}^{-1}\ \mathrm{s}^{-1}$, respectively) are calculated from [7, 65] [6]

$$D_\varepsilon N = \frac{\gamma}{3}\int_0^\infty \frac{u^2}{\Omega_x(u)} f(u) du \tag{66a}$$

$$\mu_\varepsilon N = -\frac{\gamma}{3}\int_0^\infty \frac{u^2}{\Omega_x(u)} \frac{df(u)}{du} du \quad , \tag{66b}$$

with $x = c, \mathrm{SST}$ for temporal or spatial growth of the electron density, respectively (see (11a)-(11b)).

In the case of HF simulations at $\omega \gg \nu_c$, in addition to the DC parameters (64a)-(66b) that might be of interest for fluid/global models, the code evaluates the *complex* HF mobility, with the real [7] and the imaginary parts given by [24]

$$\Re[\mu_{\mathrm{HF}} N] = -\frac{\gamma}{3}\int_0^\infty \frac{u}{\Omega_{\mathrm{PT}}(u)} \frac{df(u)}{du} du \tag{67a}$$

$$\Im[\mu_{\mathrm{HF}} N] = \frac{\gamma}{3}\int_0^\infty \frac{u}{\Omega_{\mathrm{PT}}(u)} \frac{\omega/N}{\Omega_c \gamma \sqrt{u}} \frac{df(u)}{du} du \quad . \tag{67b}$$

The rate coefficients (in $\mathrm{m}^3\ \mathrm{s}^{-1}$), for excitation / de-excitation mechanisms between states i and $j > i$, are given by [17]

$$C_{i,j} = \gamma \int_0^\infty u \sigma_{i,j}(u) f(u) du \tag{68a}$$

$$C_{j,i} = \gamma \int_0^\infty \frac{g_i}{g_j}(u + V_{i,j}) \sigma_{i,j}(u + V_{i,j}) f(u) du \quad . \tag{68b}$$

Note that (68a)-(68b) can be applied not only to the cross sections adopted for solving the EBE, but also to the *extra* set of cross sections defined by the user as additional input data of LoKI-B (see User Guide [12]).

The ionisation and the attachment rate coefficients are defined similarly to (68a) as

$$C_{i,\mathrm{ion}} \equiv \frac{\langle \nu_{i,\mathrm{ion}} \rangle}{N} = \gamma \int_0^\infty u \sigma_{i,\mathrm{ion}}(u) f(u) du \tag{69a}$$

$$C_{i,\mathrm{att}} \equiv \frac{\langle \nu_{i,\mathrm{att}} \rangle}{N} = \gamma \int_0^\infty u \sigma_{i,\mathrm{att}}(u) f(u) du \quad . \tag{69b}$$

In a DC case, the **anisotropic equations** (7b) and (8b) can be integrated after multiplication by $\gamma u/3$, to define the electron mean drift velocity (in $\mathrm{m\ s}^{-1}$)

$$v_d \equiv \frac{\gamma}{3}\int_0^\infty u f^1(u) du \quad ,$$

yielding, respectively,

$$v_d = \mu_e E \ , \ \textbf{for a PT situation} \tag{70a}$$

$$v_d = \mu_e E - D_e \frac{1}{n_e}\frac{dn_e}{dz} = \mu_e E - D_e \alpha_{\mathrm{eff}} \ , \ \textbf{for a SST situation} \quad . \tag{70b}$$

[6] These transport coefficients correspond to the thermoelectricity β and the heat diffusion G defined in [15]. Note that here, and contrary to [7, 15, 65], D_ε and μ_ε are not normalized to the electron mean energy ε.

[7] In `LoKI-B_v1.0.0`, the `swarmParameters` datafile in the output of HF simulations was writing the DC mobility, calculated from (64b) at $\omega = 0$, instead of the real part of the HF mobility. We acknowledge Ilija Stefanovic (Ruhr Universität Bochum, Germany) for pointing this out.

In an HF case, simulations can only be performed for a PT situation. In this case, the **anisotropic equation** (6) can be integrated after multiplication by $\gamma u/3$, to define the *time-dependent* electron drift velocity

$$v_d(t) \equiv \frac{\gamma}{3}\int_0^\infty u f^1(u,t)du = \Re\left[\frac{\gamma}{3}\int_0^\infty u\tilde{f}^1(u)e^{j\omega t}du\right] \equiv \Re[\mu_{\rm HF}E_p e^{j\omega t}] \quad .$$

Consistently, the parameters (70a)-(70b) are not calculated, and the user can obtain $v_d(t)$ from the HF mobility (67a)-(67b)

$$v_d(t) = \Re[\mu_{\rm HF}E_p e^{j\omega t}] = \frac{E_p}{2}\Re[\mu_{\rm HF}e^{j\omega t} + \mu^\star_{\rm HF}e^{-j\omega t}] = \{\Re[\mu_{\rm HF}N]\cos(\omega t) - \Im[\mu_{\rm HF}N]\,\sin(\omega t)\}\,\frac{\zeta E}{N} \quad .$$

For DC/HF simulations involving the **time-growth of the electron density**, the **isotropic equation** (7a) can be integrated after multiplication by γ, to define the reduced-effective frequency (in $\mathrm{m^3\ s^{-1}}$)

$$\frac{\langle\nu_{\rm eff}\rangle}{N} = \gamma\sum_{k,i}\delta_{k_i}\int_0^\infty [S_{k_i,{\rm ion}}(u) + S_{k_i,{\rm att}}(u)]\,du \quad , \tag{71}$$

where

$$\nu_{\rm eff} = \langle\nu_{\rm ion}\rangle - \langle\nu_{\rm att}\rangle \quad , \tag{72}$$

with

$$\begin{aligned}\langle\nu_{\rm ion}\rangle &\equiv \sum_{k,i}\delta_{k_i}\langle\nu_{k_i,{\rm ion}}\rangle \\ \langle\nu_{\rm att}\rangle &\equiv \sum_{k,i}\delta_{k_i}\langle\nu_{k_i,{\rm att}}\rangle \quad .\end{aligned}$$

In this case, equation (71) is solved iteratively until convergence of the parameter $\langle\nu_{\rm eff}\rangle/N$ is achieved (see section 5.1.5).

For DC simulations involving the **spatial-growth of the electron density**, the **isotropic equation** (8a) can be integrated after multiplication by γ, to define the effective Townsend coefficient

$$\alpha_{\rm eff} = \frac{\langle\nu_{\rm ion}\rangle - \langle\nu_{\rm att}\rangle}{v_d} = \frac{\langle\nu_{\rm eff}\rangle}{v_d} \quad , \tag{73}$$

with

$$\frac{\alpha}{N} \equiv \frac{\langle\nu_{\rm ion}\rangle/N}{v_d} \tag{74a}$$

$$\frac{\eta}{N} \equiv \frac{\langle\nu_{\rm att}\rangle/N}{v_d} \tag{74b}$$

the first and the second reduced Townsend coefficients (in $\mathrm{m^2}$) for ionisation and attachment, respectively. In this case, equations (70b) and (73) can be combined to yield [66]

$$D_e\alpha_{\rm eff}^2 - \mu_e E\alpha_{\rm eff} + \langle\nu_{\rm eff}\rangle = 0 \quad , \tag{75}$$

which is solved iteratively until convergence of the parameter $\alpha_{\rm eff}$ is achieved (see section 5.1.5).
Note that the initial EEDF guess may lead to a negative discriminant $\mu_e^2E^2 - 4D_e\langle\nu_{\rm eff}\rangle$ of the quadratic equation (75), in which case, it can be calculated

- either assuming that there is no electron density gradient (see (70b)): $\alpha_{\rm eff} = \langle\nu_{\rm eff}\rangle/v_d = \langle\nu_{\rm eff}\rangle/(\mu_e E)$;
- or assuming that the discriminant is zero (and using the expression for the root of the quadratic equation): $\alpha_{\rm eff} = \mu_e E/(2D_e) = 4D_e\langle\nu_{\rm eff}\rangle/(2D_e\mu_e E) = 2\langle\nu_{\rm eff}\rangle/(\mu_e E)$.

Note further that the parameters (73)-(74b) are not calculated in the case of HF simulations.

3.1.2 Power-transfer terms

The balance equation for the electron power-density, per electron at unit gas density, is obtained as follows:

i) by writing the **isotropic component** of the EBE as a gain/loss equation, *i.e.*

- for the **steady-state EBE**:
  - when adopting a **temporal-growth model** (see equation (7a))

$$0 = -\frac{1}{\gamma\sqrt{u}}\frac{\langle\nu_{\rm eff}\rangle}{N}uf(u)\bigg|_n - \frac{1}{N}\frac{1}{\gamma}\frac{dG(u)}{du}\bigg|_n + S_n \quad ; \tag{76a}$$

  - when adopting a **spatial-growth model** (see equation (8a))

$$0 = -\frac{1}{3}\frac{\alpha_{\rm eff}}{N}uf^1(u)\bigg|_n - \frac{1}{N}\frac{1}{\gamma}\frac{dG(u)}{du}\bigg|_n + S_n \quad ; \tag{76b}$$

- for the **time-dependent EBE**:
  - when adopting a **temporal-growth model** (see equation (19a))

$$\frac{\partial f(u,t)}{\partial t}\bigg|_n = -N\frac{\langle\nu_{\rm eff}\rangle}{N}f(u)\bigg|_n - \frac{1}{\sqrt{u}}\frac{dG(u)}{du}\bigg|_n + N\gamma\sqrt{u}\frac{1}{u}S_n \quad ; \tag{76c}$$

  - when adopting a **spatial-growth model**

$$\frac{\partial f(u,t)}{\partial t}\bigg|_n = -\frac{1}{3}\gamma\sqrt{u}N\frac{\alpha_{\rm eff}}{N}f^1(u)\bigg|_n - \frac{1}{\sqrt{u}}\frac{dG(u)}{du}\bigg|_n + N\gamma\sqrt{u}\frac{1}{u}S_n \quad . \tag{76d}$$

ii) by multiplying either (76a) or (76b) by γu and integrating it over all energies; or by multiplying either (76c) or (76d) by $u^{3/2}/N$ and integrating it over all energies.

The general result writes (in eV s^{-1} m^3)

$$\frac{\Theta(t)}{N} = \frac{\Theta_{\rm growth}}{N} + \frac{\Theta_E}{N} + \frac{\Theta_{\rm coll}}{N} \quad , \tag{77a}$$

where the different terms on the right-hand side represent, in order, the variation of the power density due to non-conservative mechanisms inducing a time/space net growth of the electron density, the power density gained from the applied electric field [8], and the power density gained/lost in collisional ("coll") events, which can be separately written as (note that all "gain" terms are positively defined and all "loss" terms are negatively defined).

$$\frac{\Theta_{\rm coll}}{N} \equiv \frac{\Theta_{\rm coll}^{\rm gain}}{N} + \frac{\Theta_{\rm coll}^{\rm loss}}{N} \quad . \tag{77b}$$

The expressions for the various power-transfer terms are given below (see equations (12a)-(12d); see also

[8] Note that the electron power-density gained from the applied electric field corresponds to

$$\frac{\Theta_E}{N} = \begin{cases} \mu_e N \left(\frac{E}{N}\right)^2 & \text{, for DC simulations} \\ \Re[\mu_{\rm HF} N] \left(\frac{E}{N}\right)^2 & \text{, for HF simulations} \end{cases} .$$

section 3.1.1)

$$\frac{\Theta(t)}{N} = \begin{cases} 0 \ , \text{ for stationary and steady-state simulations} \\ \dfrac{1}{N}\dfrac{\partial \varepsilon(t)}{\partial t} \ , \text{ for time-dependent simulations} \end{cases} \tag{78a}$$

$$\frac{\Theta_{\text{growth}}}{N} = \begin{cases} -\dfrac{\langle \nu_{\text{eff}} \rangle}{N}\varepsilon \ , \text{ for a PT situation} \\ -\dfrac{\alpha_{\text{eff}}}{N}\left(\mu_\varepsilon E - D_\varepsilon \alpha_{\text{eff}}\right) \ , \text{ for a SST situation} \end{cases} \tag{78b}$$

$$\begin{aligned} \frac{\Theta_E}{N} &= -\frac{1}{N}\int_0^\infty u \frac{dG_E(u)}{du} du \\ &= \gamma \begin{cases} \displaystyle\int_0^\infty u \frac{d}{du}\left[g_E^{\text{PT}}(u) \frac{df(u)}{du} \right] du \left(\frac{E}{N}\right)^2 \ , \text{ for a PT situation} \\ \displaystyle\int_0^\infty u \frac{d}{du}\left\{ g_E^{\text{SST}}(u) \left[\frac{E}{N}\frac{\alpha_{\text{eff}}}{N} f(u) + \left(\frac{E}{N}\right)^2 \frac{df(u)}{du} \right] \right\} du \ , \text{ for a SST situation} \end{cases} \end{aligned} \tag{78c}$$

$$\begin{aligned} \frac{\Theta_{x=\text{el,CAR}}}{N} &= -\frac{1}{N}\int_0^\infty u \frac{dG_x(u)}{du} du \\ &= \gamma \int_0^\infty u \frac{d}{du}\left\{ g_x(u) \left[f(u) + \frac{k_B T_g}{e}\frac{df(u)}{du} \right] \right\} du \end{aligned} \tag{78d}$$

$$\frac{\Theta_{x=\text{el,CAR}}^{\text{gain}}}{N} = \gamma \int_0^\infty u \frac{d}{du}\left[g_x(u) \frac{k_B T_g}{e}\frac{df(u)}{du} \right] du \tag{78e}$$

$$\frac{\Theta_{x=\text{el,CAR}}^{\text{loss}}}{N} = \frac{\Theta_{x=\text{el,CAR}}}{N} - \frac{\Theta_{x=\text{el,CAR}}^{\text{gain}}}{N} \tag{78f}$$

$$\begin{aligned} \frac{\Theta_{ee}}{N} &= -\frac{1}{N}\int_0^\infty u \frac{dG_{ee}(u)}{du} du \\ &= \gamma \int_0^\infty u \frac{d}{du}\left\{ g_{ee}(u) \left[I(u) f(u) + J(u)\frac{df(u)}{du} \right] \right\} du \end{aligned} \tag{78g}$$

$$\frac{\Theta_{ee}^{\text{gain}}}{N} = \gamma \int_0^\infty u \frac{d}{du}\left[g_{ee}(u) J(u) \frac{df(u)}{du} \right] du \tag{78h}$$

$$\frac{\Theta_{ee}^{\text{loss}}}{N} = \frac{\Theta_{ee}}{N} - \frac{\Theta_{ee}^{\text{gain}}}{N} \tag{78i}$$

$$\frac{\Theta_{\text{sup}}^{\text{gain}}}{N} = \gamma \sum_{i,j>i} \int_0^\infty S_{i,j}|_{\text{sup}}(u) u du = \sum_{i,j>i} \delta_j C_{j,i} V_{i,j} \tag{78j}$$

$$\frac{\Theta_{\text{inel}}^{\text{loss}}}{N} = \gamma \sum_{i,j>i} \int_0^\infty S_{i,j}|_{\text{inel}}(u) u du = -\sum_{i,j>i} \delta_i C_{i,j} V_{i,j} \tag{78k}$$

$$\frac{\Theta_{\text{ion}}}{N} = \gamma \sum_i \int_0^\infty S_{i,\text{ion}}(u) u du = -\sum_i \delta_i C_{i,\text{ion}} V_{i,\text{ion}} \tag{78l}$$

$$\frac{\Theta_{\text{att}}}{N} = \gamma \sum_i \int_0^\infty S_{i,\text{att}}(u) u du \tag{78m}$$

In these equations, the labels "inel" and "sup" refer to inelastic and superelastic collisions, respectively, associated with electronic, vibrational and rotational excitation / de-excitation mechanisms.

In time-dependent simulations the quantity $\Theta(t)/N$ is calculated by summing all the power terms on the right-hand side of (77a), and on output the value of this quantity is tagged as *Power balance* (see User Guide [12]).

In stationary and steady-state simulations, the code evaluates the relative power balance equation, dividing (77a) by the *reference power-density* (per electron at unit gas density) $\Theta_{\rm ref}/N$, with the latter defined as the sum of all power-gain terms.

As mentioned, LoKI-B can also adopt a generalized Maxwellian EEDF prescribed by the user (see section 2.4). In terms of output, the expressions in section 3.1.1 hold for all swarm parameters, but the power-transfer terms are taken in a slightly different way.

For the latter, the following applies

- the electron power density gained from the applied electric field is deduced from the (stationary) power balance equation, obtained from (77a)-(77b) at $\Theta(t)/N = \Theta_{\rm growth}/N = 0$

$$\frac{\Theta_E}{N} = -\frac{\Theta_{\rm coll}}{N} \equiv -\frac{\Theta_{\rm coll}^{\rm gain}}{N} - \frac{\Theta_{\rm coll}^{\rm loss}}{N} \quad , \tag{79a}$$

  with the $\Theta_{\rm coll}^{\rm gain}/N$ and $\Theta_{\rm coll}^{\rm loss}/N$ obtained from the different terms (78d)-(78l), calculated for the prescribed EEDF (23);

- the *equivalent* applied reduced electric-field is deduced from

$$\frac{E}{N} = \sqrt{\frac{-\Theta_{\rm coll}/N}{(\Theta_E/N)/(E/N)^2}} \quad , \tag{79b}$$

  where the expression in the denominator corresponds to (82d) (for a PT situation) divided by the square of the reduced electric-field [9];

- the electron power density transferred in collisional events is calculated assuming conservative mechanisms only, *i.e.* treating ionisation as an excitation mechanism and setting to zero the power-transfer due to attachment.
  For this reason, **selecting a prescribed EEDF for studying electronegative gases is not recommended.**

3.2 Output of LoKI-C

On output, LoKI-C calculates

- the steady-state and the time-dependent absolute densities N_{k_i} of the various states i of gas k;
- the steady-state populations $\xi_{k_i} \equiv N_{k_i}/N_{\text{parent of } k_i}$ of the various states i of gas k (see section 2.1);
- the final (self-consistent/converged) values of E/N, p, T_g, T_w, N, $Q_{\rm sccm}^{\rm outflow}$, n_e, $I_{\rm dc}$, $dP_{\rm discharge}/dV$ (see section 4);
- the steady-state values of the rate coefficients k_{r_i} for the different reactions r affecting states i;
- the steady-state values of the rates for the different reactions considered in the kinetic scheme;
- the steady-state fractional contribution (in percentage) of the different reactions to the creation/destruction of each state, as well as the final rate balance.

[9] In practice, the *equivalent* applied reduced electric-field is deduced from

$$\frac{E}{N} = \begin{cases} \sqrt{\frac{-\Theta_{\rm coll}/N}{\mu_e N}} & \text{, for DC simulations} \\ \sqrt{\frac{-\Theta_{\rm coll}/N}{\Re[\mu_{\rm HF} N]}} & \text{, for HF simulations} \end{cases} .$$

The rates R_{r,k_i} of generic reactions r, contributing to the creation/destruction of state k_i, are given by (see section 2.5)

$$R_{r,k_i} \equiv \begin{cases} a^{(2)}_{k_i r} k_r \Pi_l N_l^{a^{(1)}_{lr}} \quad , \text{ reaction gain for state } k_i \\ -a^{(1)}_{k_i r} k_r \Pi_l N_l^{a^{(1)}_{lr}} \quad , \text{ reaction loss for state } k_i \\ a^{(2)}_{k_i r} \frac{N_l}{\tau_r} \quad , \text{ transport gain for neutral state } k_i \\ -\frac{N_{k_i}}{\tau_{\text{transp}_{k_i}}} \quad , \text{ transport loss for neutral/ion state } k_i \end{cases} \tag{80}$$

From (80) it is possible to write the fractional contribution of each reaction to the creation/destruction of a specific state k_i, and the steady-state relative rate-balance for each state (which, upon convergence, should be found typically below $\sim 10^{-10}$), respectively

$$\frac{R^{\text{gain}}_{r,k_i}}{\sum_r R^{\text{gain}}_{r,k_i}} \; ; \; \frac{R^{\text{loss}}_{r,k_i}}{\sum_r R^{\text{loss}}_{r,k_i}} \tag{81a}$$

$$\sum_r \left[\frac{R^{\text{gain}}_{r,k_i} - R^{\text{loss}}_{r,k_i}}{\sum_r R^{\text{gain}}_{r,k_i}} \right] \ll 1 \quad , \tag{81b}$$

where R^{gain}_{r,k_i} and R^{loss}_{r,k_i} represent the gain/loss rates of state k_i due to reaction r.

4 Workflow of LoKI-GM

LoKI-GM comprises two modules (LoKI-B and LoKI-C) that can run self-consistently coupled or as standalone tools. In a coupled configuration, LoKI-B supplies to LoKI-C the electron macroscopic parameters (transport parameters and rate coefficients, see section 3). These parameters are then used to solve the system of nonlinear, spatially-averaged rate balance equations (24), together with the gas/plasma thermal balance equation (57), based on the specified kinetic scheme for the gas/plasma system. Figure 3 represents the general workflow of LoKI.

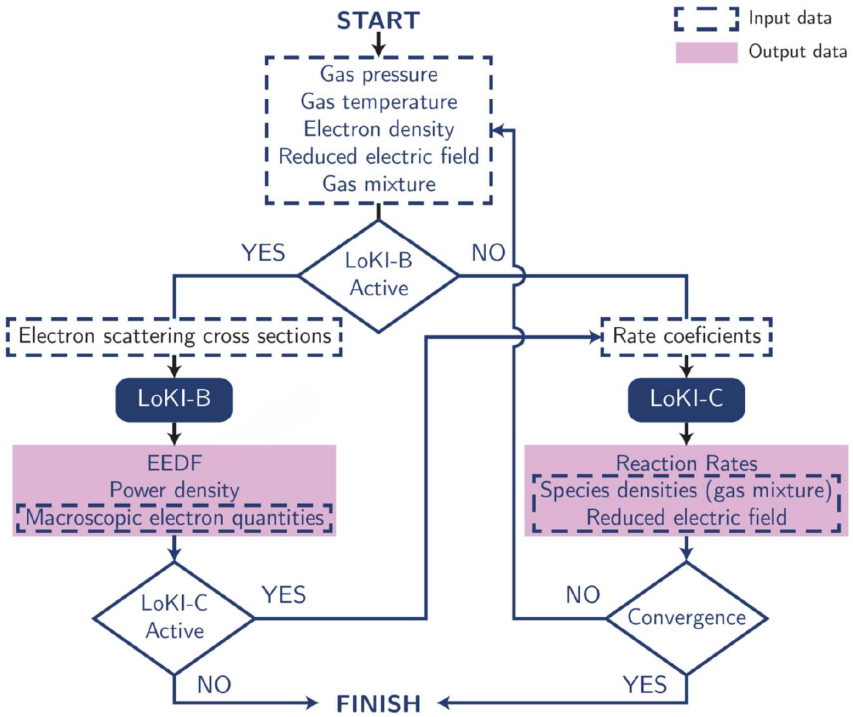


Figure 3: General workflow of LoKI-GM. For *steady-state simulations*, in which LoKI-C and LoKI-B are run in a coupled manner, the input electron density can be complemented by values of the discharge current and/or power density. In this case, the `Convergence` step includes two or three iterative cycles associated with the gas pressure (optional), charge neutrality and the populations of the heavy species relevant to solve the EBE (see text). For *quasi-stationary simulations*, LoKI-B can provide LoKI-C with lookup tables of electron macroscopic parameters tabulated as functions of E/N (for calculations adopting the LFA) and ε (for calculations adopting the LEA). In this case, LoKI-C runs as a standalone tool (see text). *Post-discharge simulations* start with a coupled LoKI-C/LoKI-B steady-state calculation, followed by a standalone LoKI-C simulation at $E/N = 0$ to describe the time-dependent post-discharge, using the steady-state values of species densities, electron transport parameters and electron-neutral collisional power transfer (see text). In the latter two cases, no `Convergence` step is required.

4.1 Steady-state simulations

When modeling *active plasmas*, LoKI-GM can implement *steady-state simulations*, in which LoKI-C and LoKI-B are run in a coupled manner. In this framework, LoKI-C updates, at steady-state, the values of the reduced electric field E/N, the populations of the various heavy species ξ_{k_j} and the gas temperature T_g (see section 3), which are then supplied to LoKI-B for the solution of the electron kinetics.

The numerical workflow involves the following iterative cycles:

1. the *pressure cycle* (optional), running within LoKI-C to ensure calculations at constant gas-pressure p, spec-

ified by the user. This cycle verifies, after the full integration of equations (24) and (57), whether particle conservation (26) is satisfied. Deviations from condition (26) can arise due to dissociation/recombination/adsorption/desorption processes, which can alter the gas pressure (or the total gas density, obtained by summing the densities of the childless states). If such deviations are detected, the initial value of the total gas density is modified accordingly, and the system of equations is integrated again. This iterative procedure is repeated until equation (26) is satisfied within a user-prescribed tolerance;

2. the *neutrality cycle*, coupling LoKI-C to LoKI-B, to ensure calculations at constant electron density (or any related quantity, such as the discharge current or discharge power density). This cycle verifies, after the full integration of equations (24) and (57), whether the charge-neutrality condition is satisfied,

$$\sum_{j_p} N_{j_p} - \sum_{j_n} N_{j_n} = n_e \quad , \tag{82a}$$

modifying the reduced electric field E/N accordingly, until convergence within a user-prescribed tolerance. The charge-neutrality condition (82a) guarantees that, at steady-state, electron production via ionization processes is exactly balanced by their losses due to volume recombination and transport to the walls, where recombination also occurs.

If the calculations are made at constant discharge current I_{dc} (for a cylindrical setup) or at constant discharge power density $dP_{\mathrm{discharge}}/dV$, this iterative cycle modifies also the electron density n_e as to satisfy, alternatively,

$$I_{\mathrm{dc}} \simeq e n_e (\mu_e N) \left(\frac{E}{N}\right) \pi R^2 \tag{82b}$$

$$\frac{dP_{\mathrm{discharge}}}{dV} = e \left(\frac{\Theta_{\mathrm{E}}}{N}\right) n_e N V \quad , \tag{82c}$$

where the power-density gained by the plasma from the applied electric field, per electron at unit gas density, is calculated in LoKI-B using [3, 4]

$$\frac{\Theta_{\mathrm{E}}}{N} = (\mu_e N) \left(\frac{E}{N}\right)^2 \quad . \tag{82d}$$

Calculations adopting these configurations are extremely useful for a direct comparison between simulations and experiments, where n_e is seldom available. For this reason, the power density $dP_{\mathrm{discharge}}/dV$ is expressed in W m^{-3} (SI units), whereas Θ_{E}/N is given in eV s^{-1} m^{-3} as all other energy-related quantities in LoKI-GM;

3. the *global cycle*, coupling LoKI-C to LoKI-B, to converge over the populations of the heavy species considered in the solution of the EBE. This iterative procedure is repeated until convergence, within a user-prescribed tolerance.

The convergence criterion for the iterative cycles is defined by requiring that certain relative errors fall below prescribed tolerances. For cycles 1 and 2, this corresponds to satisfying conditions (26) and (82a), respectively; for cycle 3, it corresponds to the relative differences in the electron swarm parameters between two consecutive iterations. Iterative cycles 1 and 2 employ the same algorithm, based on a combination of numerical methods (bisection, Newton–Raphson, and spline interpolation), with a default tolerance of 5×10^{-4}, which is also adopted for iterative cycle 3 (see section 5.2).

Figure 4 depicts the graphical user interface output for a steady-state simulation in a pure nitrogen DC cylindrical glow discharge ($R = 10^{-2}$ m and $L = 10^{-1}$ m), at $p = 133$ Pa, $n_e = 8.6 \times 10^{15}$ m^{-3} and a wall boundary condition $T_w = 320$ K. The simulation considered the initial values $E/N = 100$ Td, $T_g = 350$ K and $T_{nw} = 320$ K (see upper-right panel in Fig. 4), obtaining at steady-state the self-consistent values $E/N \simeq 91.9$ Td (indicated above the middle graph, in the lower-right panel), $T_g \simeq 422$ K and $T_{nw} \simeq 330$ K (see red and violet dashed curves of the plot in the upper-left panel). Additionally, the simulation results yielded $n_{\mathrm{N}_2^+} \simeq 3.9 \times 10^{15}$ m^{-3}, $n_{\mathrm{N}_4^+} \simeq 3.2 \times 10^{15}$ m^{-3}, $n_{\mathrm{N}_3^+} \simeq 1.5 \times 10^{15}$ m^{-3} and $n_{\mathrm{N}^+} \simeq 2.3 \times 10^{13}$ m^{-3}, thus satisfying the charge-neutrality condition (82a) for $n_e = 8.6 \times 10^{15}$ m^{-3}.

As mentioned, the pressure cycle is optional, since its activation depends on the outflow model adopted (see section 2.5). When k_{outflow} is computed by specifying a **numerical value** for the outflow $Q_{\mathrm{sccm}}^{\mathrm{outflow}}$ (top

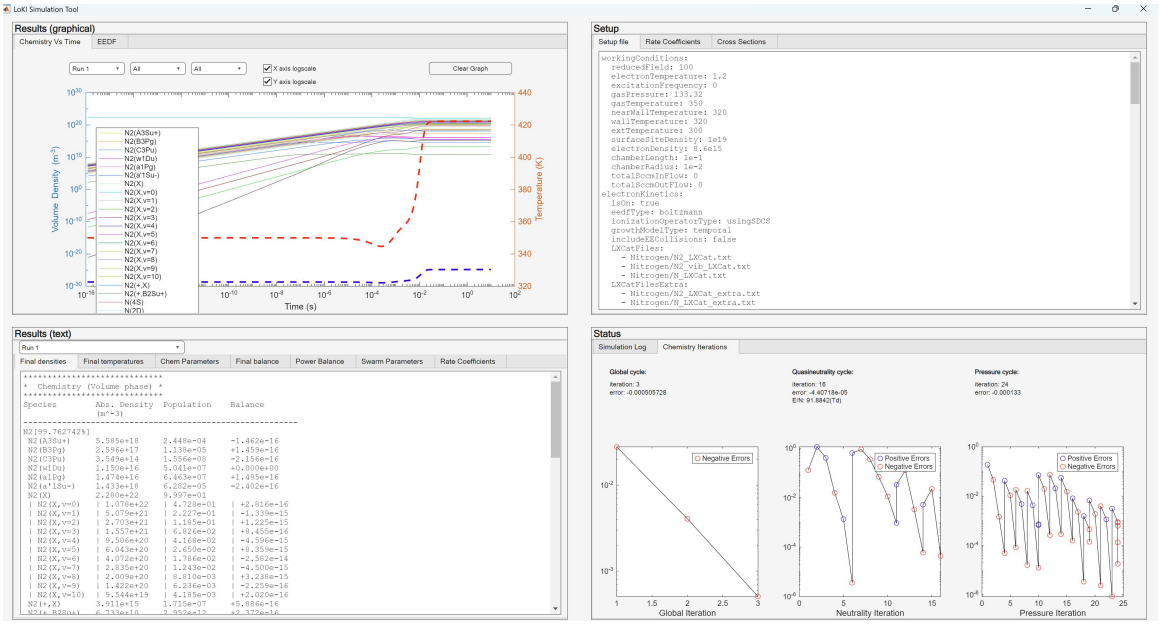


Figure 4: Graphical user interface output of LoKI-GM, for a steady-state simulation in a pure nitrogen DC cylindrical glow discharge ($R = 10^{-2}$ m and $L = 10^{-1}$ m), at $p = 133$ Pa and $n_e = 8.6 \times 10^{15}$ m^{-3}. The panels are as follows: upper-left, the results for the densities of species (solid curves), the gas temperature (red dashed curve), and the near-wall temperature (violet dashed curve), as a function of the calculation time; upper-right, the summary of the input data; lower-right, the graphs for the three convergence cycles, from right to left, for the pressure, the neutrality, and the global cycle; lower-left, the results for the densities of species in text format.

equation of (27)), the steady-state pressure is obtained by iterating on the gas density within the pressure cycle; alternatively, when an **isobaric model** is adopted (bottom equation of (27)), the steady-state pressure is maintained self-consistently during the solution of the rate-balance equations (24).

The neutrality cycle can also be deactivated (*e.g.*, by setting to 1 the number of iterations), in which case LoKI-B and LoKI-C run for the values of E/N and n_e specified by the user. Naturally, the charge conservation is not satisfied in this case, but the configuration can be of interest to assess the populations and the creation/destruction mechanisms of heavy species for certain excitation conditions.

4.2 Quasi-stationary simulations

When modeling *active plasmas*, LoKI-GM can also implement *quasi-stationary simulations*, in which LoKI-C runs as a standalone tool. In this framework, the code calculates the time-dependent populations of the various heavy species, while accounting for the prescribed temporal evolution of the electric field (or any related quantity, such as the discharge current or the discharge power-density), specified by the user. The calculations impose the charge-neutrality condition (82a) as each instant in time. The resulting electron density $n_e(t)$ can then be used into (82b) or (82c) to determine the time-dependent reduced electric field $(E/N)(t)$ from the prescribed temporal profiles of the discharge current $I(t)$ or the discharge power density $(dP_{\text{discharge}}/dV)(t)$.

Within *quasi-stationary simulations*, precomputed lookup tables of electron macroscopic parameters, tabulated as functions of E/N and ε, are used in LoKI-GM to interpolate, at selected time instants, the corresponding electron transport parameters, rate coefficients and power transfer required by the chemistry solver. The interpolation procedure can follow one of two approaches:

- it may use directly the values of the reduced electric field $(E/N)(t)$, obtained from the input data, which corresponds to the LFA;
- it may use the electron mean energy $\varepsilon(t)$, computed self-consistently within the chemistry module, which corresponds to the LEA. In the latter case, the set of rate balance equations is solved simultaneously with

the electron mean energy equation

$$\frac{d\varepsilon}{dt} = N\left[\frac{\Theta_E}{N}(t) + \left(\frac{\Theta_{\text{coll}}^{\text{gain}}}{N}\right)(t) + \left(\frac{\Theta_{\text{coll}}^{\text{loss}}}{N}\right)(t)\right] \quad , \tag{83}$$

where Θ_E/N is calculated using (82d), and $\left(\Theta_{\text{coll}}^{\text{gain}}/N\right)(t)$ and $\left(\Theta_{\text{coll}}^{\text{loss}}/N\right)(t)$ are the electron power densities lost/gained in inelastic/superelastic collisions, respectively, per electron at unit gas density. These quantities are obtained from the precomputed lookup tables employed by LoKI-GM.

4.3 Post-discharge simulations

In *post-discharge simulations*, LoKI-GM first computes the corresponding steady-state solution, determining the species densities and the gas temperature. The code then sets the inelastic/superelastic electron rate coefficients to zero (corresponding to a condition with $E/N = 0$), and integrates in time the system of equations (24) and (57), under the following assumptions:

- the steady-state species densities and gas temperature are used as initial conditions;
- the electron transport parameters and the electron power exchanged in collisions are kept constant, fixed at their steady-state values;
- the electron temperature is imposed equal to the gas temperature, consequently, both become time-dependent, if the thermal model is activated;
- quasi-neutrality is imposed through condition (82a).

4.4 Closure conditions

When coupling electron kinetics with chemical kinetics, the closure conditions adopted in LoKI-GM differ somewhat from those commonly employed in other global models [17]. First, LoKI-GM does not explicitly solve the electron particle balance equation. Instead, it imposes charge neutrality, determining the electron density from the corresponding ion balance equations. Second, LoKI-GM does not directly integrate an electron energy balance equation. For consistency, it relies on the solution of the homogeneous EBE to obtain information on the electron power distribution. In general, the electron power balance equation can be expressed as [17]

$$\frac{d(\varepsilon n_e)}{dt} = \Theta_{\text{growth}} n_e + \Theta_E n_e - \Theta_{\text{transp}} n_e + \Theta_{\text{coll}}^{\text{gain}} n_e + \Theta_{\text{coll}}^{\text{loss}} n_e \quad , \tag{84a}$$

where the terms on the right-hand side represent, in order: the variation of power density due to non-conservative processes leading to net electron density growth; the power density gained from the applied electric field; the power density lost through transport; and the power density gained/lost in collisional processes. Equation (84a) may be rewritten as

$$\frac{d\varepsilon}{dt} = \Theta_E - \Theta_{\text{transp}} + \Theta_{\text{coll}}^{\text{gain}} + \Theta_{\text{coll}}^{\text{loss}} \quad , \tag{84b}$$

since $(dn_e/dt)\,\varepsilon \equiv \Theta_{\text{growth}} n_e$. Note that, when transport losses are neglected ($\Theta_{\text{transp}} \simeq 0$), this expression reduces to (83), which in fact can be derived by integrating the two-term homogeneous EBE.

5 Numerical solution

The following sections provide essential information on the numerical solution of LoKI-B and LoKI-C. The sections about LoKI-B cover the discretisation of the EBE, the calculation and discretisation of the electron particle-balance and power-balance equations, and the solution methods employed to solve the EBE. The section about LoKI-C covers its numerical solution.

The material regarding LoKI-B is intended primarily for developers, as much of the content is too technical or detailed for the majority of users.

5.1 Numerical solution of LoKI-B

5.1.1 Discretisation of the EBE

The EBE is numerically solved in an uniform energy grid defined in the energy interval $[0, u_{\max}]$, with *cells*

$$u_n^{\text{cell}} \equiv u_n = (n-1/2)\Delta u, \;\; n = 1, 2, \cdots \mathcal{N} \tag{85}$$

of fixed step-size $\Delta u = u_{\max}/\mathcal{N}$, each cell n being limited by *nodes* $n-1/2$ and $n+1/2$, each node n having energy

$$u_n^{\text{node}} = u_{n-1/2} = (n-1)\Delta u, \;\; n = 1, 2, \cdots \mathcal{N}+1 \tag{86}$$

Figure 5 presents a graphic representation of the energy grid as defined in (85)-(86). The user can define the

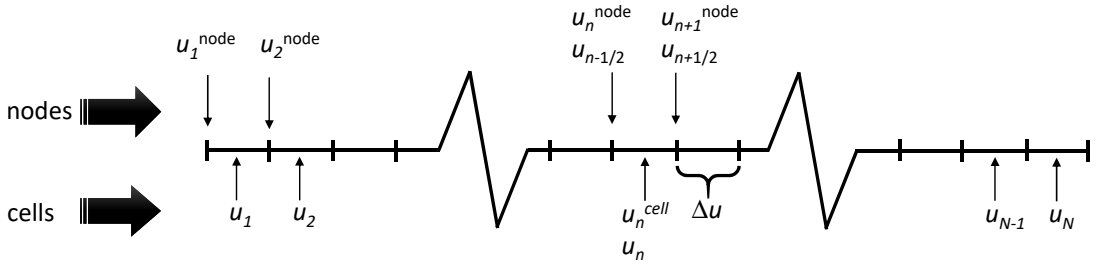


Figure 5: Graphic representation of the energy grid as defined in (85)-(86).

energy grid in two ways: (i) by setting the maximum energy $u_{\max}$ and the number of cells $\mathcal{N}$; (ii) by prescribing a number of decades in the fall of the EEDF, between f_1 and $f_{\mathcal{N}}$, as to ensure an automatic adjustment of the energy grid. Typical values for the decade-fall are within 10 to 25, considering not only the use of double precision, but also the order of magnitude of the various quantities (kinetic energy, cross sections) multiplying the EEDF in the calculations. The user is responsible for adequately defining the energy grid, namely by noticing that small energy steps are required to resolve any relevant structure in the cross sections, such as near-threshold variations and resonances, as well as situations involving the presence of a Ramsauer minimum. A significant increase in $u_{\max}$ without a corresponding increase in the number of grid points can reduce calculation accuracy. Therefore, the user should always verify the grid convergence of their simulation results.

The EBE equations (7a)-(8b) are written for each **grid-cell**, where the isotropic and anisotropic functions f and f^1 are defined, with the corresponding upfluxes G_x ($x = E, \text{el}, \text{CAR}, ee$) and cross sections σ_y ($y = k, c; k_{(i,j)}$) being set on the **grid-nodes**. The cross sections are interpolated linearly from the data provided by the user, setting to zero all the values up to the corresponding energy threshold, *i.e.* $\sigma_{y_n} = 0$ for $u_n^{\text{node}} \leq V_{i,j} \Longrightarrow n \leq \texttt{INT}\,(V_{i,j}/\Delta u) + 1$.

The EBE is discretised adopting the centered finite differences scheme presented in [67]. The result is a system of algebraic equations that can be written under matrix form, with tridiagonal elements coming from the left-hand sides of (7a) and (8a), and some sparse off-tridiagonal elements due to the electron-neutral collisional operator in the right-hand sides.

The emphasis of the discretisation is with the definition of the upflux functions $G(u)$ on the grid-nodes, i.e. on the general boundaries $u_{n\pm1/2}^{\text{cell}}$ of a cell n centered at energy u_n^{cell}. The discretisation method can be

described as a finite-volume-like procedure, by applying Gauss' theorem to an integral over a one-dimensional energy volume around $u_n^{\rm cell}$ as follows

$$-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_x(u)}{du}\right|_{n;\,x=E,\mathrm{el,CAR}} = -\frac{1}{N}\frac{1}{\gamma}\frac{1}{\Delta u}\int_{u_{n-1/2}}^{u_{n+1/2}}\frac{dG_x(u)}{du}du = -\frac{1}{N}\frac{1}{\gamma}\frac{G_{x_{n+1/2}}-G_{x_{n-1/2}}}{\Delta u} \quad , \tag{87a}$$

and therefore (see equations (13a)-(13d))

$$\begin{aligned}
-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_x(u)}{du}\right|_{n;\,x=E,\mathrm{el,CAR}} &= \left\{\frac{d}{du}\left[g_x(u)\left(d_x f(u) + c_x\frac{df(u)}{du}\right)\right]\right\}_n \\
&\simeq \frac{g_{x_{n+1/2}}}{\Delta u}\left(\frac{d_x}{\Delta u}\int_{u_n^{\rm cell}}^{u_{n+1}^{\rm cell}} f(u)du + \frac{c_x}{\Delta u}\int_{u_n^{\rm cell}}^{u_{n+1}^{\rm cell}}\frac{df(u)}{du}du\right) \\
&- \frac{g_{x_{n-1/2}}}{\Delta u}\left(\frac{d_x}{\Delta u}\int_{u_{n-1}^{\rm cell}}^{u_n^{\rm cell}} f(u)du + \frac{c_x}{\Delta u}\int_{u_{n-1}^{\rm cell}}^{u_n^{\rm cell}}\frac{df(u)}{du}du\right) \\
&\simeq \frac{g_{x_{n+1/2}}}{\Delta u}\left(\frac{d_x}{2}(f_n+f_{n+1}) + \frac{c_x}{\Delta u}(f_{n+1}-f_n)\right) \\
&- \frac{g_{x_{n-1/2}}}{\Delta u}\left(\frac{d_x}{2}(f_{n-1}+f_n) + \frac{c_x}{\Delta u}(f_n-f_{n-1})\right) \\
&= \left[\frac{g_{x_{n-1/2}}}{\Delta u}\left(\frac{c_x}{\Delta u}-\frac{d_x}{2}\right)\right] f_{n-1} \\
&- \left[\frac{g_{x_{n+1/2}}}{\Delta u}\left(\frac{c_x}{\Delta u}-\frac{d_x}{2}\right) + \frac{g_{x_{n-1/2}}}{\Delta u}\left(\frac{c_x}{\Delta u}+\frac{d_x}{2}\right)\right] f_n \\
&+ \left[\frac{g_{x_{n+1/2}}}{\Delta u}\left(\frac{c_x}{\Delta u}+\frac{d_x}{2}\right)\right] f_{n+1} \quad ,
\end{aligned} \tag{87b}$$

where c_x and d_x are constants that can be obtained straightforwardly from (12a)-(12c), and

$$\begin{aligned}
f_n &\equiv f(u_n^{\rm cell}) \\
g_{x_{n+1/2}} &\equiv g_x(u_{n+1/2}^{\rm cell}) \equiv g_x(u_{n+1}^{\rm node}) \quad .
\end{aligned}$$

In addition to the terms proportional to $dG_{x=E,\mathrm{el,CAR}}/du$, equations (7a)-(8b) are further composed by terms proportional to f, df/du and dG_{ee}/du whose discretised forms can be generally summarised as follows (the latter receiving a treatment similar to equations (87a)-(87b); see also equations (12d) and (13e))

$$[a(u)f(u)]_n = a_n f_n \tag{88a}$$

$$[a'(u)f(u\pm V_{i,j})]_n = a'_n f_{n\,\pm\,\mathrm{INT}(V_i,j/\Delta u)} \tag{88b}$$

$$\left[b(u)\frac{df(u)}{du}\right]_n = b_n\frac{f_{n+1}-f_{n-1}}{2\Delta u} \tag{88c}$$

$$-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_{ee}(u)}{du}\right|_n = \left\{\frac{d}{du}\left[g_{ee}(u)\left(I(u)f(u)+J(u)\frac{df(u)}{du}\right)\right]\right\}_n \quad , \tag{88d}$$

where $a(u)$, $a'(u)$ and $b(u)$ are generic functions of u; and $I(u)$ and $J(u)$ are the relevant Spitzer integrals [15,24] for the electron-electron collision operator.

The detailed discretized form of each term is given below[10], with the cross sections defined on the grid-nodes as

$$\begin{aligned}
\sigma_{y_n} &\equiv \sigma_y(u_n^{\rm node}) \\
\sigma_y(u_n^{\rm cell}) &= \frac{1}{2}\left(\sigma_{y_n}+\sigma_{y_{n+1}}\right) \quad .
\end{aligned}$$

- **Time-dependent term (only for time-dependent simulations, see section 2.3)**

$$\frac{1}{N}\frac{1}{\gamma}\sqrt{u}\left.\frac{\partial f(u,t)}{\partial t}\right|_n = \frac{1}{N}\frac{1}{\gamma}u_n^{1/2}\frac{\partial f_n(t)}{\partial t} \quad . \tag{89}$$

[10] The discretized expressions aim to replicate as closely as possible those employed in the code's numerical implementation.

- **Temporal growth term**

$$\left.\frac{1}{\gamma\sqrt{u}}\frac{\langle\nu_{\rm eff}\rangle}{N}uf(u)\right|_n = \frac{1}{\gamma}\frac{\langle\nu_{\rm eff}\rangle}{N}u_n^{1/2}f_n \quad . \tag{90}$$

- **Spatial growth term (see equation (13b))** [11].

$$\begin{aligned}\left.\frac{1}{3}\frac{\alpha_{\rm eff}}{N}uf^1(u)\right|_n &\simeq -\frac{\alpha_{\rm eff}}{N}\left\{\frac{\alpha_{\rm eff}}{N}\left(\frac{u}{3\Omega_{\rm SST}(u)}f(u)\right)_n\right.\\ &\left.+\frac{E}{N}\frac{1}{2}\left[\left(\frac{u}{3\Omega_{\rm SST}(u)}\frac{df(u)}{du}\right)_{n+1}^{\rm node}+\left(\frac{u}{3\Omega_{\rm SST}(u)}\frac{df(u)}{du}\right)_n^{\rm node}\right]\right\}\\ &= -\frac{\alpha_{\rm eff}}{N}\left\{\frac{\alpha_{\rm eff}}{N}D_nf_n+\frac{E}{N}\left[D_{n+1}^{\rm node}\frac{f_{n+1}-f_n}{2\Delta u}+D_n^{\rm node}\frac{f_n-f_{n-1}}{2\Delta u}\right]\right\}\\ &= \left[\frac{\alpha_{\rm eff}}{N}\frac{E}{N}\frac{D_{n-1/2}}{2\Delta u}\right]f_{n-1}\\ &-\left[\left(\frac{\alpha_{\rm eff}}{N}\right)^2D_n+\frac{\alpha_{\rm eff}}{N}\frac{E}{N}\left(\frac{D_{n-1/2}}{2\Delta u}-\frac{D_{n+1/2}}{2\Delta u}\right)\right]f_n\\ &-\left[\frac{\alpha_{\rm eff}}{N}\frac{E}{N}\frac{D_{n+1/2}}{2\Delta u}\right]f_{n+1}\end{aligned} \quad , \tag{91}$$

with (see equation (13b))

$$D_n = \frac{u_n}{\frac{3}{2}\left[\Omega_{{\rm SST}_n}+\Omega_{{\rm SST}_{n+1}}\right]} \tag{92a}$$

$$D_n^{\rm node} = D_{n-1/2} = \frac{u_{n-1/2}}{3\Omega_{{\rm SST}_n}} \quad . \tag{92b}$$

- **Electric field operator (see equations (12a) and (13a)-(13b))**

$$g_E^{x={\rm PT,SST}}(u) = \frac{u}{3\Omega_x(u)} \tag{93a}$$

$$c_E = \left(\frac{E}{N}\right)^2 \tag{93b}$$

$$d_E = \begin{cases} 0 \ , \text{ for a PT situation} \\ \dfrac{\alpha_{\rm eff}}{N}\dfrac{E}{N} \ , \text{ for a SST situation} \end{cases} \tag{93c}$$

For a PT situation

$$-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_E^{\rm PT}(u)}{du}\right|_n = \left(\frac{E}{N}\right)^2\left\{\left[\frac{g_{E_{n-1/2}}^{\rm PT}}{(\Delta u)^2}\right]f_{n-1}-\left[\frac{g_{E_{n-1/2}}^{\rm PT}+g_{E_{n+1/2}}^{\rm PT}}{(\Delta u)^2}\right]f_n+\left[\frac{g_{E_{n+1/2}}^{\rm PT}}{(\Delta u)^2}\right]f_{n+1}\right\} \quad . \tag{94a}$$

For an SST situation

$$\begin{aligned}-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_E^{\rm SST}(u)}{du}\right|_n &= \left(\frac{E}{N}\right)^2\left\{\left[\frac{g_{E_{n-1/2}}^{\rm SST}}{(\Delta u)^2}\right]f_{n-1}-\left[\frac{g_{E_{n-1/2}}^{\rm SST}+g_{E_{n+1/2}}^{\rm SST}}{(\Delta u)^2}\right]f_n+\left[\frac{g_{E_{n+1/2}}^{\rm SST}}{(\Delta u)^2}\right]f_{n+1}\right\}\\ &+\frac{\alpha_{\rm eff}}{N}\frac{E}{N}\left\{-\left[\frac{g_{E_{n-1/2}}^{\rm SST}}{2\Delta u}\right]f_{n-1}-\left[\frac{g_{E_{n-1/2}}^{\rm SST}-g_{E_{n+1/2}}^{\rm SST}}{2\Delta u}\right]f_n+\left[\frac{g_{E_{n+1/2}}^{\rm SST}}{2\Delta u}\right]f_{n+1}\right\}\end{aligned} \quad . \tag{94b}$$

- **Elastic operator**

$$g_{\rm el}(u) = 2u^2\sum_k\frac{m_e}{M_k}\chi_k\sigma_{k,c}^{\rm el}(u) \tag{95a}$$

$$c_{\rm el} = \frac{k_BT_g}{e} \tag{95b}$$

$$d_{\rm el} = 1 \tag{95c}$$

[11] The spatial growth term is discretized as follows: (i) directly in the cells, for the component proportional to $f(u)$; (ii) in the nodes, for the component proportional to $df(u)/du$. This discretisation allows satisfying the particle and the energy conservation equations - see sections 5.1.2 and 5.1.3

$$-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_{\rm el}(u)}{du}\right|_n = \left[\frac{g_{{\rm el}_{n-1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}-\frac{1}{2}\right)\right]f_{n-1} - \left[\frac{g_{{\rm el}_{n-1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}+\frac{1}{2}\right)+\frac{g_{{\rm el}_{n+1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}-\frac{1}{2}\right)\right]f_n + \left[\frac{g_{{\rm el}_{n+1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}+\frac{1}{2}\right)\right]f_{n+1} \quad . \tag{96}$$

- **CAR operator**

$$g_{\rm CAR}(u) = 4u\sum_k B_k\chi_k\sigma_{0,k}(u) \tag{97a}$$

$$c_{\rm CAR} = \frac{k_BT_g}{e} \tag{97b}$$

$$d_{\rm CAR} = 1 \tag{97c}$$

$$-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_{\rm CAR}(u)}{du}\right|_n = \left[\frac{g_{{\rm CAR}_{n-1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}-\frac{1}{2}\right)\right]f_{n-1} - \left[\frac{g_{{\rm CAR}_{n-1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}+\frac{1}{2}\right)+\frac{g_{{\rm CAR}_{n+1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}-\frac{1}{2}\right)\right]f_n + \left[\frac{g_{{\rm CAR}_{n+1/2}}}{\Delta u}\left(\frac{k_BT_g}{e\Delta u}+\frac{1}{2}\right)\right]f_{n+1} \quad . \tag{98}$$

- **Electron-electron collisions operator**

[In the case of time-dependent simulations, see section 2.3, the electron density $n_e(t)$ and the EEDF $f(u,t)$ introduce a time-dependence into $g_{ee}(u,t)$, $I(u,t)$ and $J(u,t)$, which are written for a given t.]

$$g_{ee}(u) = \frac{2}{\gamma}\frac{\nu_{ee}(u)}{N}u^{3/2} \tag{99a}$$

$$I(u) = \int_0^u f(u')u'^{1/2}du' \tag{99b}$$

$$J(u) = \frac{2}{3}\left[\int_0^u f(u')u'^{3/2}du' + u^{3/2}\int_u^\infty f(u')du'\right] \tag{99c}$$

$$-\frac{1}{N}\frac{1}{\gamma}\left.\frac{dG_{ee}(u)}{du}\right|_n = \left\{\frac{d}{du}\left[g_{ee}(u)\left(I(u)f(u)+J(u)\frac{df(u)}{du}\right)\right]\right\}_n \quad . \tag{100}$$
$$= A_{n-1}f_{n-1} - (A_n+B_n)f_n + B_{n+1}f_{n+1}$$

$$A_n \equiv A_{I_n}+A_{J_n} = -\frac{2}{\gamma}\frac{\nu_{ee_{n+1/2}}}{N}u_{n+1/2}^{3/2}\left[\frac{I_{n+1/2}}{2\Delta u}-\frac{J_{n+1/2}}{(\Delta u)^2}\right] \equiv \sum_{m=1}^{\mathcal{N}} a_{n,m}f_m \tag{101a}$$

$$B_n \equiv B_{I_n}+B_{J_n} = \frac{2}{\gamma}\frac{\nu_{ee_{n-1/2}}}{N}u_{n-1/2}^{3/2}\left[\frac{I_{n-1/2}}{2\Delta u}+\frac{J_{n-1/2}}{(\Delta u)^2}\right] \equiv \sum_{m=1}^{\mathcal{N}} b_{n,m}f_m \tag{101b}$$

$$A_{I_n} = -\frac{2}{\gamma}\frac{\nu_{ee_{n+1/2}}}{N}u_{n+1/2}^{3/2}\frac{I_{n+1/2}}{2\Delta u} \quad , \quad \text{etc} \tag{101c}$$

$$I_{n+1/2} = \sum_{m=1}^{n} f_m u_m^{1/2}\Delta u \tag{101d}$$

$$J_{n+1/2} = \frac{2}{3}\left[\sum_{m=1}^{n} f_m u_m^{3/2}\Delta u + u_{n+1/2}^{3/2}\sum_{m=n+1}^{\mathcal{N}} f_m\Delta u\right] \tag{101e}$$

$$a_{n,m} = -\frac{2}{\gamma}\frac{\nu_{ee_{n+1/2}}}{N}u_{n+1/2}^{3/2}\times\begin{cases}\frac{1}{2}u_m^{1/2}-\frac{2}{3}\frac{u_m^{3/2}}{\Delta u} & ,\ m\le n\\ -\frac{2}{3}\frac{u_{n+1/2}^{3/2}}{\Delta u} & ,\ m>n\end{cases} \tag{101f}$$

$$b_{n,m} = \frac{2}{\gamma}\frac{\nu_{ee_{n-1/2}}}{N}u_{n-1/2}^{3/2} \times \begin{cases} \frac{1}{2}u_m^{1/2} + \frac{2}{3}\frac{u_m^{3/2}}{\Delta u} & , \; m \le n-1 \\ \frac{2}{3}\frac{u_{n-1/2}^{3/2}}{\Delta u} & , \; m > n-1 \end{cases} . \tag{101g}$$

- **Discrete collision operator**

$$S_n = \sum_{i,j>i} S_{(i,j)_n} + \sum_i S_{(i,\mathrm{ion})_n} + \sum_i S_{(i,\mathrm{att})_n} \tag{102a}$$

$$l_{i,j} \equiv \mathtt{INT}\left(V_{i,j}/\Delta u\right) \tag{102b}$$

$$\sigma_{i,\mathrm{ion}}(u_n^{\mathrm{cell}}) = \sum_{m=1}^{(n-l_{i,\mathrm{ion}})/2} \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(n,m)}} \tag{102c}$$

$$\begin{aligned} S_{(i,j)_n} &= \delta_i \left[u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_{n+l_{i,j}} - u_n \frac{\sigma_{(i,j)_n} + \sigma_{(i,j)_{n+1}}}{2} f_n \right] \\ &+ \delta_j \frac{g_i}{g_j} \left[u_n \frac{\sigma_{(i,j)_n} + \sigma_{(i,j)_{n+1}}}{2} f_{n-l_{i,j}} - u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_n \right] \end{aligned} \tag{103a}$$

$$\begin{aligned} S_{(i,\mathrm{ion})_n} &= \delta_i \left\{ \sum_{m=n+l_{i,\mathrm{ion}}+1}^{2n+l_{i,\mathrm{ion}}} u_m \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,m-n-l_{i,\mathrm{ion}}+1/2)}} f_m \Delta u \right. \\ &+ \left. \sum_{m=2n+l_{i,\mathrm{ion}}}^{\mathcal{N}} u_m \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,n+1/2)}} f_m \Delta u - u_n \sigma_{(i,\mathrm{ion})_{n+1/2}} f_n \right\} \end{aligned} \tag{103b}$$

$$S_{(i,\mathrm{att})_n} = -\delta_i u_n \frac{\sigma_{(i,\mathrm{att})_n} + \sigma_{(i,\mathrm{att})_{n+1}}}{2} f_n \quad , \tag{103c}$$

where

$$\sigma_{i,\mathrm{ion}}(u) = \int_0^{(u-V_{i,\mathrm{ion}})/2} \sigma^{\mathrm{sec}}_{i,\mathrm{ion}}(u,u')du' \longrightarrow \sigma_{i,\mathrm{ion}_{m+1}} = \sum_{n=1}^{(m-l_{i,\mathrm{ion}})/2} \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1,n+1/2)}} \Delta u \tag{104a}$$

$$\sigma_{i,\mathrm{ion}_{m+1/2}} = \frac{\sigma_{i,\mathrm{ion}_m} + \sigma_{i,\mathrm{ion}_{m+1}}}{2} \tag{104b}$$

The integrals in the ionisation operator are discretized using the mid-point quadrature rule, for example

$$\begin{aligned} \int_{a(u)}^{b(u)} u'\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})}(u',u)f(u')du' &\longrightarrow \int_{a(u_{k+1}^{\mathrm{node}})}^{b(u_{n+1}^{\mathrm{node}})\Delta u} u'\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})}(u',u_{n+1/2}^{\mathrm{node}})f(u')du' \\ &= \int_{k\Delta u}^{n\Delta u} u'\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})}(u',u_n^{\mathrm{cell}})f(u')du' \\ &= \sum_{m=k}^{n} u_m \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,n+1/2)}} f_m \Delta u \quad . \end{aligned} \tag{105}$$

such that the cross section values, usually defined at the grid-nodes, are **now taken at the grid-cells** (with the indexes m and n in $\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})}$ referring to node numbers).

In particular,

$$u + V_{i,\mathrm{ion}} \longrightarrow u_{n+1}^{\mathrm{node}} + l_{i,\mathrm{ion}}\Delta u = (n + l_{i,\mathrm{ion}})\,\Delta u \tag{106a}$$

$$2u + V_{i,\mathrm{ion}} \longrightarrow 2u_{n+1}^{\mathrm{node}} + l_{i,\mathrm{ion}}\Delta u = (2n + l_{i,\mathrm{ion}})\,\Delta u \tag{106b}$$

$$\frac{u - V_{i,\mathrm{ion}}}{2} \longrightarrow \frac{u_{n+1}^{\mathrm{node}} - l_{i,\mathrm{ion}}\Delta u}{2} = \frac{(n - l_{i,\mathrm{ion}})\,\Delta u}{2} \quad . \tag{106c}$$

Note that, in the first term of equation (103b), the lower index in the summation becomes $n + l_{i,\mathrm{ion}} + 1$, because for $m = n + l_{i,\mathrm{ion}}$

$$\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,1/2)}} = 0$$

since it would involve an undefined 0-cell.

The previous equations are subjected to the following conditions [67]:

- the flux boundary conditions $G(0) = G(u_{\max}) = 0$, which are implemented here for *each* operator as $g_{x_{1/2}} = g_{x_{\mathcal{N}+1/2}} = 0$ and $A_0 = A_{\mathcal{N}} = B_1 = B_{\mathcal{N}+1} = 0$. [12]
  Note that the boundary conditions $g^{\text{SST}}_{E_{1/2}} = g^{\text{SST}}_{E_{\mathcal{N}+1/2}} = 0$ result in $D_{1/2} = D_{\mathcal{N}+1/2} = 0$ (see equation (13b));
- the imposition of a Maxwellian EEDF for $dG_{ee}/du = 0$, which is ensured by setting $a_{n,m} = \sqrt{a_{n,m} \times a_{m-1,n+1}}$ (see section 5.1.4).
- the energy conservation condition for dG_{ee}/du, which yields $b_{m,n} = a_{n,m}$ (see section 5.1.4).

5.1.2 The particle balance equation

The EBE (7a)-(8b) and (19a)-(19b) can be integrated in energy space to yield the particle balance equation, *i.e.*

- For a PT situation

$$\frac{1}{\gamma}\frac{\langle\nu_{\text{eff}}\rangle}{N}\int_0^\infty \sqrt{u}f(u)du + \frac{1}{N}\frac{1}{\gamma}\int_0^\infty \frac{dG(u)}{du}du = \int_0^\infty S(u)du$$

  and therefore

$$\frac{\langle\nu_{\text{eff}}\rangle}{N} = \gamma\int_0^\infty S(u)du \quad , \tag{107}$$

  where we have used the normalisation condition and the boundary conditions for the upflux $G(u)$.

- For a SST situation

$$-\frac{1}{3}\left(\frac{\alpha_{\text{eff}}}{N}\right)^2\int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)}f(u)du - \frac{1}{3}\frac{\alpha_{\text{eff}}}{N}\frac{E}{N}\int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)}\frac{df(u)}{du}du + \frac{1}{N}\frac{1}{\gamma}\int_0^\infty \frac{dG(u)}{du}du = \int_0^\infty S(u)du$$

  and therefore (cf. equation (75))

$$-(D_eN)\left(\frac{\alpha_{\text{eff}}}{N}\right)^2 + (\mu_eN)\frac{E}{N}\frac{\alpha_{\text{eff}}}{N} = \frac{\langle\nu_{\text{eff}}\rangle}{N} \quad , \tag{108}$$

  where we have used equation (107), the boundary conditions for the upflux $G(u)$, and the expressions (64a) and (64b) of the DC reduced transverse diffusion coefficient and the DC reduced mobility, respectively.

- For a time-dependent situation

$$\frac{1}{N}\frac{1}{\gamma}\frac{d}{dt}\int_0^\infty \sqrt{u}f(u,t)du + \frac{1}{\gamma}\frac{\langle\nu_{\text{eff}}\rangle(t)}{N}\int_0^\infty \sqrt{u}f(u,t)du + \frac{1}{N}\frac{1}{\gamma}\int_0^\infty \frac{\partial G(u,t)}{\partial u}du = \int_0^\infty S(u,t)du$$

  and therefore

$$\frac{\langle\nu_{\text{eff}}\rangle(t)}{N} = \gamma\int_0^\infty S(u,t)du \quad , \tag{109}$$

  where, similarly to the PT situation, we have used the normalisation condition and the boundary conditions for the upflux $G(u)$.

[12] Note that the *strict* boundary conditions should be $G(0) = G(u_{\max}) = \sum_x G_x(0) = \sum_x G_x(u_{\max}) = 0$. Work on their numerical implementation is underway.

Note that the integration of the right-hand sides of equations (7a), (8a) and (19a) yields

$$\begin{aligned}
\int_0^\infty S_{\text{inel}}(u)du &= \sum_{i,j>i}\int_0^\infty S_{(i,j),\text{inel}}(u)du = \\
&= \sum_{i,j>i}\int_0^\infty \delta_i\left[\sigma_{i,j}(u+V_{i,j})f(u+V_{i,j})(u+V_{i,j}) - \sigma_{i,j}(u)f(u)u\right]du \\
&= \sum_{i,j>i}\delta_i\left[\int_{V_{i,j}}^\infty \sigma_{i,j}(u')f(u')u'du' - \int_0^\infty \sigma_{i,j}(u)f(u)udu\right] \\
&= \sum_{i,j>i}\delta_i\left[\int_0^\infty \sigma_{i,j}(u)f(u)udu - \int_0^\infty \sigma_{i,j}(u)f(u)udu\right] \\
&= 0 \quad , \qquad (110)
\end{aligned}$$

where we have considered that $\sigma_{i,j}(u) = 0$ for $u \leq V_{i,j}$.

Discretisation of the particle balance equation

The numerical discretisation of the EBE must ensure the electron-particle conservation, *i.e.* it must allow retrieving the numerical version of equations (107)-(108). In particular:

- normalisation condition
  The EEDF is calculated as to satisfy the normalisation condition

$$\int_0^\infty f(u)\sqrt{u}du = 1 \longrightarrow \sum_{n=1}^{\mathcal{N}} f_n\sqrt{u_n}\Delta u_n = 1 \quad ; \qquad (111)$$

- DC free-diffusion coefficient
  The DC reduced transverse free-diffusion coefficient is calculated using (see equations (64a), (91) and (92a))

$$\begin{aligned}
D_eN &= \frac{\gamma}{3}\int_0^\infty \frac{u}{\Omega_{\text{SST}}(u)}f(u)du \\
&\longrightarrow \frac{\gamma}{3}\sum_{n=1}^{\mathcal{N}}\int_{u_{n-1/2}}^{u_{n+1/2}} \frac{u}{\Omega_{\text{SST}}(u)}f(u)du \\
&\longrightarrow \gamma\Delta u\sum_{n=1}^{\mathcal{N}}\frac{u_n}{3\Omega_{\text{SST}_n}}f_n \\
&\longrightarrow \gamma\Delta u\sum_{n=1}^{\mathcal{N}} D_nf_n \quad ; \qquad (112)
\end{aligned}$$

- DC reduced mobility
  The DC reduced mobility is calculated using (see equations (64b), (91) and (92b); recall that $D_{1/2} =$

$D_{\mathcal{N}+1/2} = 0$)

$$
\begin{aligned}
\mu_e N &= -\frac{\gamma}{3}\int_0^\infty \frac{u}{\Omega_{\mathrm{SST}}(u)}\frac{df(u)}{du}du \\
&\longrightarrow -\frac{\gamma}{3}\sum_{n=1}^{\mathcal{N}}\int_{u_{n-1/2}}^{u_{n+1/2}} \frac{u}{\Omega_{\mathrm{SST}}(u)}\frac{df(u)}{du}du \\
&\longrightarrow -\gamma\sum_{n=1}^{\mathcal{N}}\frac{1}{2}\left[\left(\frac{u}{3\Omega_{\mathrm{SST}}(u)}\frac{df(u)}{du}\right)_{n+1}^{\mathrm{node}} + \left(\frac{u}{3\Omega_{\mathrm{SST}}(u)}\frac{df(u)}{du}\right)_{n}^{\mathrm{node}}\right]\Delta u \\
&\longrightarrow -\gamma\sum_{n=1}^{\mathcal{N}}\frac{1}{2}\left[D_{n+1/2}\left(\frac{f_{n+1}-f_n}{\Delta u}\right) + D_{n-1/2}\left(\frac{f_n-f_{n-1}}{\Delta u}\right)\right]\Delta u \\
&\longrightarrow -\gamma\frac{1}{2}\left[\sum_{n=1}^{\mathcal{N}-1} D_{n+1/2}\frac{f_{n+1}-f_n}{\Delta u} + \sum_{n=2}^{\mathcal{N}} D_{n-1/2}\frac{f_n-f_{n-1}}{\Delta u}\right]\Delta u \\
&\longrightarrow -\gamma\sum_{n=2}^{\mathcal{N}} D_{n-1/2}\frac{f_n-f_{n-1}}{\Delta u}\Delta u = -\gamma\sum_{n=1}^{\mathcal{N}} D_{n-1/2}\frac{f_n-f_{n-1}}{\Delta u}\Delta u \quad ;
\end{aligned}
\tag{113}
$$

- flux conservation
  The integration of each upflux term ($x = E, \mathrm{el}, \mathrm{CAR}$) yields a zero-identity (see equation (87b); recall that $f_0 = f_{\mathcal{N}+1} = 0$ and $g_{x_{1/2}} = g_{x_{\mathcal{N}+1/2}} = 0$)

$$
\begin{aligned}
-\frac{1}{N}\frac{1}{\gamma}\int_0^\infty \frac{dG_x(u)}{du}du = 0 \longrightarrow -\frac{1}{N}\frac{1}{\gamma}\sum_{n=1}^{\mathcal{N}}\left(\frac{dG_x(u)}{du}\right)_n \Delta u \\
&= \sum_{n=1}^{\mathcal{N}}\left\{\frac{g_{x_{n-1/2}}}{\Delta u}\left[\frac{c_x}{\Delta u}-\frac{d_x}{2}\right]\right\} f_{n-1}\Delta u \\
&- \sum_{n=1}^{\mathcal{N}}\left\{\frac{g_{x_{n+1/2}}}{\Delta u}\left[\frac{c_x}{\Delta u}-\frac{d_x}{2}\right] + \frac{g_{x_{n-1/2}}}{\Delta u}\left[\frac{c_x}{\Delta u}+\frac{d_x}{2}\right]\right\} f_n\Delta u \\
&+ \sum_{n=1}^{\mathcal{N}}\left\{\frac{g_{x_{n+1/2}}}{\Delta u}\left[\frac{c_x}{\Delta u}+\frac{d_x}{2}\right]\right\} f_{n+1}\Delta u \\
&= \sum_{n=1}^{\mathcal{N}-1}\left\{g_{x_{n+1/2}}\left[\frac{c_x}{\Delta u}-\frac{d_x}{2}\right]\right\} f_n \\
&- \sum_{n=1}^{\mathcal{N}-1}\left\{g_{x_{n+1/2}}\left[\frac{c_x}{\Delta u}-\frac{d_x}{2}\right]\right\} f_n - \sum_{n=2}^{\mathcal{N}}\left\{g_{x_{n-1/2}}\left[\frac{c_x}{\Delta u}+\frac{d_x}{2}\right]\right\} f_n \\
&+ \sum_{n=2}^{\mathcal{N}}\left\{g_{x_{n-1/2}}\left[\frac{c_x}{\Delta u}+\frac{d_x}{2}\right]\right\} f_n \\
&= 0 \quad ;
\end{aligned}
\tag{114}
$$

- flux conservation for electron-electron collisions
  The integration of the upflux term for electron-electron collisions yields also a zero-identity (see equa-

tion (100); recall that $f_0 = f_{\mathcal{N}+1} = 0$ and $A_0 = A_{\mathcal{N}} = B_1 = B_{\mathcal{N}+1} = 0$)

$$
\begin{aligned}
-\frac{1}{N}\frac{1}{\gamma}\int_0^\infty \frac{dG_{ee}(u)}{du}du = 0 \longrightarrow -\frac{1}{N}\frac{1}{\gamma}\sum_{n=1}^{\mathcal{N}}\left(\frac{dG_{ee}(u)}{du}\right)_n \Delta u \\
&= \sum_{n=1}^{\mathcal{N}} \left[A_{n-1}f_{n-1} - (A_n + B_n)\, f_n + B_{n+1}f_{n+1}\right] \\
&= \sum_{n=2}^{\mathcal{N}} A_{n-1}f_{n-1} - \sum_{n=1}^{\mathcal{N}-1} A_n f_n - \sum_{n=2}^{\mathcal{N}} B_n f_n + \sum_{n=1}^{\mathcal{N}-1} B_{n+1}f_{n+1} \\
&= \sum_{n=1}^{\mathcal{N}-1} A_n f_n - \sum_{n=1}^{\mathcal{N}-1} A_n f_n - \sum_{n=1}^{\mathcal{N}-1} B_{n+1}f_{n+1} + \sum_{n=1}^{\mathcal{N}-1} B_{n+1}f_{n+1} \\
&= 0 \quad .
\end{aligned}
\tag{115}
$$

- inelastic/superelastic collisions
  The integration of the inelastic and superelastic collision terms yields a zero identify (see equations (103a))

$$
\begin{aligned}
\int_0^\infty S_{\text{inel}}(u)du = 0 \longrightarrow \sum_{n=1}^{\mathcal{N}} S_n^{\text{inel}}\Delta u = \sum_{i,j>i}\sum_{n=1}^{\mathcal{N}} S_{(i,j)_n}^{\text{inel}}\Delta u \\
&= \sum_{i,j>i}\sum_{n=1}^{\mathcal{N}} \delta_i \left[u_{n+l_{i,j}}\sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_{n+l_{i,j}} - u_n \sigma_{(i,j)_{n+1/2}}\, f_n\right]\Delta u \\
&= \sum_{i,j>i} \delta_i \left[\sum_{n=1}^{\mathcal{N}-l_{i,j}} u_{n+l_{i,j}}\sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_{n+l_{i,j}} - \sum_{n=1+l_{i,j}}^{\mathcal{N}} u_n \sigma_{(i,j)_{n+1/2}}\, f_n\right]\Delta u \\
&= \sum_{i,j>i} \delta_i \left[\sum_{n=1}^{\mathcal{N}-l_{i,j}} u_{n+l_{i,j}}\sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_{n+l_{i,j}} - \sum_{n=1}^{\mathcal{N}-l_{i,j}} u_{n+l_{i,j}} \sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_{n+l_{i,j}}\right]\Delta u \\
&= 0 \quad \text{for } \sigma_{(i,j)_{n+1/2}} = 0, n \le l_{i,j} \quad ;
\end{aligned}
\tag{116a}
$$

$$
\begin{aligned}
\int_0^\infty S_{\text{sup}}(u)du = 0 \longrightarrow \sum_{n=1}^{\mathcal{N}} S_n^{\text{sup}}\Delta u = \sum_{i,j>i}\sum_{n=1}^{\mathcal{N}} S_{(i,j)_n}^{\text{sup}}\Delta u \\
&= \sum_{i,j>i}\sum_{n=1}^{\mathcal{N}} \delta_j \frac{g_i}{g_j} \left[u_n\sigma_{(i,j)_{n+1/2}}\, f_{n-l_{i,j}} - u_{n+l_{i,j}} \sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_n\right]\Delta u \\
&= \sum_{i,j>i} \delta_j \frac{g_i}{g_j} \left[\sum_{n=1+l_{i,j}}^{\mathcal{N}} u_n\sigma_{(i,j)_{n+1/2}}\, f_{n-l_{i,j}} - \sum_{n=1}^{\mathcal{N}-l_{i,j}} u_{n+l_{i,j}} \sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_n\right]\Delta u \\
&= \sum_{i,j>i} \delta_j \frac{g_i}{g_j} \left[\sum_{n=1}^{\mathcal{N}-l_{i,j}} u_{n+l_{i,j}}\sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_n - \sum_{n=1}^{\mathcal{N}-l_{i,j}} u_{n+l_{i,j}} \sigma_{(i,j)_{n+l_{i,j}+1/2}}\, f_n\right]\Delta u \\
&= 0 \quad \text{for } \sigma_{(i,j)_{n+1/2}} = 0, n \le l_{i,j} \quad .
\end{aligned}
\tag{116b}
$$

- ionisation

The integration of the ionisation collision terms yields the reduced-ionisation frequency (see equation (103b))

$$\begin{aligned}
\int_0^\infty S_{\rm ion}(u)du = 0 \longrightarrow \sum_{n=1}^{\mathcal{N}} S_n^{\rm ion}\Delta u &= \sum_i \sum_{n=1}^{\mathcal{N}} S_{(i,{\rm ion})_n}\Delta u \\
&= \sum_i \sum_{n=1}^{\mathcal{N}} \delta_i \left\{ \sum_{m=n+l_{i,\rm ion}+1}^{2n+l_{i,\rm ion}} u_m \sigma^{\rm sec}_{(i,{\rm ion})_{(m+1/2,m-n-l_{i,\rm ion}+1/2)}} f_m \Delta u \right. \\
&\left. + \sum_{m=2n+l_{i,\rm ion}}^{\mathcal{N}} u_m \sigma^{\rm sec}_{(i,{\rm ion})_{(m+1/2,n+1/2)}} f_m \Delta u - u_n \sigma_{(i,{\rm ion})_{n+1/2}} f_n \right\} \Delta u \\
&= \sum_i \delta_i \left\{ \sum_{m=1}^{\mathcal{N}} u_m f_m \left[\sum_{k=1}^{(m-l_{i,\rm ion})/2} \sigma^{\rm sec}_{(i,{\rm ion})_{(m+1/2,k+1/2)}} \Delta u \right] \right. \\
&\left. + \sum_{m=1}^{\mathcal{N}} u_m f_m \left[\sum_{n=1}^{(m-l_{i,\rm ion})/2} \sigma^{\rm sec}_{(i,{\rm ion})_{(m+1/2,n+1/2)}} \Delta u \right] - \sum_{n=1}^{\mathcal{N}} u_n f_n \sigma_{(i,{\rm ion})_{n+1/2}} \right\} \Delta u \\
&= \sum_i \delta_i \left\{ \sum_{m=1}^{\mathcal{N}} u_m f_m \sigma_{(i,{\rm ion})_{m+1/2}} + \sum_{m=1}^{\mathcal{N}} u_m f_m \sigma_{(i,{\rm ion})_{m+1/2}} - \sum_{n=1}^{\mathcal{N}} u_n f_n \sigma_{(i,{\rm ion})_{n+1/2}} \right\} \Delta u \\
&= \sum_i \delta_i \sum_{n=1}^{\mathcal{N}} u_n f_n \sigma_{(i,{\rm ion})_{n+1/2}} = \frac{1}{\gamma} \sum_i \delta_i \frac{\langle \nu_{i,\rm ion} \rangle}{N} = \frac{1}{\gamma} \frac{\langle \nu_{\rm ion} \rangle}{N} \quad ,
\end{aligned} \tag{117}$$

where we have used equations (69a) and (104a), and considered

$$\begin{aligned}
& n + l_{i,\rm ion} + 1 \le m \le 2n + l_{i,\rm ion} \\
& \quad \Longrightarrow 1 \le m - n - l_{i,\rm ion} \equiv k \le n \le \frac{m - l_{i,\rm ion}}{2} \quad , \quad (\text{for } m = 2n + l_{i,\rm ion}) \\
& 2n + l_{i,\rm ion} \le m \le \mathcal{N} \Longrightarrow 1 \le n \le \frac{m - l_{i,\rm ion}}{2} \le \mathcal{N} \quad .
\end{aligned}$$

- attachment
  The integration of the attachment collision term yields the reduced-attachment frequency (see equation (103c))

$$\begin{aligned}
\int_0^\infty S_{\rm att}(u)du = 0 \longrightarrow \sum_{n=1}^{\mathcal{N}} S_n^{\rm att}\Delta u &= \sum_i \sum_{n=1}^{\mathcal{N}} S_{(i,{\rm att})_n}\Delta u \\
&= -\sum_i \delta_i \sum_{n=1}^{\mathcal{N}} u_n \frac{\sigma_{(i,{\rm att})_n} + \sigma_{(i,{\rm att})_{n+1}}}{2} f_n \Delta u \\
&= \frac{1}{\gamma} \sum_i \delta_i \frac{\langle \nu_{i,\rm att} \rangle}{N} = \frac{1}{\gamma} \frac{\langle \nu_{\rm att} \rangle}{N} \quad .
\end{aligned} \tag{118}$$

The discretized forms of the particle balance equations are verified by using equations (111)-(118) in the expressions (107), (108) and (109).

5.1.3 The power balance equation

The balance equation for the electron power-density, per electron at unit gas density, is obtained (depending on the particular PT/SST situation considered) by multiplying either (76a) or (76b) by γu and integrating it over all energies; or by multiplying either (76c) or (76d) by $u^{3/2}/N$ and integrating it over all energies (see section 3.1.2). The general result writes (in eV s^{-1} m^3)

$$\frac{\Theta(t)}{N} = \frac{\Theta_{\rm growth}}{N} + \frac{\Theta_E}{N} + \frac{\Theta_{\rm coll}}{N} \quad , \tag{119a}$$

where the term in the left-hand side represents the intrinsic time-variation of the power density, and the different terms on the right-hand side represent, in order, the variation of the power density due to non-conservative

mechanisms inducing a time/space net growth of the electron density, the power density gained from the applied electric field, and the power density gained/lost in collisional ("coll") events, which can be separately written as (note that all "gain" terms are positively defined and all "loss" terms are negatively defined).

$$\frac{\Theta_{\rm coll}}{N} \equiv \frac{\Theta_{\rm coll}^{\rm gain}}{N} + \frac{\Theta_{\rm coll}^{\rm loss}}{N} \quad . \tag{119b}$$

Discretisation of the power balance equation
The discretized form of (119a) can be obtained using the same finite-volume-like procedure as in section 5.1.1, applying Gauss' theorem to integrals over one-dimensional energy volumes around each n-cell centered at energy u_n (see (87a)-(87b)).

The different terms in (119a) are written as follows.

- intrinsic time-variation term
  The left-hand side of (77a) is

$$\frac{\Theta(t)}{N} = \begin{cases} 0 \ , \text{ for stationary and steady-state simulations} \\ \dfrac{1}{N}\displaystyle\int_0^\infty u^{3/2}\frac{\partial f(u,t)}{\partial t}du = \frac{1}{N}\frac{d\varepsilon(t)}{dt} \ , \text{ for time-dependent simulations} \end{cases} . \tag{120}$$

  The quantity $\Theta(t)/N$ is numerically calculated by summing all the power terms on the right-hand side of (119a), and on output the value of this quantity is tagged as *Power balance* (see User Guide [12]).

- time/space net growth term

$$\frac{\Theta_{\rm growth}}{N} = \begin{cases} -\dfrac{\langle\nu_{\rm eff}\rangle}{N}\displaystyle\int_0^\infty \frac{1}{\sqrt{u}}u^2 f(u)du = -\frac{\langle\nu_{\rm eff}\rangle}{N}\varepsilon \\ \quad \to -\dfrac{\langle\nu_{\rm eff}\rangle}{N}\displaystyle\sum_{n=1}^{\mathcal{N}} u_n^{3/2} f_n \Delta u \ , \\ \quad \text{for a PT situation} \\ -\dfrac{\alpha_{\rm eff}}{N}\dfrac{\gamma}{3}\displaystyle\int_0^\infty u^2 f^1(u)du \\ \quad = -\dfrac{\alpha_{\rm eff}}{N}\dfrac{\gamma}{3}\displaystyle\int_0^\infty u^2\left[-\frac{E/N}{\Omega_{\rm SST}(u)}\frac{df(u)}{du} - \frac{\alpha_{\rm eff}/N}{\Omega_{\rm SST}(u)}f(u)\right]du = -\dfrac{\alpha_{\rm eff}}{N}\left(\mu_\varepsilon E - D_\varepsilon\alpha_{\rm eff}\right) \\ \quad \to -\dfrac{\alpha_{\rm eff}}{N}\gamma\displaystyle\sum_{n=1}^{\mathcal{N}}\left[-\frac{E}{N}u_{n-1/2}D_{n-1/2}\frac{f_n - f_{n-1}}{\Delta u} - \frac{\alpha_{\rm eff}}{N}u_n D_n f_n\right]\Delta u, \\ \quad \text{for a SST situation (see equations (66a)-(66b) and (112)-(113))} \end{cases} \tag{121}$$

- terms involving upflux functions $G_{x=E,\rm el,CAR}(u)$
  The $\Theta_{x=E,\rm el,CAR}/N$ terms involving upflux functions $G_x(u)$ are handled as follows (see equations (13a)-(13b) and (93b)-(93c); (13c) and (95b)-(95c); (13d) and (97b)-(97c))

$$\begin{aligned} \frac{\Theta_{x=E,\rm el,CAR}}{N} &= -\frac{1}{N}\sum_n\int_{u_{n-1/2}}^{u_{n+1/2}} u\frac{dG_x(u)}{du}du = -\frac{1}{N}\sum_n u_n\left[G_{x_{n+1/2}} - G_{x_{n-1/2}}\right] \\ &= \gamma\sum_n\left\{u_n\left[g_{x_{n+1/2}}\left(\frac{d_x}{2}(f_n + f_{n+1}) + \frac{c_x}{\Delta u}(f_{n+1} - f_n)\right)\right.\right. \\ &- \left.\left. g_{x_{n-1/2}}\left(\frac{d_x}{2}(f_{n-1} + f_n) + \frac{c_x}{\Delta u}(f_n - f_{n-1})\right)\right]\right\} \\ &= \gamma\left\{\sum_{n=1}^{\mathcal{N}} u_n\left[g_{x_{n+1/2}}\left(\frac{d_x}{2} - \frac{c_x}{\Delta u}\right) - g_{x_{n-1/2}}\left(\frac{d_x}{2} + \frac{c_x}{\Delta u}\right)\right]f_n\right. \end{aligned}$$

$$
\begin{aligned}
&+ \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_{n+1/2}} \left(\frac{d_x}{2} + \frac{c_x}{\Delta u} \right) \right] f_{n+1} - \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_{n-1/2}} \left(\frac{d_x}{2} - \frac{c_x}{\Delta u} \right) \right] f_{n-1} \Bigg\} \\
&= \gamma \Bigg\{ \sum_{n=1}^{\mathcal{N}} u_n \left[g_{x_{n+1/2}} \left(\frac{d_x}{2} - \frac{c_x}{\Delta u} \right) - g_{x_{n-1/2}} \left(\frac{d_x}{2} + \frac{c_x}{\Delta u} \right) \right] f_n \\
&+ \sum_{n=2}^{\mathcal{N}+1} u_{n-1} \left[g_{x_{n-1/2}} \left(\frac{d_x}{2} + \frac{c_x}{\Delta u} \right) \right] f_n - \sum_{n=0}^{\mathcal{N}-1} u_{n+1} \left[g_{x_{n+1/2}} \left(\frac{d_x}{2} - \frac{c_x}{\Delta u} \right) \right] f_n \Bigg\} \\
&= \gamma \left[\sum_{n=1}^{\mathcal{N}} (u_n - u_{n+1}) \, g_{x_{n+1/2}} \left(\frac{d_x}{2} - \frac{c_x}{\Delta u} \right) - (u_n - u_{n-1}) \, g_{x_{n-1/2}} \left(\frac{d_x}{2} + \frac{c_x}{\Delta u} \right) \right] f_n \\
&= \gamma \left[\sum_{n=1}^{\mathcal{N}} g_{x_{n+1/2}} \left(c_x - d_x \frac{\Delta u}{2} \right) - g_{x_{n-1/2}} \left(c_x + d_x \frac{\Delta u}{2} \right) \right] f_n \quad , \qquad (122)
\end{aligned}
$$

where we have used the boundary conditions $g_{x_{1/2}} = g_{x_{\mathcal{N}+1/2}} = 0$.

Note that the power absorbed from the electric field can also be written as

- for a PT situation (see equations (13a), (93b)-(93c), (92a) and (122))

$$
\begin{aligned}
\frac{\Theta_E^{\mathrm{PT}}}{N} &= \gamma \left(\frac{E}{N} \right)^2 \sum_{n=1}^{\mathcal{N}} \left[g^{\mathrm{PT}}_{E_{n+1/2}} - g^{\mathrm{PT}}_{E_{n-1/2}} \right] f_n \\
&= \gamma \left(\frac{E}{N} \right)^2 \left[\sum_{n=2}^{\mathcal{N}+1} g^{\mathrm{PT}}_{E_{n-1/2}} f_{n-1} - \sum_{n=1}^{\mathcal{N}} g^{\mathrm{PT}}_{E_{n-1/2}} f_n \right] \\
&= -\gamma \left(\frac{E}{N} \right)^2 \sum_{n=1}^{\mathcal{N}} g^{\mathrm{PT}}_{E_{n-1/2}} (f_n - f_{n-1}) \\
&= \left[-\gamma \sum_{n=1}^{\mathcal{N}} \frac{u_{n-1/2}}{3\Omega_{\mathrm{PT}_{n-1/2}}} (f_n - f_{n-1}) \right] \left(\frac{E}{N} \right)^2 \\
&= \left[-\gamma \sum_{n=1}^{\mathcal{N}} D^{\mathrm{PT}}_{n-1/2} (f_n - f_{n-1}) \right] \left(\frac{E}{N} \right)^2 = (\mu_e N) \left(\frac{E}{N} \right)^2 \quad , \qquad (123a)
\end{aligned}
$$

where we have used the boundary conditions $g^{\mathrm{PT}}_{E_{1/2}} = g^{\mathrm{PT}}_{E_{\mathcal{N}+1/2}} = 0$, and equation (113).

- for a SST situation (see equations (13b), (93b)-(93c), (92a)-(92b) and (122))

$$
\begin{aligned}
\frac{\Theta_E^{\mathrm{SST}}}{N} &= \gamma \left(\frac{E}{N} \right)^2 \sum_{n=1}^{\mathcal{N}} \left[g^{\mathrm{SST}}_{E_{n+1/2}} - g^{\mathrm{SST}}_{E_{n-1/2}} \right] f_n - \gamma \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \frac{1}{2} \sum_{n=1}^{\mathcal{N}} \left[g^{\mathrm{SST}}_{E_{n+1/2}} + g^{\mathrm{SST}}_{E_{n-1/2}} \right] f_n \Delta u \\
&= \gamma \Bigg[\left(\frac{E}{N} \right)^2 \sum_{n=2}^{\mathcal{N}+1} g^{\mathrm{SST}}_{E_{n-1/2}} f_{n-1} - \left(\frac{E}{N} \right)^2 \sum_{n=1}^{\mathcal{N}} g^{\mathrm{SST}}_{E_{n-1/2}} f_n \\
&\quad - \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \frac{1}{2} \sum_{n=2}^{\mathcal{N}+1} g^{\mathrm{SST}}_{E_{n-1/2}} f_{n-1} \Delta u - \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \frac{1}{2} \sum_{n=1}^{\mathcal{N}} g^{\mathrm{SST}}_{E_{n-1/2}} f_n \Delta u \Bigg] \\
&= -\gamma \left(\frac{E}{N} \right)^2 \sum_{n=1}^{\mathcal{N}} g^{\mathrm{SST}}_{E_{n-1/2}} (f_n - f_{n-1}) - \gamma \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \sum_{n=1}^{\mathcal{N}} g^{\mathrm{SST}}_{E_{n-1/2}} \frac{f_{n-1} + f_n}{2} \Delta u \\
&= \left[-\gamma \sum_{n=1}^{\mathcal{N}} \frac{u_{n-1/2}}{3\Omega_{\mathrm{SST}_{n-1/2}}} (f_n - f_{n-1}) \right] \left(\frac{E}{N} \right)^2 - \left[\gamma \sum_{n=1}^{\mathcal{N}} \frac{u_{n-1/2}}{3\Omega_{\mathrm{SST}_{n-1/2}}} f_{n-1/2} \Delta u \right] \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \\
&= \left[-\gamma \sum_{n=1}^{\mathcal{N}} D^{\mathrm{SST}}_{n-1/2} (f_n - f_{n-1}) \right] \left(\frac{E}{N} \right)^2 - \left[\gamma \sum_{n=1}^{\mathcal{N}} D^{\mathrm{SST}}_{n-1/2} f_{n-1/2} \Delta u \right] \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \\
&= (\mu_e N) \left(\frac{E}{N} \right)^2 - (D_e N) \frac{E}{N} \frac{\alpha_{\mathrm{eff}}}{N} \quad , \qquad (123b)
\end{aligned}
$$

where we have used the boundary conditions $g^{\rm SST}_{E_{1/2}} = g^{\rm SST}_{E_{\mathcal{N}+1/2}} = 0$, and equations (112)-(113).

Furthermore, the discretized form of the collisional gain-terms writes (see equation (78e))

$$\frac{\Theta^{\rm gain}_{x={\rm el,CAR}}}{N}\gamma\int_0^\infty u\frac{d}{du}\left[g_x(u)c_x\frac{df(u)}{du}\right]du \to \gamma\sum_{n=1}^{\mathcal{N}} c_x\left[g_{x_{n+1/2}} - g_{x_{n-1/2}}\right]f_n \quad . \tag{124}$$

- terms involving the upflux function $G_{\rm ee}(u)$
  The $\Theta_{\rm ee}/N$ terms involving the upflux function $G_{\rm ee}(u)$ are specifically handled as follows (see equation (100))

$$\begin{aligned}
\frac{\Theta_{ee}}{N} &= -\frac{1}{N}\sum_n\int_{u_{n-1/2}}^{u_{n+1/2}} u\frac{dG_{\rm ee}(u)}{du}du \\
&= \gamma\sum_{n=1}^{\mathcal{N}} u_n\left[A_{n-1}f_{n-1} - (A_n+B_n)f_n + B_{n+1}f_{n+1}\right]\Delta u \\
&= \gamma\left[\sum_{n=0}^{\mathcal{N}-1} u_{n+1}A_nf_n - \sum_{n=1}^{\mathcal{N}} u_n(A_n+B_n)f_n + \sum_{n=2}^{\mathcal{N}+1} u_{n-1}B_nf_n\right]\Delta u \\
&= \gamma\left[\sum_{n=1}^{\mathcal{N}-1}(u_{n+1}-u_n)A_nf_n - \sum_{n=2}^{\mathcal{N}}(u_n-u_{n-1})B_nf_n\right]\Delta u \\
&= \gamma\sum_{n=1}^{\mathcal{N}}(A_n-B_n)f_n(\Delta u)^2 \qquad (125a) \\
\frac{\Theta^{\rm gain}_{ee}}{N} &= \gamma\sum_n\int_{u_{n-1/2}}^{u_{n+1/2}} u\frac{d}{du}\left[g_{ee}(u)J(u)\frac{df(u)}{du}\right]du \\
&= \gamma\sum_{n=1}^{\mathcal{N}}(A_{J_n}-B_{J_n})f_n(\Delta u)^2 \quad , \qquad (125b)
\end{aligned}$$

  where we have used the boundary conditions $A_0 = A_{\mathcal{N}} = B_1 = B_{\mathcal{N}+1} = 0$.

- inelastic/superelastic collision terms
  The terms $\Theta^{\rm loss}_{\rm inel}/N$ and $\Theta^{\rm gain}_{\rm sup}/N$, relative to the power lost/gained in inelastic/superelastic collisions are obtained as follows (see equation (103a))

$$\begin{aligned}
\frac{\Theta^{\rm loss}_{(i,j),{\rm inel}}}{N} &= \gamma\sum_n\int_{u_{n-1/2}}^{u_{n+1/2}} S_{(i,j)}\big|_{\rm inel}(u)udu \\
&= \delta_i\gamma\sum_{n=1}^{\mathcal{N}} u_n\left[u_{n+l_{i,j}}\frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2}f_{n+l_{i,j}} - u_n\frac{\sigma_{(i,j)_n}+\sigma_{(i,j)_{n+1}}}{2}f_n\right]\Delta u \\
&= \delta_i\gamma\left[\sum_{n=l_{i,j}+1}^{l_{i,j}+\mathcal{N}} u_{n-l_{i,j}}u_n\frac{\sigma_{(i,j)_n}+\sigma_{(i,j)_{n+1}}}{2}f_n - \sum_{n=1}^{\mathcal{N}} u_nu_n\frac{\sigma_{(i,j)_n}+\sigma_{(i,j)_{n+1}}}{2}f_n\right]\Delta u \\
&= \delta_i\gamma\sum_{n=1}^{\mathcal{N}}\left(u_{n-l_{i,j}}-u_n\right)u_n\frac{\sigma_{(i,j)_n}+\sigma_{(i,j)_{n+1}}}{2}f_n\Delta u \\
&= -\delta_i\gamma\sum_{n=1}^{\mathcal{N}}\left[u_n\frac{\sigma_{(i,j)_n}+\sigma_{(i,j)_{n+1}}}{2}f_n\Delta u\right]V_{i,j} = -\delta_iC_{i,j}V_{i,j} \qquad (126a) \\
\frac{\Theta^{\rm loss}_{\rm inel}}{N} &= \sum_{i,j>i}\frac{\Theta^{\rm loss}_{(i,j),{\rm inel}}}{N} = \sum_{i,j>i} -\delta_iC_{i,j}V_{i,j} \quad ; \qquad (126b)
\end{aligned}$$

$$
\begin{aligned}
\frac{\Theta^{\text{gain}}_{(i,j),\text{sup}}}{N} &= \gamma \sum_n \int_{u_{n-1/2}}^{u_{n+1/2}} S_{(i,j)}\big|_{\text{sup}}(u) u du \\
&= \delta_j \frac{g_i}{g_j} \gamma \sum_{n=1}^{\mathcal{N}} u_n \left[u_n \frac{\sigma_{(i,j)_n} + \sigma_{(i,j)_{n+1}}}{2} f_{n-l_{i,j}} - u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_n \right] \Delta u \\
&= \delta_j \frac{g_i}{g_j} \gamma \left[\sum_{n=l_{i,j}+1}^{\mathcal{N}} u_n u_n \frac{\sigma_{(i,j)_n} + \sigma_{(i,j)_{n+1}}}{2} f_{n-l_{i,j}} - \sum_{n=1}^{\mathcal{N}-l_{i,j}} u_n u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_n \right] \Delta u \\
&= \delta_j \frac{g_i}{g_j} \gamma \sum_{n=1}^{\mathcal{N}} \left(u_{n+l_{i,j}} - u_n \right) u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_n \Delta u \\
&= \delta_j \frac{g_i}{g_j} \gamma \sum_{n=1}^{\mathcal{N}} \left[u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_n \Delta u \right] V_{i,j} = \delta_j C_{j,i} V_{i,j} \qquad (127a) \\
\frac{\Theta^{\text{gain}}_{\text{sup}}}{N} &= \sum_{i,j>i} \frac{\Theta^{\text{gain}}_{(i,j),\text{sup}}}{N} = \sum_{i,j>i} \delta_j C_{j,i} V_{i,j} \quad . \qquad (127b)
\end{aligned}
$$

In the previous equations we have used the discretized form of the inelastic / superelastic rate coefficients (see equations (68a)-(68b))

$$
\begin{aligned}
C_{i,j} &= \gamma \sum_{n=1}^{\mathcal{N}} u_n \frac{\sigma_{(i,j)_n} + \sigma_{(i,j)_{n+1}}}{2} f_n \Delta u \qquad (128a) \\
C_{j,i} &= \gamma \frac{g_i}{g_j} \sum_{n=1}^{\mathcal{N}} u_{n+l_{i,j}} \frac{\sigma_{(i,j)_{n+l_{i,j}}} + \sigma_{(i,j)_{n+l_{i,j}+1}}}{2} f_n \Delta u \quad , \qquad (128b)
\end{aligned}
$$

as well as the boundary conditions $\sigma_{(i,j)_n} = 0$ for $n \leq l_{i,j}$, and $f_n = 0$ for $n > \mathcal{N}$ and $n < 1$.

- ionisation collision term
  The term Θ_{ion}/N, relative to the power lost in ionisation collisions is obtained as follows (see equations (103b) and (117))

$$
\begin{aligned}
\frac{\Theta_{i,\text{ion}}}{N} &= \gamma \sum_n \int_{u_{n-1/2}}^{u_{n+1/2}} S_{i,\text{ion}}(u) u du \\
&= \delta_i \gamma \sum_{n=1}^{\mathcal{N}} u_n \left\{ \sum_{m=n+l_{i,\text{ion}}+1}^{2n+l_{i,\text{ion}}} u_m \sigma^{\text{sec}}_{(i,\text{ion})_{(m+1/2, m-n-l_{i,\text{ion}}+1/2)}} f_m \Delta u \right. \\
&+ \left. \sum_{m=2n+l_{i,\text{ion}}}^{\mathcal{N}} u_m \sigma^{\text{sec}}_{(i,\text{ion})_{(m+1/2,n+1/2)}} f_m \Delta u - u_n \sigma_{(i,\text{ion})_{n+1/2}} f_n \right\} \Delta u \\
&= \delta_i \gamma \left\{ \sum_{m=1}^{\mathcal{N}} u_m f_m \left[\sum_{k=1}^{(m-l_{i,\text{ion}})/2} u_{m-k-l_{i,\text{ion}}} \sigma^{\text{sec}}_{(i,\text{ion})_{(m+1/2,k+1/2)}} \Delta u \right] \right. \\
&+ \left. \sum_{m=1}^{\mathcal{N}} u_m f_m \left[\sum_{n=1}^{(m-l_{i,\text{ion}})/2} u_n \sigma^{\text{sec}}_{(i,\text{ion})_{(m+1/2,n+1/2)}} \Delta u \right] - \sum_{n=1}^{\mathcal{N}} u_n^2 f_n \sigma_{(i,\text{ion})_{n+1/2}} \right\} \Delta u \\
&= \delta_i \gamma \left\{ \sum_{m=1}^{\mathcal{N}} u_m f_m \left[\sum_{k=1}^{(m-l_{i,\text{ion}})/2} \left(m - \frac{1}{2} - k + \frac{1}{2} - l_{i,\text{ion}} - \frac{1}{2} \right) \Delta u \, \sigma^{\text{sec}}_{(i,\text{ion})_{(m+1/2,k+1/2)}} \Delta u \right] \right. \\
&+ \left. \sum_{m=1}^{\mathcal{N}} f_m \left[\sum_{n=1}^{(m-l_{i,\text{ion}})/2} u_m u_n \sigma^{\text{sec}}_{(i,\text{ion})_{(m+1/2,n+1/2)}} \Delta u \right] - \sum_{n=1}^{\mathcal{N}} u_n^2 f_n \sigma_{(i,\text{ion})_{n+1/2}} \right\} \Delta u
\end{aligned}
$$

$$
\begin{aligned}
&= \delta_i\gamma\left\{\sum_{m=1}^{\mathcal{N}} u_m^2 f_m\left[\sum_{k=1}^{(m-l_{i,\mathrm{ion}})/2}\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,k+1/2)}}\Delta u\right] - \sum_{m=1}^{\mathcal{N}} f_m\left[\sum_{k=1}^{(m-l_{i,\mathrm{ion}})/2} u_m u_k \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,k+1/2)}}\Delta u\right]\right.\\
&- \sum_{m=1}^{\mathcal{N}} u_m f_m\left[\sum_{k=1}^{(m-l_{i,\mathrm{ion}})/2}\left(l_{i,\mathrm{ion}}+\frac{1}{2}\right)\Delta u\,\sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,k+1/2)}}\Delta u\right]\\
&+ \left.\sum_{m=1}^{\mathcal{N}} f_m\left[\sum_{n=1}^{(m-l_{i,\mathrm{ion}})/2} u_m u_n \sigma^{\mathrm{sec}}_{(i,\mathrm{ion})_{(m+1/2,n+1/2)}}\Delta u\right] - \sum_{n=1}^{\mathcal{N}} u_n^2 f_n \sigma_{(i,\mathrm{ion})_{n+1/2}}\right\}\Delta u\\
&= \delta_i\gamma\left\{\sum_{m=1}^{\mathcal{N}} u_m^2 f_m \sigma_{(i,\mathrm{ion})_{m+1/2}} - u_{l_{i,\mathrm{ion}}+1}\sum_{m=1}^{\mathcal{N}} u_m f_m \sigma_{(i,\mathrm{ion})_{n+1/2}} - \sum_{n=1}^{\mathcal{N}} u_n^2 f_n \sigma_{(i,\mathrm{ion})_{n+1/2}}\right\}\Delta u\\
&= -\delta_i u_{l_{i,\mathrm{ion}}+1}\gamma\sum_{m=1}^{\mathcal{N}} u_m f_m \sigma_{(i,\mathrm{ion})_{n+1/2}}\Delta u = -\delta_i C_{i,\mathrm{ion}} V^{\star}_{i,\mathrm{ion}} \qquad (129)
\end{aligned}
$$

$$
\frac{\Theta_{\mathrm{ion}}}{N} = \sum_i \frac{\Theta_{i,\mathrm{ion}}}{N} = \sum_i -\delta_i C_{i,\mathrm{ion}} V^{\star}_{i,\mathrm{ion}} \quad . \qquad (130)
$$

In the previous equations, the ionisation potential is modified by an extra shift of $\Delta u/2$, due to the discretization of the ionisation collisional operator,

$$
V^{\star}_{i,\mathrm{ion}} \equiv u_{l_{i,\mathrm{ion}}+1} = l_{i,\mathrm{ion}}\Delta u + \frac{\Delta u}{2} \quad , \qquad (131)
$$

and we have used the discretized form of the ionisation rate coefficient (see equations (69a) and (128a))

$$
C_{i,\mathrm{ion}} = \gamma\sum_{n=1}^{\mathcal{N}} u_n \sigma_{(i,\mathrm{ion})_{n+1/2}} f_n \Delta u \quad . \qquad (132)
$$

- attachment collision term
  The term Θ_{att}/N, relative to the power lost in attachment collisions is obtained as follows (see equation (103c))

$$
\begin{aligned}
\frac{\Theta_{i,\mathrm{att}}}{N} &= \gamma\sum_n\int_{u_{n-1/2}}^{u_{n+1/2}} S_{i,\mathrm{att}}(u)u\,du\\
&= -\gamma\sum_{n=1}^{\mathcal{N}}\delta_i\,(u_n)^2\,\frac{\sigma_{(i,\mathrm{att})_n}+\sigma_{(i,\mathrm{att})_{n+1}}}{2} f_n\Delta u \qquad (133)\\
\frac{\Theta_{\mathrm{att}}}{N} &= \sum_i \frac{\Theta_{i,\mathrm{att}}}{N} \quad . \qquad (134)
\end{aligned}
$$

The term Θ_{coll}/N in equation (119a) is numerically obtained from expressions (122), (125b), (126b), (127b), (130) and (134) as

$$
\frac{\Theta_{\mathrm{coll}}}{N} = \frac{\Theta_{\mathrm{el}}}{N} + \frac{\Theta_{\mathrm{CAR}}}{N} + \frac{\Theta_{\mathrm{ee}}}{N} + \frac{\Theta_{\mathrm{inel}}}{N} + \frac{\Theta_{\mathrm{sup}}}{N} + \frac{\Theta_{\mathrm{ion}}}{N} + \frac{\Theta_{\mathrm{att}}}{N} \quad . \qquad (135)
$$

In stationary and steady-state simulations, the code evaluates the relative power balance equation, dividing (77a) by the *reference power-density* (per electron at unit gas density) Θ_{ref}/N, with the latter defined as the sum of all power-gain terms. Usually, the stationary / steady-state power balance is satisfied to at least $\sim 10^{-10}\%$, and the user can define a threshold value for the relative power balance above which the code returns a warning message (see User Guide [12]).

In time-dependent simulations, equation (77a) is as accurate as the numerical solution of the EBE, being satisfied to at least $\sim 1\%$ (according to the relative tolerance imposed as parameter to the ode solver, see User Guide [12]).

Note that equation (77a) is directly satisfied by solving the EBE. Thus, for calculations using the LoKI-B+C simulation tool, the power balance for the electrons is updated at every call of LoKI-B, as part of the strategy for the correct closure of the global model (see section 4).

5.1.4 Conservation conditions for the electron-electron collision operator

Energy conservation for electron-electron collisions
Electron-electron collisions should not affect the net gain/loss of energy by the electrons, which means that

$$\begin{aligned}\frac{\Theta_{ee}}{N} &= -\frac{1}{N}\int_0^\infty u\frac{dG_{ee}(u)}{du}du = \frac{1}{N}\int_0^\infty G_{ee}(u)du \\ &= -\gamma\int_0^\infty\left[g_{ee}(u)\left(I(u)f(u)+J(u)\frac{df(u)}{du}\right)\right]du \\ &= -\gamma g_{ee}\left\{\int_0^\infty\left[I(u)f(u)-\frac{dJ(u)}{du}f(u)\right]du+\left[J(u)f(u)\right]_0^\infty\right\}=0 \quad , \end{aligned} \tag{136a}$$

where in the last expression we have used the fact that

$$g_{ee}\propto\frac{u^{3/2}}{v^3}=\text{const} \quad , \tag{136b}$$

and performed an integration by parts of the last term.

The Spitzer integral J, after an integration by parts, can be written as (see (99c))

$$J(u)=\frac{2}{3}\left[\int_0^u f(u')u'^{3/2}du'+u^{3/2}\int_u^\infty f(u')du'\right]=\int_0^u u'^{1/2}\left[\int_{u'}^\infty f(u'')du''\right]du' \quad , \tag{136c}$$

which yields

$$\frac{dJ(u)}{du} = u^{1/2}\int_u^\infty f(u')du' \tag{136d}$$

$$\left[J(u)f(u)\right]_0^\infty = \int_0^\infty u'^{1/2}\left[\int_{u'}^\infty f(u'')du''\right]du'\times f(\infty)-0\longrightarrow 0 \quad , \tag{136e}$$

where we have used Leibnitz law and the fact that the EEDF vanishes at ∞.

Moreover, noting that the derivative of the Spitzer integral I yields (see (99b))

$$I(u)=\int_0^u f(u')u'^{1/2}du' \Longrightarrow \frac{dI(u)}{du}=f(u)u^{1/2} \quad , \tag{136f}$$

one can write using (136d),

$$\begin{aligned}\int_0^\infty\frac{dJ(u)}{du}f(u)du &= \int_0^\infty f(u)u^{1/2}\int_u^\infty f(u')du'du \\ &= \int_0^\infty\frac{dI(u)}{du}\int_u^\infty f(u')du'du \\ &= \left[I(u)\int_u^\infty f(u')du'\right]_0^\infty+\int_0^\infty I(u)f(u)du \quad , \end{aligned} \tag{136g}$$

where the last expression was obtained with an integration by parts.

Finally, using (136e) and (136g) in (136a) one obtains

$$\begin{aligned}\frac{\Theta_{ee}}{N} &= -\gamma g_{ee}\left\{\int_0^\infty\left[I(u)f(u)-I(u)f(u)\right]du-\left[I(u)\int_u^\infty f(u')du'\right]_0^\infty\right\} \\ &= \gamma g_{ee}\left\{\int_0^\infty f(u)u^{1/2}du\times 0-0\times\int_0^\infty f(u)du\right\}\longrightarrow 0 \quad , \end{aligned} \tag{136h}$$

thus confirming the result of equation (136a).

Numerically, this result introduces a relationship between the matrix terms of the electron-electron collision operator. From (125b) and (101a)- (101b), one has

$$\gamma \sum_{n=1}^{\mathcal{N}} (A_n - B_n) f_n (\Delta u)^2 = 0 \implies \gamma \sum_{n=1}^{\mathcal{N}} \left[\sum_{m=1}^{\mathcal{N}} a_{n,m} - \sum_{m=1}^{\mathcal{N}} b_{n,m} \right] f_m f_n (\Delta u)^2 = 0$$
$$\implies \sum_{n,m=1}^{\mathcal{N}} (a_{n,m} - b_{n,m}) f_m f_n (\Delta u)^2 = 0 \quad , \tag{137a}$$

which means that the matrix multiplying the EEDFs should be antisymmetric satisfying

$$a_{n,m} - b_{n,m} = -a_{m,n} + b_{m,n} \tag{137b}$$

or

$$a_{n,m} - b_{m,n} = -a_{m,n} + b_{n,m} = 0 \Longrightarrow b_{m,n} = a_{n,m} \quad . \tag{137c}$$

Maxwellian solution for collisional equilibrium

Under the exclusive action of electron-electron collisions, the EBE (7a), (8a) or (19a) writes

$$\frac{dG_{ee}(u)}{du} = 0 \implies G_{ee}(u) = 0$$
$$\implies g_{ee}(u) \left(I(u) f(u) + J(u) \frac{df(u)}{du} \right) = 0 \quad , \tag{138a}$$

where we have used the boundary conditions $G_{ee}(0) = G_{ee}(\infty) = 0$ and equation (100).

Equation (138a) can be rewritten using (99b), (136b) and (136c) to obtain

$$f(u) \int_0^u u'^{1/2} f(u') du' + \frac{df(u)}{du} \int_0^u u'^{1/2} \left[\int_{u'}^{\infty} f(u'') du'' \right] du' = 0 \quad . \tag{138b}$$

This identity is satisfied if f is a Maxwellian distribution function, since in this case

$$f(u) = a \int_u^{\infty} f(u') du' \Longrightarrow \frac{df(u)}{du} = -a f(u) \Longrightarrow f(u) = C \exp(-au) \quad , \tag{138c}$$

with a and C constants.

Numerically, this equilibrium situation implies that (see (100) and (137c))

$$\sum_{m=1}^{\mathcal{N}} (a_{n-1,m} f_m) f_{n-1} - \sum_{m=1}^{\mathcal{N}} (a_{n,m} f_m + b_{n,m} f_m) f_n + \sum_{m=1}^{\mathcal{N}} (b_{n+1,m} f_m) f_{n+1} = 0$$
$$\Longrightarrow \sum_{m=1}^{\mathcal{N}} (a_{n-1,m} f_m) f_{n-1} - \sum_{m=1}^{\mathcal{N}} (a_{n,m} f_m + a_{m,n} f_m) f_n + \sum_{m=1}^{\mathcal{N}} (a_{m,n+1} f_m) f_{n+1} = 0$$
$$\Longrightarrow \sum_{m=1}^{\mathcal{N}} a_{n-1,m} f_m f_{n-1} - \sum_{m=1}^{\mathcal{N}} a_{n,m} f_m f_n - \sum_{m=1}^{\mathcal{N}-1} a_{m,n} f_m f_n + \sum_{m=1}^{\mathcal{N}-1} a_{m,n+1} f_m f_{n+1} = 0 \tag{139a}$$
$$\Longrightarrow \sum_{m=1}^{\mathcal{N}} a_{n-1,m} f_m f_{n-1} - \sum_{m=2}^{\mathcal{N}} a_{m-1,n} f_{m-1} f_n - \sum_{m=1}^{\mathcal{N}} a_{n,m} f_m f_n + \sum_{m=2}^{\mathcal{N}} a_{m-1,n+1} f_{m-1} f_{n+1} = 0$$
$$\Longrightarrow \sum_{m=1}^{\mathcal{N}} [a_{n-1,m} f_m f_{n-1} - a_{m-1,n} f_{m-1} f_n] - \sum_{m=1}^{\mathcal{N}} [a_{n,m} f_m f_n - a_{m-1,n+1} f_{m-1} f_{n+1}] = 0 \quad ,$$

where we have used the boundary conditions $A_0 = A_{\mathcal{N}} = 0$ or, equivalently, $a_{0,k} = a_{\mathcal{N},k} = 0$, and where the EEDFs are Maxwellian distribution functions satisfying

$$\begin{aligned} f_{m-1} f_{n+1} &= C^2 \exp(-a u_{m-1}) \exp(-a u_{n+1}) \\ &= C^2 \exp(-a u_m) \exp(-a u_n) \exp(a \Delta u) \exp(-a \Delta u) \\ &= f_m f_n \quad . \end{aligned} \tag{139b}$$

Equations (139a)-(139b) allow defining a relationship between the upper/lower matrix terms a_{mn} of the electron-electron collision operator

$$a_{n,m} = a_{m-1,n+1}\,,\ 1 \le n \le \mathcal{N}-1 \quad ;\, 1 \le m \le \mathcal{N}$$

or to balance out the contributions from the upper/lower matrix terms

$$a_{n,m} = \sqrt{a_{n,m} \times a_{m-1,n+1}}\,,\ 1 \le n \le \mathcal{N}-1 \quad ;\, 1 \le m \le \mathcal{N}\,. \tag{140}$$

5.1.5 Solution of LoKI-B

After discretisation, the isotropic part of the EBE can be written as $\sum_{m=1}^{\mathcal{N}} C_{n,m}(f) f_m = 0$, which is solved coupled to the normalisation condition $\sum_{m=1}^{\mathcal{N}} f_m \sqrt{u_m} \Delta u = 1$.

The matrix $C(f)$ is nonlinear due to the presence of the following terms.

- The temporal/spatial growth-term of the electron density, associated with non-conservative mechanisms, involves either the effective ionisation rate coefficient $\langle \nu_{\text{eff}} \rangle / N$ or the reduced-effective Townsend coefficient α_{eff}/N, both being functions of the EEDF (see section 3.1.1).
- The expression of the electron-electron upflux G_{ee} involves linear functions of the EEDF, as confirmed by (12d) and (101a)-(101b).

The EBE is solved as a balance equation, written as in (76a), (76b), (76c), (76d), using the solution methods indicated below (see User Guide [12] for details about the numerical calling of the solution methods)

- direct solution method (successively applied in the presence of nonlinear terms), using `mixingDirectSolutions` as `algorithm`
  - for steady-state simulations with a temporal growth model (equation (76a))
  - for steady-state simulations with a spatial growth model (equation (76b));
- relaxation method, using `temporalIntegration` as `algorithm`
  - for steady-state simulations with temporal growth model (equation (76a))
  - for time-dependent simulations with temporal or spatial growth models (equations (76c), (76d)).

For electropositive gases, when choosing `ionizationOperatorType` as `conservative` (see section 2.2 and User Guide [12]), the first term on the right-hand side of equations (76a), (76b), (76c), (76d) is set to zero.

The solution method adopted in solving the EBE depends also on its linear/nonlinear features.

- In the absence of nonlinear effects (*e.g.*, for electropositive gases, with the ionisation taken as a conservative mechanism and no electron-electron collisions), the EBE is solved adopting a very effective direct solution method. In this case the first equation of the Boltzmann system is added to the normalisation condition of the EEDF, *i.e.* $\sum_{m=1}^{\mathcal{N}} C_{1,m} f_m = \sum_{m=1}^{\mathcal{N}} \sqrt{u_m} \Delta u f_m = 1$, and the resulting non-homogeneous equation system becomes ready for direct inversion.
- When the matrix $C(f)$ contains nonlinear terms, the EBE is solved using an iterative method.
  - In the case of weak nonlinear effects (*e.g.*, low electronegativity and low ionisation degree), the solution can be obtained by using two iterative algorithms, based on successive direct solutions. The algorithm is represented in the simplified flowchart of figure 6, and it adopts:
    - a mixing of solutions to converge over the temporal growth parameter $\langle \nu_{\text{eff}} \rangle / N$ or the spatial growth parameter α_{eff}/N (see sections 3.1.1 and 5.1.2). In this case the convergence criterion looks for relative differences, between two consecutive iterations, below 10^{-10} for the growth parameter and below a user-prescribed value (typically 10^{-9}) for the EEDF;
    - a Newton-Raphson-based approach to converge over the electron-electron collision operator. In this case the convergence criterion looks for relative differences, between two consecutive iterations, below $\sim 10^{-9} - 10^{-10}$ for the ratio of the power "dissipated" in electron-electron collisions to the reference power $\Theta_{ee}/\Theta_{\text{ref}}$ (see section 3.1.2 for the definition of Θ_{ref}), and below a user-prescribed value (typically 10^{-9}) for the EEDF.

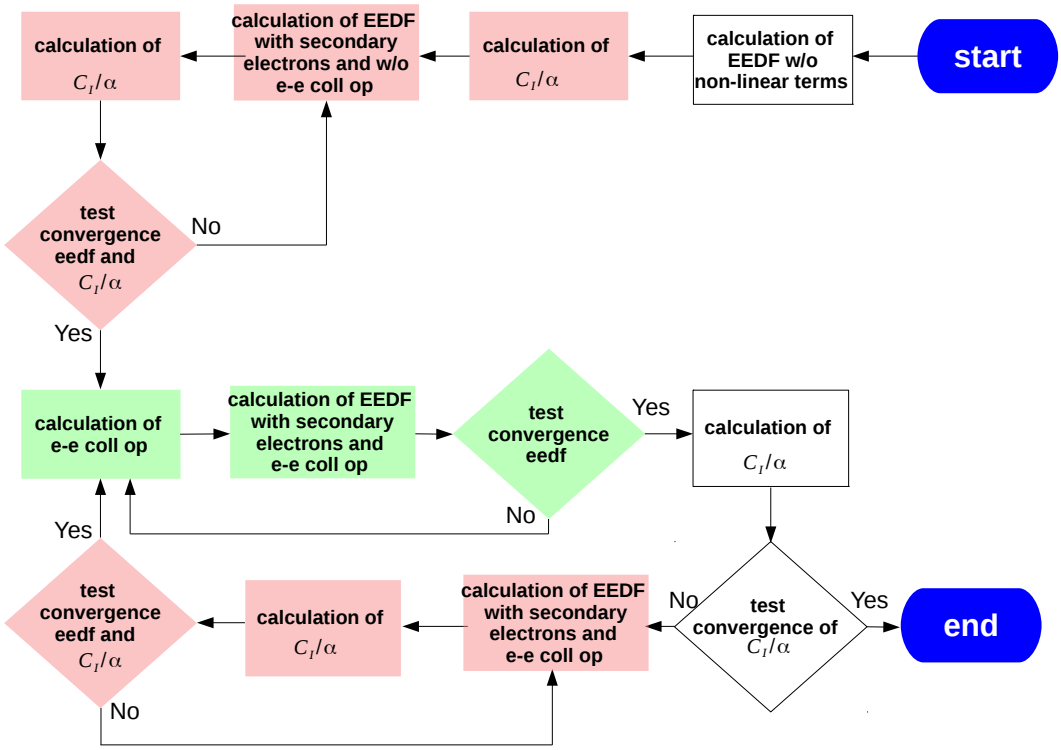


Figure 6: Simplified flowchart of LoKI-B, in the case of weak nonlinear effects. In red are the steps done within the ionisation routine. In green are the steps done within the electron-electron collisions routine.

When both algorithms are needed, the code ensures that the relative differences between the final EEDFs, obtained in each of the algorithms, are below a user-prescribed value for the EEDF.

– in the presence of strong nonlinear effects, the user is recommended to choose a relaxation method. In this case the EBE is solved with a time-dependent algorithm, using MATLAB's ordinary differential equation solver `ode15s` [68], and the user can specify the (optional) parameters contained in the structure of the solver, *e.g.* maximum step-size (typical value 10^{-7}). The default convergence criterion demands relative variations of the EEDF below a user-prescribed value (typically 10^{-9}), between two consecutive iterations.

The solution of the time-dependent EBE adopts also a relaxation method, using MATLAB's ordinary differential equation solver `ode15s` [68]. When electron-electron collisions are included in the calculations, LoKI-B can solve the system of equations

$$\frac{\partial f(t)}{\partial t} = C[f(t)]f(t) \tag{141a}$$

$$\frac{dn_e(t)}{dt} = \langle \nu_{\text{eff}}(t) \rangle n_e(t) \quad , \tag{141b}$$

subjected to some initial conditions $f(t=0) \equiv f_0$ and n_0. **Although this is implemented in LoKI-B, the option is not available by default. If needed, it can be easily activated by the user directly in the code.** The solver adopts a variable-step, variable-order algorithm suited for stiff problems, based on a set of highly-stable implicit integration formulas of orders 1 to 5 (optionally, it can also use the backward differentiation formulas). Specifically, the integrator estimates the error at each iteration and then adjusts the timestep Δt to improve either the accuracy (*i.e.* by decreasing the timestep) or the efficiency (*i.e.* by increasing the timestep). At each time t the solution is updated using $f(t) = f(t-\Delta t) + (\partial f(t)/\partial t)\Delta t$ and $n_e(t) = n_e(t-\Delta t) + (dn_e(t)/dt)\Delta t$. Again, the user can specify the (optional) parameters contained in the structure of the solver, *e.g.* absolute tolerance (typical value 10^{-300}), relative tolerance (typical value 10^{-6}) and maximum step-size (typical value 10^{-7}), which are applied by the solver to adjust Δt for each instant t. The relative tolerance works as convergence criterion, imposing the maximum value of the relative variations of the EEDF, between two consecutive iterations.

5.2 Numerical solution of LoKI-C

LoKI-C solves the system of non-linear rate-balance equations, describing the evolution in time of the densities of states involved in the gas/plasma kinetic scheme. The system of equations can be generally written as

$$\frac{\partial N_j}{\partial t} = \sum_r R_{r,j}(N_l, E/N, T_g, t, ...) \tag{142a}$$

$$\frac{\partial T_g}{\partial t} = \frac{\sum_{\text{mech}} P_{\text{mech}}(R_r, n_e, T_g, T_{\text{bound}}, t, ...)}{\sum_k N_k C_k / N_A} \quad , \tag{142b}$$

where the rates $R_{r,j}$ of reactions r, contributing to the creation/destruction of state j, can depend, explicitly or implicitly, on the absolute density of any state N_l, the reduced electric field E/N, the gas temperature T_g, the instant of time t, etc; and the power densities P_{mech}, due to different power-transfer mechanisms mech, can depend on the rates R_r for the creation/destruction of all the states, the electron density n_e, the gas and the boundary temperatures T_g and T_{bound}, the instant of time t, etc.

The system of equations (142a)-(142b) is solved using MATLAB's ordinary differential equation (ODE) solvers [69], which adopt time-dependent stiff algorithms (the rhs of equation (142b) is set to zero when the thermal model is not activated). The choice of the solver is defined by the user but, at the time of this writing, we strongly recommend the solver `ode15s`, because it has provided the best results in terms of accuracy and performance for the cases tested. The integration of the rate balance equations proceeds:

- for default initial conditions (adopted if the user does not provide other conditions) $N_0 = N$, $N_{l\neq 0} = 0$ and $T_{nw} = T_g$ (and additionally $T_w = T_{\text{ext}}$, when adopting the *external* boundary condition in the thermal model);
- until the final time(s) defined by the user, considering the (optional) parameters contained in the structure of the solver, *e.g.* relative tolerance (typical value 5×10^{-4}), absolute tolerance (10^{-10}) and maximum step-size (0.1).

LoKI describes the time evolution of discharges until steady-state, as well as post-discharges. The user must define adequate final integration times, according to the situations.

- active discharges are characterized by a *discharge time*. The discharge time should be carefully chosen, as to allow the densities of species to attain a steady-state situation, considering also the inflow/outflow conditions and the associative/dissociative nature of the gas mixture and the surface kinetics. Typical value is `dischargeTime`: 10 (s);
- post-discharges are characterized by both, a *discharge time* and a *post-discharge time*. The post-discharge time should be carefully chosen, as to yield a decrease in the densities of species within reasonable values. Typical value is `postDischargeTime`: 0.1 (s).

When simulating an active discharge, the algorithm performs a series of iterative cycles (see section 4) until convergence is achieved. The following values are recommended for configuring these cycles (see User Guide [12]).

- `pressureRelError`: 2× `RelTol`: 10^{-3} (default value); `pressureMaxIterations`:1 (to eliminate the pressure cycle for an isobaric flow model);
- `neutralityRelError`: 2× `RelTol`: 10^{-3} (default value); `neutralityMaxIterations`:500;
- `globalRelError`: 2× `RelTol`: 10^{-3} (default value); `globalMaxIterations`:100.

6 Validity limits and control (for LoKI-B)

As any numerical tool, LoKI-B provides a mere representation of the energy distribution of the electrons in a LTP, because it is based on a specific formulation with embedded approximations. Therefore, the tool should be used only in situations fitting its validity limits, and users are advised to carefully read the extensive literature on the two-term EBE [17], for obtaining details about its solution.

The two-term approximation adopted in LoKI-B to solve the EBE is valid in a situation of small anisotropies [17], corresponding to electron–neutral mean-free-paths much smaller than any characteristic dimension of the plasma container, and an energy gain from the electric field, between collisions, much smaller than the electron kinetic energy. For a homogeneous scenario like the one described here, the latter condition is the most relevant and can be approximately written as (see (70a) and (82d))

$$\Theta_E \sim \mu_e E^2 \ll \varepsilon \nu_c \quad , \tag{143}$$

which yields, for a Maxwellian EEDF at 1 eV temperature and a typical reduced collision frequency $\nu_c/N \sim \sigma_c \langle v \rangle \sim 10^{-19} \times 10^6 = 10^{-13}$ m^3 s^{-1}, $E/N \ll \sqrt{m_e \varepsilon / e}(\nu_c/N) \sim 300$ Td. This is a conservative limit for the reduced electric-field, often exceeded in calculations because the two-term approximation yields fairly good results still.
However, users are advised to check the EEDF results when taking E/N values much above the estimated validity limit, namely by comparing the solution obtained for the isotropic and the anisotropic parts of the electron distribution function.

Also, HF calculations are valid only if the oscillation frequency of the electric field is much larger than the typical frequency for the electron energy relaxation, which can be approximately written as (see (12b))

$$\omega \gg (m_e/M) N (\nu_c/N) \quad , \tag{144}$$

corresponding to $\omega/(2\pi) \gg 20$ MHz - 20 kHz for $\nu_c/N \sim 10^{-13}$ m^3 s^{-1}, $m_e/M \sim 5 \times 10^{-5}$, $T_g = 300$ K and $p \sim 10^5$ Pa - 130 Pa.
Again, users are advised to check the validity of the HF approximation for the particular working conditions considered.

LoKI-B contains several control features upon the input data and the calculation results, and in some cases it **returns warning messages** for the benefit of users. In particular:

- the code controls the maximum value $u_{\max}$ of the energy grid adopted, comparing it with the cutoff energy u_{cutoff} of the electron-neutral elastic momentum-transfer cross section.
  When $u_{\text{cutoff}} < u_{\max}$ a warning message is returned, and if the user accepts to proceed with the calculations these are carried out by setting to zero the values of any cross section from its corresponding u_{cutoff} and up to $u_{\max}$. **LoKI-B does not extrapolate cross sections beyond the cutoff energies defined in the input data-files**, again because it adopts an ontology that separates the tool and the data. The user is recommended to always careful examine the cross sections being used and, if needed, he/she should adopt adequate laws to extrapolate them, chosen according to the specific processes considered (*e.g.* dipole allowed or dipole forbidden electron excitation, ionisation, etc);

- the code **returns a warning message** if the calculation of the elastic momentum-transfer cross section from the corresponding effective **produces negative values for** $\sigma_{k,c}^{\text{el}}$ (see User Guide [12]), which is then set to zero. **In this case, users are advised to check the cross section data adopted**;

- **in stationary and steady-state simulations**, the code controls the quality of the Boltzmann-matrix inversion, by monitoring the error upon the fractional electron power-balance. The latter is obtained from (77a), by normalizing this equation with respect to a *reference power-density* Θ_{ref}, **defined for each calculation as the sum of all power-gain terms** (see section 3.1).
  The relative error $\left[\Theta_E + \Theta_{\text{coll}}^{\text{gain}} + \Theta_{\text{coll}}^{\text{loss}} + \Theta_{\text{growth}}\right]/\Theta_{\text{ref}}$ **(with typical values** $\sim 10^{-9} - 10^{-10}$**, and user-defined** `maxPowerBalanceRelError` **threshold for producing a warning message, see User Guide [12]) degrades for very-low** E/N**s**, due to the smallness of the power-density values for all the power-transfer channels, **and for very-high** E/N**s**, if the $u_{\max}$ grid-limit adopted in the calculations is not large enough to prevent ill-conditioned EBE-matrices.

Concerning cross section data, the user is further advised to carefully check

- if the electron-scattering cross sections are set to zero at the threshold energy, as it is coherently assumed by LoKI-B. Note that, excluding errors, the LXCat files obtained from the IST-Lisbon database comply with this recommendation;
- the maximum value $u_{\max}$ of the energy grid adopted, comparing it with the highest threshold $u_{\mathrm{th-max}}$ of the collisional cross sections associated with reactions for which superelastic mechanisms are activated. Note that $u_{\max} \gg u_{\mathrm{th-max}}$ should be ensured, to properly account for the superelastic effects.

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